Exploring the Potential to Reduce Aviation's Non-CO2 Climate Impact through Controlled Plume Intersections

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Abstract

Acknowledgements

Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes/Research Degree Programmes (delete as appropriate) and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

A. M. Turing

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1. Introduction

Aircraft act as high-altitude emissions vectors, transporting a number of radiatively and chemically active substances across vast regions of the globe. These substances induce a net global warming effect that constitutes 3.5% of the global climate impact due to anthropogenic sources [116]. Two-thirds of this impact result from aircraft non-CO_2 emissions, which have a variable impact depending on when and where they are released. This means that avoiding flying in climate-sensitive regions of the atmosphere has the potential to reduce aviation's climate impact significantly [145]. Furthermore, the environmental conditions within the aircraft exhaust plume result in additional chemical and microphysical processing that is not accounted for in global chemistry models. In high-density airspace, such as along the North Atlantic Flight Corridor, aircraft often fly in close proximity, giving rise to the superposition of their plumes and the accumulation of emissions contained within them [168]. It has been shown in [28] that the overlap of four consecutive plumes can influence the nonlinear chemistry and microphysics significantly, leading to saturation effects that can potentially provide a net climate impact reduction.

1.1. Motivation

It is the aim of this research to simulate this plume overlap concept over a range of atmospheric conditions and aircraft frequencies, to investigate the sensitivity of the climate response. The simulation results can then be used to test the mitigation potential of plume superposition through the application of aircraft trajectory optimisation and formation flight.

1.1.1. Aircraft Emissions

For aircraft to achieve and maintain steady level flight, a propulsion system is often required, providing the necessary thrust force to counteract aerodynamic drag and generate sufficient lift to overcome the aircraft's weight. In the commercial aviation industry, which is responsible for 88% of global aviation fuel usage [69], it is commonplace to use a gas turbine propulsion system operating on kerosene-based jet fuel. The stoichiometric combustion of kerosene with air produces the mechanical energy required to operate the engine and generate thrust, however it also emits a number of additional chemical species into the aircraft wake in the process. The emission species which are of great chemical and radiative importance are carbon dioxide (CO_2), carbon monoxide (CO), water vapour (H_2O), nitrogen oxides ($NO_x = NO + NO_2$), sulphur dioxide (SO_2) and particulate matter (PM).

1.1.2. Radiative Forcing of Aircraft Emissions

The most commonly used metric to compare the magnitudes of climate impact from a range of emission species is radiative forcing (RF). RF is defined as the perturbation to the net energy balance of the Earth-atmosphere system due to natural or anthropogenic factors of climate change, measured in watts per square metre [W/m2].

The RF of aviation CO_2 emissions is direct and well understood. Due to its thermodynamic and photochemical stability, CO_2 has a relatively long atmospheric lifetime on the order of 100 to 1,000 years. This means that aviation CO_2 emitted into the atmosphere simply serves to increase its atmospheric concentration, leading to the eventual distribution over global spatial scales and a steady, uniform increase in the planetary greenhouse effect. The non-CO_2 aircraft emission species however, are much more chemically active and hence have much shorter lifetimes and a more localised impact. They primarily contribute to climate change through two indirect radiative forcing mechanisms. Firstly, chemically active species such as NO_x and CO react with the surrounding air to modify chemical concentrations of key climate agents such as ozone (O_3) and methane (CH_4). Secondly, emissions of H2O and PM are responsible for the formation of condensation trails (contrails) and may even trigger the generation of additional cirrus clouds, both of which can affect the Earth's radiative balance. With the chemical and meteorological state of the atmosphere being highly variable with respect to both location and time, this implies that aircraft non-CO_2 emissions have a variable climate impact depending on when and where they are released. The spatio-temporal sensitivity of the climate to non-CO_2 aircraft emissions becomes particularly important when observing the chemical and physical effects that occur in the aircraft exhaust plume.

1.1.3. Aircraft Exhaust Plume Treatment

Following expulsion into the free atmosphere, exhaust species become entrained in the aircraft wake, forming a plume of elevated chemical concentrations which persist for 2 to 12 hours [1], before fully dispersing into ambient air. The build-up of non-CO_2 emissions in the plume gives rise to a number of nonlinear chemical and microphysical effects, which influence the ensuing atmospheric response by altering the net production rates of radiatively active gases and affecting contrail formation and persistence. It is often the case however, that in large-scale atmospheric models, emissions are instantaneously diluted into the volume of the smallest resolved grid cell, with dimensions according to the model's spatial resolution. The instantaneous dilution (ID) approach neglects the plume-scale processes, thus leading to discrepancies in the calculated climate response such as the overestimation of O3 production, CH_4 and CO destruction, and the increased rate of NO_x conversion to nitrogen reservoir species [159]. Furthermore, it is stated in [60] that the formation of ice in aircraft exhaust plumes may result in additional heterogeneous chemical reactions, that are not captured in global atmospheric models.

1.1.4. Plume Superposition in High-Density Airspace

Air traffic levels correlate with aviation demand, giving rise to a spatial and temporal inhomogeneity in the distribution of aircraft, and hence their emissions, across the globe. In a spatial sense, demand tends to be highest around and between densely-populated regions such as the United States, Europe and Asia, meaning flights often take place over mainland airspace and along the flight corridors connecting such regions. In a temporal sense, demand varies both diurnally (time of day) and seasonally (time of year), favouring daylight hours and the summer months [150].

Airspace regions that are particularly susceptible to high demand may see aircraft fly at minimum separation distances from one another. This close proximity flying leads to the superposition of aircraft exhaust plumes, and the accumulation of emissions entrained within them. [168] carries out an in-situ investigation into air traffic emissions signatures in the North Atlantic Flight Corridor (NAFC), finding that the superposition of 2-5 plumes lead to the levels of NO_x, SO_2 and PM exceeding background concentrations by factors of 30, 5 and 3, respectively. This highlights the severity with which amassing aircraft emission species can alter the local chemical composition of the atmosphere. Such changes to atmospheric conditions can significantly affect the non-CO_2 climate response of a flight through further chemical and microphysical processing.

1.1.5. Atmospheric Response to Plume Superposition

In [28], the notion of plume superposition and the extent to which it influences the atmospheric response was explored, by comparing results from an expanding plume model against an atmospheric model which assumes instant dilution. The analysis was carried out firstly on a single aircraft at cruise altitude, then on four aircraft flying in the same direction each separated by 1 hr. Observing the single aircraft's plume 10 hrs after emission, it was found that the instantaneous dilution approach overestimated O3 production by 33%, CH_4 destruction by 30% and CO destruction by 32%. When observing the response due to the four overlapping plumes however, the overestimation of O3 production increases to 77%, CH_4 destruction to 68% and CO destruction to 74%.

The marked difference in measured response for four overlapping plumes compared to a single plume serves as evidence that emissions saturation in superimposed plumes can significantly alter the chemical response of the atmosphere. However, further work needs to be done to clarify the direction and magnitude of the net change in climate impact resulting from these changes to the chemical composition. Moreover, the sensitivity of the chemical and climate response to plume superposition needs to be tested for a range of aircraft transit frequencies and ambient atmospheric conditions, so as to build up a visual representation over the phase space of the atmosphere.

In addition to chemical changes in the atmosphere, plume overlap can also influence the formation and persistence of aircraft contrails and the subsequent generation of aviation-induced cirrus clouds. [174] explored the effect of recurrent water vapour emission on contrail formation and persistence at typical aircraft cruising altitudes. It was discovered that continual contrail generation actually leads to dehydration of the surrounding atmosphere and a diminishing radiative forcing effect of ~15% in the affected areas. This outcome provides further impetus to explore the potential effects of plume superposition on the global warming induced by aviation.

1.1.6. Mitigation Potential

The significance of the plume overlap effect and its potential for influencing the atmospheric response brings forth the potential for controlled implementation of this effect, as a means of aviation climate impact mitigation. A near-term fuel burn reduction measure that is currently popular in the literature is the concept of formation flight, which has the potential to provide a complementary non-CO_2 climate reduction by exploiting the benefits of plume superposition. Formation flight involves the flight of two or more aircraft in aerodynamic formation, with the follower aircraft positioned in the smooth updraft of the leader aircraft's wake, reducing required lift and thrust, hence reducing fuel and CO_2 emissions by 5-8% per trip [215]. A fortuitous outcome of formation flight is however, the saturation of emissions in the trailing exhaust plumes, which can lead to the aforementioned nonlinear chemical and microphysical response that reduces ozone and contrail formation.

The non-CO_2 climate effects of a twin-aircraft formation flight are observed in [36], where a climate model was used to determine the changes in NO_x-related ozone production and contrail formation processes. The case study found that, despite a 1-3% increase in flown distance, CO_2 was reduced by 6% and NO_x was reduced by 11%. This resulted in a 5% reduction in ozone production efficiency due to NO_x saturation and a 48% contrail reduction due to mutual competition for available water vapour, resulting in a total climate impact reduction of approximately 23%. While part of the alleviated climate impact can be related to the decrease in total emissions due to wake energy retrieval of the follower aircraft, there is also further emphasis due to the saturation of emissions in the overlapping plumes of both aircraft involved. This demonstrates great potential for reducing both CO_2- and non-CO_2-induced climate effects from aviation, if such a scheme were to be carried out on a global scale.

Fleet-wide implementation of formation flight would however, pose a number of research obstacles, that must be overcome to maximise its cost-benefit potential, whilst ensuring the safe and orderly operation of aircraft in controlled airspace. Optimising aircraft trajectories to minimise fuel burn requires the consideration of nonlinear aircraft performance, wind and weather, payload, fuel load, and constraints set out by air traffic control [187]. Formation flight adds another variable to the fuel burn optimisation problem, maximising the time spent flying in formation whilst minimising deviation from the true optimal route. This is a research topic covered extensively in the literature [78], [147]. Optimising formation flight trajectories to minimise climate impact on the other hand, is a relatively novel concept, which requires deeper understanding of the sensitivity of the climate response to emissions under a range of different atmospheric states.

1.2. Thesis Objectives

The studies performed in [28] and [174] provide a thought-provoking insight into the notion of aircraft plume superposition, and [36] demonstrates its potential utilisation for aviation climate impact mitigation through formation flight. This thesis aims to expand the scope of analysis conducted into chemical saturation effects that occur in high-density airspace. It intends to do so by satisfying the following objectives:

- To conduct an expansive literature review into the state-of-the-art research surrounding aircraft emissions, aircraft exhaust plume dynamics, air traffic management and distribution, the local and global climate effects induced by aviation, and lastly, the mitigation of aviation climate impact, with a focus on operational strategies targeting non-CO_2 emissions.

- To utilise real-world air traffic data and first-order plume dispersion methods to assess the operational feasibility of aircraft exhaust plume overlap in today's airspace.

- To investigate the global spatial and temporal variability of NO_x saturation effects, using a world-leading atmospheric chemistry transport model (CTM) STOCHEM-CRI, which incorporates the chemistry scheme CRIv2.2-R5 [108].
- To develop a modelling framework to robustly investigate the chemical and physical saturation effects that influence the non-CO_2 climate impact of formation flights.
- To carry out a localised study on aircraft NO_x saturation effects using this modelling framework, in which parameters such as aircraft transit frequency and air temperature will be varied.

1.3. Thesis Structure

In essence, the thesis is aiming to solve one fundamental question:

"What is the sensitivity of the climate response to the superposition of two or more aircraft exhaust plumes, under a variety of atmospheric conditions?"

As such, the document is structured as follows: Chapter 2 introduces the science behind aviation climate impact, and explores a wide range of literature pertaining to aircraft emissions, their plume-scale effects, and the spatio-temporal sensitivity of the atmospheric response. Chapter 3 investigates the feasibility of aircraft exhaust plume intersections in modern airspace, through consideration of real-world air traffic data and rudimentary dispersion calculations, to estimate degree of plume overlap in high-density regions. In chapter 4, the potential to offset climate change through formation flying is investigated in more detail, with a focus on gas-phase chemical kinetics associated with NO_x emissions and their varying impact on ozone formation. This leads to the development of a localised plume chemistry modelling framework in chapter 5. In this chapter, the methodology behind a novel aircraft plume analysis tool is outlined and supporting models and methods are introduced. The model is also validated against its predecessor. In chapter 6, this modelling framework is then utilised in a formation flight study, to investigate the most critical parameters affecting the severity of the NO_x saturation phenomenon, and the benefits and drawbacks of formation flying in the context of aviation climate impact mitigation.

2. Aircraft Emissions, their Plume-Scale Effects, and the Spatio-Temporal Sensitivity of the Atmospheric Response

The content of this chapter is based on the following published journal article:

Tait, K. N.; Khan, M. A. H.; Bullock, S.; Lowenberg, M. H.; Shallcross, D. E., "Aircraft Emissions, their Plume-Scale Effects, and the Spatio-Temporal Sensitivity of the Atmospheric Response: a Review," *Aerospace*, vol. 9, no. 7, pp. 355, 2022. DOI: https://doi.org/10.3390/aerospace9070355.

This chapter incorporates material from that work with permission from the publisher and co-authors, where necessary.

2.1. Background

Whilst CO_2 emissions are often portrayed as the leading contributor to aviation-induced climate change, they are only responsible for a mere third of the climate impact. The remaining two-thirds are attributed to more reactive non-CO_2 emissions, NO_x, H_2O and PM [116]. These emission species interact with ambient air through chemical and microphysical processing, giving rise to the production and depletion of radiatively active substances that perturb the net energy balance of the atmosphere (e.g. NO_x-induced ozone production, contrail generation through H_2O and PM emissions etc.). The degree to which aircraft emissions induce a climatic response varies depending on the state of the background atmosphere (i.e. its chemical composition and meteorology) and the time of day and year in which emissions are released. This means that aviation climate impact is spatio-temporally sensitive, i.e. the same emissions released at different times and/or locations can lead to very different climate effects.

The dispersion of aircraft emissions occurs over great distance and time scales, with emissions entrained in the aircraft exhaust plume which spreads hundreds of kilometres over its lifetime of up to 12 hours [1], [112]. The elevated concentrations of emitted chemical species present within the plume result in additional nonlinear chemical (gas-phase and heterogeneous) and microphysical processes which are not accounted for in global chemistry models, due to the inherent assumption of instantaneous dispersion (ID) of emissions. The ID assumption models emissions as homogeneously mixed into the volume of the computational grid cell to which they are released, thus neglecting any subgrid scale nonlinear processing that may occur throughout the plume lifetime [151]. Plume-scale nonlinear effects are also further augmented in high-density airspace regions, where plumes intersect and the emissions contained within them accumulate, leading to chemical saturation. Two key saturation effects have been noted to be of particular significance with respect to aviation climate impact: (1) the saturation of NO_x emissions, which leads to decreasing ozone production with increasing NO_x concentrations [92], and (2) the dehydration of water vapour leading to diminished contrail climate impact [174]. Therefore, the atmospheric response to non-CO_2 aircraft emissions is not only sensitive to natural variations in the atmospheric state, but it is also affected by heightened emissions concentrations due to the presence of lingering plumes from previous aircraft.

Optimising flight routes with respect to minimum climate impact (instead of minimum fuel burn) is a concept that has become prevalent in the literature in recent years.

This involves re-routing aircraft to avoid particularly climate-sensitive regions of the atmosphere, based on provision of en-route chemical and meteorological information. Niklaß et al. [145] have shown through simulation efforts that it is possible to achieve a 12% reduction in climate impact at virtually no additional cost to fuel burn, evidencing the practicability of climate-optimised routing. Other studies approach this issue purely from the perspective of superimposing aircraft plumes through formation flight. The aerodynamic benefits of formation flight due to wake energy retrieval are already well established [10], [106], however it has become increasingly evident that formation flight can also be used to exploit climate-beneficial saturation effects due to the superposition of plumes from aircraft involved. In Dahlmann et al. [36] a preliminary study exploring the potential for climate impact reduction due to saturation effects in formation flight and found that NO_x saturation led to a reduction in ozone production efficiency of ~5% and a mutual inhibition of contrail growth, quelling their radiative properties by 20 to 60%.

This review chapter outlines the current state of literature on aviation's atmospheric effects and the influence of nonlinear plume-scale processing on the global net climate impact, to bring to light the notion that aviation-induced perturbations to atmospheric chemistry can simply be viewed as another constraint in the climate-optimised routing problem. Section 2 covers the emissions generation process and the methods used to model aircraft performance and emissions. Section 3 explores the dispersion characteristics of aircraft emissions on the subgrid scale, alluding to the various modelling methods developed to investigate the dynamical characteristics of aircraft plumes. In section 4, air traffic management principles and their effect on aviation emissions distribution is considered, on both the local and global scale, followed by section 5 which explores aircraft climate impact, based on the emissions released and the chemical and physical processes that occur on the grid and subgrid scale of global models. Section 6 looks into the effect of subgrid scale processes that occur due to emissions accumulation in high-density airspace, and finally section 7 explores potential mitigation efforts to reduce aviation-induced climate change by modifying aircraft operations, such as climate-optimal routing and formation flight.

2.2. Aircraft emissions

Aircraft propulsion systems provide the essential driving force component to enable powerful and efficient flight. Modern commercial aircraft rely on gas turbine propulsion systems, which operate on kerosene-based jet fuel, derived from fossil fuels. Aviation's environmental impact stems from the emission and dispersion of chemically and radiatively active substances that are generated during jet fuel combustion. This section details the emissions generation process and existing aircraft performance/emissions modelling methods, for fuel burn and emissions estimation.

2.2.1. The Generation of Aircraft Emissions

The widespread usage of kerosene-based jet fuels stems from the need to satisfy strict power-to-weight and safety requirements demanded by commercial aircraft; this is because kerosene has an exceptionally high energy density and wide operating temperature range at a low financial cost, compared with alternative fuel types [79], [82]. The combustion reaction that occurs between kerosene and air in the aircraft engine, required to generate thrust, does however produce a number of chemically- and radiatively-active emission species which are emitted from the aircraft into its wake. These aircraft emissions interact with the surrounding atmosphere, perturbing the natural balance of chemistry and contributing to air quality issues and anthropogenic climate change. The mass of emissions produced per unit mass of fuel burnt is often referred to as the emission index (EI), measured in grams of emission per kilogram of fuel [g/kg].

Primary jet fuel combustion products

The products of jet fuel combustion are often divided into two categories: primary and secondary, alluding to the way in which they are formed inside the aircraft engine. The primary products of jet fuel combustion are CO_2, H_2O and, due to fuel impurities, a relatively small amount of sulphur oxides (SO_x). These products are generated as a direct result of the combustion reaction that takes place, and hence depends on the carbon-hydrogen-sulphur composition of the fuel. The direct coupling of the production of these species to fuel consumption means that they have a constant EI throughout all phases of flight. Example values based on the typical chemical composition of jet fuel are given in table 2.1 [156].

Table 2.1.: Emission indices estimates for CO_2, H_2O and SO_2, averaged from a range of existing studies testing various aviation fuel types [30].

$\mathbf{CO_2}$	$\mathbf{H_2O}$	$\mathbf{SO_2}$
3149 g/kg fuel	1230 g/kg fuel	0.84 g/kg fuel

Secondary jet fuel combustion products

Further to this, a number of secondary combustion products are generated in the aircraft exhaust, namely oxides of nitrogen ($NO_x \equiv NO$ & NO_2), carbon monoxide (CO), unburnt hydrocarbons (HC), PM and trace levels of volatile organic compounds (VOCs) [20]. These products are termed secondary because their production levels differ depending on the nature of the combustion process and the engine load condition [39]. This means that their EIs are variable throughout flight, depending on aircraft engine type, engine operating conditions, and the atmospheric conditions of the surrounding environment [44].

NO_x emissions originate from the entry of atmospheric nitrogen (N_2) into the high temperature combustion chamber. The level of NO_x increases with increasing temperature and pressure, as it is coupled to the thermal reaction processes that occur in the primary combustion zone. Therefore, with the assumption of constant polytropic and combustion efficiencies, the emission index of NO_x ($EINO_x$) can be correlated with aircraft fuel flow [39], [43]. As a result of inefficiencies in the combustion process, products such as CO and HC are formed. In contrast to NO_x, these emissions are direct products of incomplete combustion, meaning their concentrations are inversely proportional to combustion efficiency. Since combustion efficiency correlates with thrust for sea level static (SLS) conditions, and thrust correlates with fuel flow, this means that EICO and EIHC decrease with increasing fuel flow.

PM emissions from aircraft can be categorised by their volatility: non-volatile PM (nvPM) and volatile PM (vPM). The primary source of nvPM present in aircraft exhaust is soot, which constitutes the greatest warming effect of all particles released from aircraft [68]. Soot is generated in the fuel-rich regions of the combustor, where the condensation of unburnt aromatic hydrocarbons takes place, converting low carbon content HC fuel molecules into carbonaceous agglomerates containing millions of carbon atoms [3], [17]. The extent of soot formation therefore depends on the fuel-air-ratio and the mixing processes that take place in the combustor, which vary with combustor design and are influenced by the non-homogeneous flow and temperature fields. Accurate measurements of these parameters are rare, making it very difficult to directly acquire quantitative data on soot emissions. Instead, soot EI can be estimated based on correlation with the so-called smoke number measured in engine certification procedures [43].

The formation of vPM from aircraft largely derives from the emission of sulphur derivatives, lubrication oil and VOCs [210]. During combustion, the sulphur content in the fuel is mostly oxidised to sulphur dioxide (SO_2), some of which is then further oxidised to sulphuric acid (H_2SO_4) when emitted into the atmosphere. In the presence of sufficient water vapour, sulphate aerosols (SO_4) can be generated, which exhibit the largest cooling effect from aircraft PM emissions [156]. The sulphate EI is a factor of fuel sulphur content, which varies depending on fuel composition and the specific emissions characteristics of the engine [171]. Due to the low volatility of lubricant oil, the emitted oil vapour from aircraft will add to the condensed mass of VOCs and contribute to vPM concentrations. The VOCs produced either from the oxidized fuel fragments or due to the pyrolysis in the combustion chamber can also act as vPM, which may have sufficiently low vapour pressure to allow condensation in the atmosphere, forming a coating on the surface of the nvPM and impacting cloud formation, precipitation and climate [156]. The particulate matter emitted near the exit nozzle plane of the combustor consists only of nvPM, but vPM are produced through nucleation and condensation downstream. Thus, it is difficult to estimate the total amount of vPM produced within the exhaust plume, as the formation of vPM is dependent on the concentration of sulphates and VOCs present in the exhaust, and the distance from the combustor [125].

2.2.2. Aircraft emissions modelling

Accurate quantification of aircraft emissions for any given flight requires the calculation of aircraft performance to estimate the total energy consumed, and hence the total fuel burnt throughout the flight duration. With knowledge of fuel flow rates experienced throughout flight, flow rates of aircraft emission species can be deduced based on empirical engine performance datasets and emissions models.

Fuel burn estimation

Computational modelling of aircraft performance allows the simulation of aircraft trajectories and the quantification of forces experienced throughout flight, enabling the approximation of thrust and fuel flow across all phases of flight. Aircraft performance models that are prevalent in academia and industry, such as the Base of Aircraft DAta (BADA) method [149] and Piano-X [160], approximate aircraft behaviour by coupling a database of aircraft-specific performance datasets to mathematical models to calculate useful flight characteristics, such as fuel flow and fuel burn, at discrete time steps during flight. This iterative approach accounts for the time-varying nature of aircraft properties like mass, speed and climb rate, thus leading to a more accurate estimate of performance and fuel burn estimation. Further to this, models such as the Aviation Environment Design Tool (AEDT) [115] exist, which carry out four-dimensional (4-D) physics-based simulations of aircraft trajectories at an exceptionally high spatial and temporal resolution, providing highly accurate predictions of fuel consumption and localised emissions impacts. Such tools do however come at the expense of high computational and financial cost, and often require proprietary data that is not in the public domain. An alternative state-of-the-art open-source performance model that has been made available in recent years is the OpenAP model [195]. This consists of four main components: aircraft and engine properties, kinematic performance and dynamic performance, and utility libraries; and can describe the characteristics of 27 common aircraft and 400 turbofan engines. The authors present a favourable comparison to established methods, matching BADA's fidelity across many characteristics, and exceeding it in several, deeming it a feasible alternative for researchers who do not have access to data and licensing for proprietary models.

Estimating aircraft emissions at reference conditions

Aircraft performance models provide estimates of aviation fuel burn, but to determine the particular chemical speciation of emissions released due to fuel combustion aircraft emissions models must be implemented. As mentioned earlier, the primary combustion products, CO_2, H_2O and SO_2, have a direct relation to the amount of fuel burnt and hence a constant EI for a given fuel composition, with standard estimates provided in table 2.1. Secondary combustion products such as NO_x, CO and HC however, largely depend on operational and atmospheric conditions. Empirical relationships between EI and fuel flow for secondary combustion products have been determined at reference operating conditions using engine performance datasets, such as the International Civil Aviation Organisation (ICAO) engine emissions databank [88]. This databank was developed for the purpose of engine certification and compliance with landing and take-off (LTO) cycle emissions standards, outlined in ICAO Annex 16 Vol. II [86]. Engine test data have been collected at sea-level static (SLS), International Standard Atmosphere (ISA) conditions, for four reference operating conditions (thrust settings) relevant to the LTO cycle: take-off (100% thrust), climb out (85%), approach (30%) and taxi in/out (7%). For every engine at each of the four LTO modes, a reference fuel flow and corresponding emission index has been derived, allowing emissions to be estimated to a reasonable degree of accuracy for aircraft operating under any of the four modes. An exemplary ICAO EI dataset is provided in table 2.2.

Table 2.2.: Example ICAO engine emissions data for Rolls-Royce Trent 970-84 engine [88].

	Take-off	**Climb out**	**Approach**	**Idle**
Fuel flow [kg/s]	2.605	2.157	0.720	0.255
EI NO_x [g/kg]	38.29	29.42	12.09	5.44
EI CO [g/kg]	0.32	0.31	1.16	13.38
EI HC [g/kg]	0.02	0.12	0.08	0.04

Estimating aircraft emissions at non-reference conditions

In reality, aircraft spend the majority of flight outside the LTO vicinity (above 3,000 ft), and the operating conditions and atmospheric conditions vary considerably. To enable the accurate analysis of aircraft emissions outside of reference conditions, a number of emissions modelling methods have been developed. Such methods apply the necessary adjustments and interpolations to the LTO-limited engine performance datasets, to generate more realistic estimates of aircraft emissions across the whole flight profile. SAE International Aerospace Information Report 5715 (AIR5715) [164] describes a number of methods for calculating aircraft emissions throughout all modes of operation and compares their relative merits. For the primary combustion species, the Fuel Composition method is presented, which determines $EICO_2$, EIH_2O and $EISO_x$ from the proportions of carbon, hydrogen and sulphur in the fuel. The set of methods concerning the estimation of $EINO_x$, EICO and EIHC include the ICAO reference method, the Boeing fuel flow method 2 (BFFM2), the DLR fuel flow method and the P3T3 method, in order of increasing fidelity. The ICAO reference method serves as the simplest and least accurate approach, computing emissions purely based on the ICAO reference conditions, without applying corrections to account for atmospheric effects at altitude. Therefore, it is only applicable for emissions analysis of aircraft flying in the LTO region. The remaining methods, BFFM2, DLR and P3T3 can be applied at all aircraft operating conditions, including at cruise, as they apply interpolations to engine performance datasets to determine emissions indices throughout the whole duration of a flight. The BFFM2 method and the DLR method (NO_x only) are the mid-tier models, as they provide reasonable estimates of EIs purely from interpolating the EI reference data and applying corrections for atmospheric effects based on ambient meteorological data, aircraft fuel flow and Mach number. The gold standard for modelling of NO_x, CO and HC emissions is the P3T3 method, as it utilises detailed thermodynamic modelling data to determine the precise emission indices at any given point throughout flight. Required data includes the combustor inlet temperature (T3), pressure (P3) and the fuel-air ratio (FAR) at both reference and operational conditions, all of which are difficult to obtain without access to proprietary engine-specific performance data, which limits the accessibility of this model to open-access researchers [21], [45].

Figure 2.1 shows an example of how EIs are interpolated based on the logarithmic relationship between EI and fuel flow, using the BFFM2 method [209]. Furthermore, methods are also presented to account for the remaining key emission species, such as the Derivative Factor method [115] used to approximate the EI values for VOCs such as non-methane HC (NMHC) throughout flight and the First Order Approximation method [210] used to estimate PM emission indices. The choice of method generally depends on the emission species to be observed, the compromise between modelling resources and data availability, and the level of accuracy required.

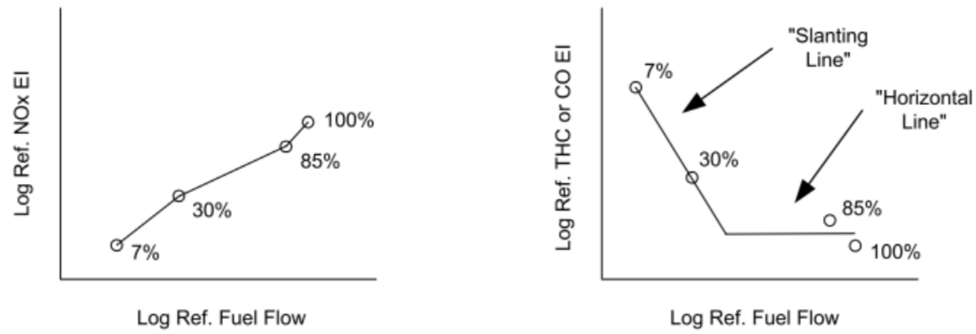


Figure 2.1.: Example log-log plots of EI against fuel flow at reference conditions, as prescribed by the BFFM2. The percentage values refer to the thrust as a percentage of maximum thrust at SLS conditions [164].

Emissions inventories and integration into large-scale climate models

The estimation of aircraft emissions for a specific flight involves the simulation of aircraft performance across the entire flight profile, to estimate fuel flow and engine operating conditions throughout flight. Knowledge of fuel flows and engine performance characteristics permits the estimation of aircraft emissions, based on the coupling of empirical engine certification data and emissions modelling methods, which interpolate the data to determine the emissions at non-reference conditions. This emissions estimation procedure is commonly carried out on a regional and global scale to determine emissions from a whole range of flights, which are then stored in so-called emissions inventories [208]. Aircraft emissions inventories collate data from all flights in the desired range and populate three-dimensional (3-D) latitude-longitude-altitude grid cells (e.g. $1° \times 1° \times$ 1000 ft) with total emissions quantities [151].

Aircraft emissions inventories are utilised to model the atmospheric effects of aviation, using large-scale models that capture the chemistry, physics and dynamics of the Earth-atmosphere system (see section 2.5.3 for further detail on climate modelling approaches). Such models allow one to simulate the perturbation to the state of the atmosphere due to an input of emissions and, in turn, provide quantitative indicators that enable modellers to determine the resultant climate impact (e.g. concentrations of key greenhouse gases, aerosol formation and distribution, cloud processes etc.) [123]. One underlying issue with this conventional approach to aviation climate modelling is that the use of gridded emissions data inherently assumes the instantaneous dispersion (ID) of emissions into the latitude-longitude-altitude computational grid cell in which they were released [159]. The dimensions of this grid cell are solely determined by the spatial resolution of the global model, and hence do not serve as an accurate physical representation of emissions dispersion.

In reality, emissions released from aircraft are confined to the aircraft exhaust plume, which inhibits mixing with the surrounding atmosphere for up to a day after emission. Throughout this time, a number of nonlinear chemical and microphysical processes occur, due to the elevated concentrations of emissions species in the plume. These nonlinear plume-scale processes affect the eventual climate response to these emissions, yet most regional and global aviation climate impact studies [29] often neglect the presence of the aircraft exhaust plume and opt for the simplified ID method. The following section explores the dynamical evolution of emissions following their release into ambient air, and discusses modelling approaches present in the literature which can be implemented to represent plume-scale effects in large-scale models. Such modelling is, however, often set aside due to computational issues associated with resolving consistency between the two model resolutions.

2.3. The Dispersion of Aircraft Emissions and the Aircraft Exhaust Plume

Following their expulsion into the free atmosphere throughout flight, aircraft exhaust gases are confined to a plume that undergoes a series of dynamical regimes (**jet**, **vortex**, **dispersion** and **diffusion**), before becoming fully diluted in the surrounding air. The entrainment of emissions within the plume throughout these dynamical regimes leads to initial species concentrations that are several orders of magnitude higher than background levels [37], giving rise to a number of nonlinear chemical and microphysical effects. These plume-scale effects have considerable implications on the eventual chemical composition of the surrounding atmosphere and lead to the formation of aerosols and ice crystals in the aircraft wake. Therefore, inclusion of plume-scale effects is vital for high fidelity modelling of aviation's impact on the climate.

2.3.1. Plume-Scale Dynamical Regimes

In order to accurately account for nonlinear effects experienced in the aircraft exhaust plume, one must first understand the dynamical response of the plume after combustion, to gauge the length and time scales over which aircraft emissions are entrained within it. Figure 2.2 is a visual representation of the typical dynamical evolution of an aircraft exhaust plume, throughout its lifetime.

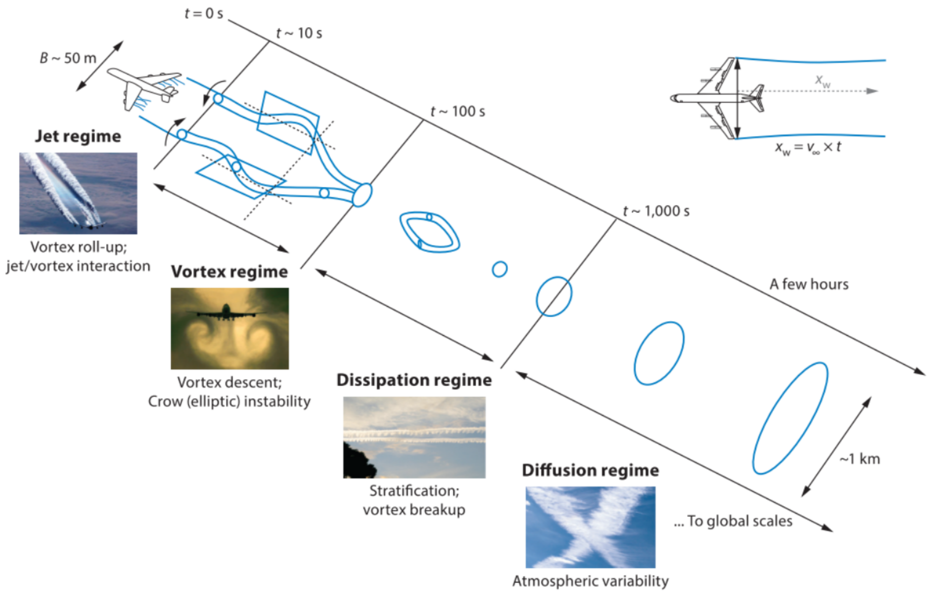


Figure 2.2.: Aircraft exhaust plume dynamical regimes. Wake distance behind aircraft $x_w = v_\infty \times t$ where v_∞ is aircraft speed and t is the post emission time [66], [154].

Exhaust gas temperatures range from around 1500 K post combustion, 600 K at engine tailpipe, followed by mixing with bypass air cooling the flow to around 300 K [20] as the dispersion process begins. During the first 1–20 s post emission, an axisymmetric jet is formed, which rapidly diffuses into ambient air and cools to ambient temperatures.

Over this period, known as the **jet** regime, the airflow passing over the wings is diverted downwards to generate lift, thus creating a vortex sheet at the trailing edge of the aircraft. This vortex sheet rolls up into a pair of counter-rotating vortices which are shed at the wing tips. The evolving vortex pair then merge together and propagate downwards, due to their mutually induced downwash velocity, trapping the exhaust plume within their cores and signalling the beginning of the vortex regime.

Throughout the **vortex** regime, which occurs between 20 s and 2 minutes after combustion, the primary wake containing the vortices and trapped exhaust plume sinks by around 150–200 m, resulting in a slight temperature increase of 1–3 K, due to the adiabatic heating of exhaust constituents in the sinking vortices. Further to this, the organised vortical structure means the wake does not grow significantly during this time, and hence the concentrations of entrained chemical species remain relatively constant. The adiabatic heating of the exhaust does however lead to baroclinicity at the border between each vortex and the ambient air, which detrains some momentum, heat and exhaust constituents from the primary wake, to form a secondary wake. This secondary wake trails upwards as it is warmer than the surrounding ambient air, and it escapes the influence of the vortex structure, resulting in enhanced mixing with ambient air and thus experiences different chemical and microphysical processes compared with the primary wake [66].

Following this is the **dispersion** regime, in which the aircraft-induced dynamics subside due to the growth of Crow instability [34], which dissipates and disintegrates the primary and secondary wake vortices [154]. The breakdown of the organised vortical structure and the production of turbulent motion leads to a sudden increase in the rate of entrainment between the exhaust plume and the ambient air by a factor of 10, therefore giving rise to a continuous decay of concentration and temperature within the plume. This regime lasts for 2–5 minutes after combustion, however this varies as the strength of aircraft induced vortices is proportional to the weight and span, and inversely proportional to the speed of the aircraft [66].

Lastly, the plume undergoes its final dynamical event, known as the **diffusion** regime. This regime is characterised by the aircraft-induced dynamics becoming negligible (after about 6 minutes [203]), followed by the subsequent dominance of atmospheric processes in the spreading of the aircraft exhaust plume and its constituents. Atmospheric turbulence, radiation transport and stratification are examples of natural phenomena that contribute to the diffusion of the plume, with total dilution to ambient concentrations often occurring over timescales of 2–12 h post emission [1]. During this time, the plume may spread up to a few kilometres through atmospheric turbulence and shear in the ambient air, diluting the exhaust species over vast volumes of airspace [172].

Plume-scale climate effects that result from the confinement of emissions to the aircraft exhaust plume during the four dynamical regimes considerably alter the eventual global warming effect of a particular flight, and therefore should be appropriately accounted for in modelling efforts to estimate aviation's climate impact.

2.3.2. Plume-Scale Modelling

To tackle the issue of neglected plume-scale effects in the computational analysis of aviation-induced climate change, a number of plume modelling methods of varying fidelity have been theorised in the literature. Subgrid resolution plume models simulate the dynamical response of the aircraft plume, to capture the nonlinear chemistry and microphysical effects that occur within it. The outputs of plume models can then be parametrised into low-resolution global models, to increase the accuracy of climate impact calculations through better accounting of the emissions dispersion process.

Empirical Dilution Model

Plume dynamics control the rate at which aircraft emissions mix and dilute into the surrounding atmosphere, directly affecting the resulting climate impact of aircraft emissions due to the nonlinear effects experienced in the plume, before emissions become homogeneously mixed into the ambient air. Quantifying the rate of dilution and modelling the climate effects that occur within aircraft plumes is therefore an essential process in the accurate analysis of aviation's climate impact [172]. In Schumann et al. [175], an empirical dilution model was developed to investigate the mixing rate of plumes throughout their typical lifetimes, based on data collated from over 70 aircraft exhaust plume encounters with research aircraft. The characteristic property observed in this study is the plume dilution ratio, N, which is defined as the amount of air mass that the exhaust plume generated from a unit mass of fuel burn mixes with, per unit flight distance within the bulk of the plume.

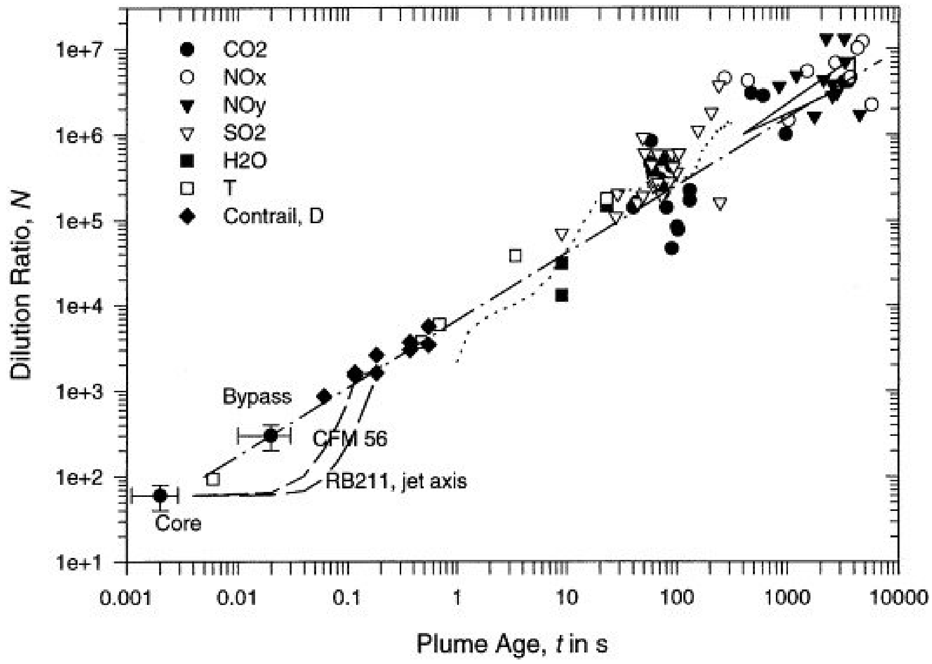


Figure 2.3.: Dilution ratio against plume age derived from empirical data. Marker shapes correspond to tracer species as displayed in legend. Markers with error bars correspond to characteristic values for the engine core (leftmost and lower) and bypass exits (rightmost and upper). The dashed curves near the core represent dilution on the jet axis, calculated for two engine types: CFM56 and RB211. The dotted curve represents large eddy simulation results from Gerz and Ehret [65]. The range of dilution values computed for large plumes ages by Dürbeck and Gerz [47] are bounded by the triangle, top right of the figure. The dash-dotted line shows the interpolation that is represented by equation (2.1). From Schumann et al. [175].

In figure 2.3, measured dilution ratios from the plume encounters are plotted against plume age. The data was generated by measuring the concentrations of a range of chemical sources across a variety of aircraft types, within the time interval of 0.001–10,000 s. The significance of these findings is that, in spite of the diverse range of chemical species and aircraft types observed, throughout all four dynamical regimes, a relatively consistent logarithmic relationship emerges between dilution ratio and increasing plume age. When interpolating the regression line fitted to the data in figure 2.3, the following equation can be obtained where N represents dilution ratio, t is plume age in seconds, and t_0 serves as an arbitrary reference scale.

$$N = 7000(t/t_0)^{0.8},\ t_0 = 1\ s \tag{2.1}$$

It is evident from the figure however, that encounters with plumes older than 50–100 s tend to diverge from the line fit, indicating a reduction in accuracy of the logarithmic approximation over time. This is likely due to the transition from the organised vortex structure present in the vortex regime, to the turbulent dispersion regime, where more unpredictable atmospheric processes begin to take place and become the primary influence on the evolution of the plume. Therefore, this empirical model can only be used reliably up to the vortex regime. Beyond this, a Gaussian approximation to the distribution of species concentrations is typically employed, accounting for dispersion effects experienced at cruising altitudes, such as advection, gravitational sedimentation, anisotropic diffusion, wind shear and stable stratification [110]. Two popular modelling methods which implement Gaussian approximation and the two-dimensional diffusion equation to model aircraft plumes over their whole lifetime are the Single Plume model and its discretised counterpart, the Multi-layered Plume model.

Single and Multi-layered Plume Models

The Single Plume (SP) model, first presented in Petry et al. [159], approximates the time-evolving concentration field of an aircraft exhaust plume using a Gaussian distribution [207]. Petry represents diffusion through eq. (2.2), a differential equation that describes the temporal variation of exhaust concentration C of species i in the plume

$$\frac{\partial C_{\mathrm{i}}}{\partial t} = -sz\frac{\partial}{\partial y}C_{\mathrm{i}} + D_{\mathrm{v}}\frac{\partial^2}{\partial^2 z}C_{\mathrm{i}} + D_{\mathrm{h}}\frac{\partial^2}{\partial^2 y}C_{\mathrm{i}} + 2D_{\mathrm{s}}\frac{\partial^2}{\partial y \partial z}C_{\mathrm{i}} \tag{2.2}$$

This two-dimensional diffusion equation is a function of wind shear (s), horizontal (y) and vertical distance from plume centre (z), and the horizontal (D_{h}), vertical (D_{v}) and shear (D_{s}) diffusion coefficients, as calculated based on empirical data recorded under typical atmospheric conditions at aircraft cruising altitudes and assuming a horizontal flight path [46]. The solution to the diffusion equation is a time-varying Gaussian function, with standard deviations σ_{h}, σ_{v} and σ_{s} that depend on s, D_{h}, D_{v}, D_{s}, time t and the respective initial values ${\sigma_{0\mathrm{h,v}}}^2$

$$\sigma_{\mathrm{h}}^2(t) = \frac{2}{3}s^2 D_{\mathrm{v}} t^3 + (2D_{\mathrm{s}} + s{\sigma_{0\mathrm{v}}}^2)st^2 + 2D_{\mathrm{h}}t + {\sigma_{0\mathrm{h}}}^2, \tag{2.3}$$

$$\sigma_{\mathrm{v}}^2(t) = 2D_{\mathrm{v}}t + {\sigma_{0\mathrm{v}}}^2, \tag{2.4}$$

$$\sigma_{\mathrm{s}}^2(t) = sD_{\mathrm{v}}t^2 + (2D_{\mathrm{s}} + s{\sigma_{0\mathrm{v}}}^2)t. \tag{2.5}$$

The standard deviations of the Gaussian function can then be used to deduce useful parameters such as plume cross-sectional areas and concentrations. Outputs such as these can serve as input to atmospheric models, enabling the simulation of the dynamical evolution of the plume, and its entrained emissions throughout its lifetime.

The main drawback of the SP model however, is the assumption of homogeneous concentration distribution throughout the plume at any given time. This homogeneity assumption is sufficiently accurate up to the vortex regime, where plume cross-sectional areas are relatively small, entrainment rates are low, and the mixing ratio is relatively consistent across the plume diameter [60]. However, beyond this, the spike in entrainment rates following the breakdown of vortices causes rapid plume expansion, and the drop in concentration from plume core to outer edges becomes increasingly significant [130], [133].

This spatial concentration gradient along the plume cross-section cannot be captured using the SP approach, so an alternative model is proposed, known as the Multi-layered Plume (MP) model, as seen in figure 2.4. The MP model builds upon the SP model by discretising the plume cross-section into a number of concentric rings, enabling the inhomogeneous concentration profile to be represented by varying the mean concentration in each ring.

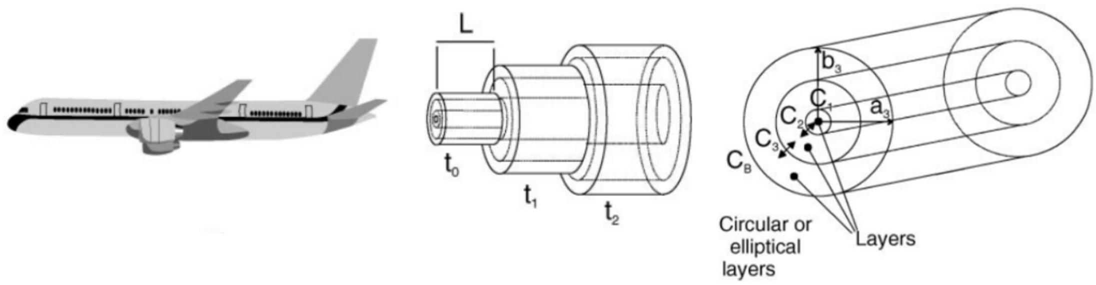


Figure 2.4.: MP model visual representation [112]. Plume length L was set to the distance travelled over 1 s (247 m in this scenario) with three out of eight concentric rings shown for each time step. $C_1 - C_3$ indicate the concentrations in each ring and the arrows represent mixing between the layers.

As Kraabol et al. [112] states, the Gaussian approximation to emissions dispersion only applies to the dynamical evolution of passive species, and is not suitable for modelling the evolution of chemically active species, due to the nonlinear chemical response in the plume. To counteract this issue, this paper implements an adapted version of the MP model, in which the plume is divided into 8 concentric rings, each with a chemistry module incorporated to estimate the chemical production and loss mechanisms of all species present. Applying this model under assumed turbulent conditions derived from Dürbeck and Gerz [47], graphs of horizontal and vertical plume radius are plotted against plume age, as shown in figure 2.5. After just 1 hour, it is predicted that plumes can spread between 1 and 10 km horizontally, whilst only reaching around 50–100 m vertically due to atmospheric stratification [46]. As the plume approaches the end of its typical lifetime, between 10 and 15 h, plume cross-sections can reach 100–200 km horizontally and 200–400 m vertically. The vast length scales over which plumes span throughout their lifetime provides ample evidence to suggest that, in high-density airspace, plumes can overlap. The overlapping of plumes can thus lead to spikes in emissions concentrations that exceed that of single aircraft plumes, thus augmenting non-linearities in the climate response and further propagating discrepancies between plume and global model outputs [168].

(a) (b)

Figure 2.5.: Evolution of **(a)** horizontal and **(b)** vertical plume radius over time, for MP model under assumed turbulent conditions [112].

Aircraft Plume Chemistry, Emissions, and Microphysics Model (APCEMM)

Models such as APCEMM from Fritz [59] further increase the accuracy of the SP and MP models by capturing additional effects that affect plume evolution and hence the resultant climate response once the plume is fully diluted. This includes effects such as plume anisotropy and asymmetry which impact the eventual spatial distribution of the plume, and modelling of the microphysical processes that strongly influence contrail formation and persistence. The model is similar to the MP model in that a chemistry module is simulated in a number of concentric rings which have differing concentration fields, that decrease with increasing radial distance from the plume core. The rest of the computation (i.e. mixing and microphysics) is performed on a high-resolution Cartesian grid [58]. The primary aim of this model development was to bridge the gap between the simplified Gaussian approximation and more comprehensive large eddy simulations, as elaborated on in the following subsection.

Large Eddy Simulations (LESs)

Finally, the plume dynamical evolution can be most accurately captured using high-resolution LESs over the entire lifetime of the plume. LESs can model dynamics on a scale of several millions of grid points, for a few seconds to a few minutes of plume age, providing unmatched levels of accuracy at the cost of extremely high computational demand. For this reason, LESs are usually limited to case studies from which the data obtained can be used to derive and calibrate plume parametrisations for use in the lower fidelity methods [120].

In Dürbeck and Gerz [46], LES data are used to calculate effective diffusion coefficients and plume cross-sectional properties for plume modelling purposes. The data obtained from the simulations is said to have agreed with Schumann et al. [172], where the horizontal and vertical plume scales and respective diffusivities are estimated from experimental data captured by a research aircraft measuring NO concentrations. Moreover, LESs have been used to determine plume properties on much shorter timescales in Unterstrasser et al. [203]. In this paper, plume evolution is analysed for the jet and vortex period up to 6 minutes, whilst aircraft-induced dynamics dominate; concentration profiles and plume cross-sectional areas are determined for a range of atmospheric conditions, varying stratification, turbulence, wind shear and aircraft properties. Similarly, in Paoli [152], detailed LES numerical simulations are carried out for the jet and vortex phase, confirming hypotheses surrounding aerosol and microphysics modelling. These studies exemplify the use of LES methods to validate experimental findings, and serve as a means of calibration for plume model parameters that represent real-life plume dilution characteristics.

2.4. Air Traffic and Emissions Distribution

The previous section discussed the forces of flight and the generation and dispersion of aircraft emissions, from the perspective of a single aircraft. However, the state of the atmosphere is inhomogeneous with respect to space and time, meaning that the climate sensitivity to aircraft emissions differs considerably, depending on the time and location of their release. Furthermore, in airspace regions where traffic density is high, aircraft fly in close proximity and their exhaust plumes may intersect, giving rise to nonlinear local climate effects that must also be considered in aviation climate analysis. Therefore, it is useful to explore the influence of air traffic management and spatial and temporal demand on the distribution of air traffic, and hence associated emissions.

2.4.1. Air Traffic Management

Air traffic management (ATM) is the system of services responsible for overseeing the network-wide implementation of safe, orderly and efficient air traffic flows, providing assistance to aircraft in transit from departure to destination aerodrome. It is the role of the air traffic control (ATC) team to manage and monitor air traffic in their respective airspace in real time, ensuring that optimum safety, order and efficiency of aircraft operations are maintained at all times [183].

Air Traffic Safety

The inherent risk involved in the transportation of vast numbers of passengers at near transonic speeds through the upper atmosphere means that aviation safety is of paramount importance. A safe aircraft operation takes the path of least danger, primarily influenced by the need to avoid unfavourable atmospheric conditions and to prevent conflicts with other aircraft [11], [216]. In-flight atmospheric conditions susceptible to icing, turbulence or the presence of hazardous convective weather can all be classified as unfavourable for aircraft, with the latter presenting the greatest constraint on aircraft routing [114], [137]. The increased risk resulting from flight through weather-affected regions means that aircraft must re-route, leading to restrictions on available airspace and deviations from the optimal flight profile, thus increasing flight-times, fuel burn and delays [52].

The safety risks associated with aircraft-to-aircraft collisions necessitate air traffic controllers to impose safe separation standards between aircraft in the lateral, longitudinal and vertical direction, as specified in [87]. The stated minimum separation distances between aircraft are 5 nautical miles (NM) laterally, 20 NM longitudinally and 1,000 ft in the vertical direction under the most lenient scenarios. A breach of separation laws in more than one direction is known as a conflict and must be resolved as quickly as possible. The enforcement of separation minima therefore introduces a theoretical upper limit on airspace density, i.e. the number of aircraft that occupy a fixed volume of airspace at any one time.

Air Traffic Order

To keep air traffic flows organised within controlled airspace, aircraft are ordered to follow the traditional fixed-route air traffic network, constructed from four key airspace elements that facilitate the air traffic management process [183]:

- **airports/aerodromes** - an area of land or water intended to be used for the arrival, departure and surface movement of aircraft;
- **waypoints** - a specified geographical location used to define the flight path of an aircraft, representing either a navigational aid (navaid) or a reference coordinate that the aircraft must fly by or fly over;

- **airways** - a controlled portion of airspace established in the form of a corridor (usually 8-10NM wide) between two waypoints;
- **sectors** - a region of airspace managed by a single ATC team, stratified into various levels to accommodate a wide variety of traffic.

In [199], the notion of optimising air traffic flows for a given demand and capacity is explored, in which air traffic flows are represented using these four key elements. **Airports** represent the sources and sinks of the flow, **airways** are the arcs along which the flow travels, **waypoints** are the network nodes at which airways intersect, merge or diverge, and **sectors** are a collection of waypoints and contiguous segments of airways. The fixed-route network restricts airspace availability even further, due to the discretisation of flight levels and the requirement to pass specified waypoints [13], [51]. This can lead to particularly high frequencies of aircraft passing through high-density airspace, potentially leading to congestion along busy airways and waypoints where airways intersect resulting in inhomogeneities in the distribution of air traffic and potential congestion along high-density airways.

Air Traffic Efficiency

The third and final component of effective air traffic management is the optimisation of flight trajectories, subject to the prioritisation of safety and the compliance with the fixed-route airspace structure. Flight trajectory optimisation is an essential step in ensuring maximum airspace utilisation and efficiency, so that revenue is maximised and demand levels are sufficiently met. Trajectory optimisation is a multi-faceted problem, requiring consideration of nonlinear aircraft performance, wind and weather forecasts, payload, departure fuel load, reserve fuel load and ATM constraints that restrict aircraft operations and routing [187]. This requires an exhaustive assessment to be carried out at the flight planning stage, to test all possible combinations of route, payload, fuel load and operating approach, involving tens to hundreds of thousands of calculations per flight. The most optimal scenarios are then ranked in order of optimality, with the final route selected based on operator preference and/or the occurrence of unexpected circumstances, such as sudden adverse weather conditions or aircraft conflicts [4].

In an ideal airspace situation where the atmosphere is calm and constant; aircraft are not constrained to a fixed route; and there is no risk of conflict with other aircraft, the least-time and least-energy aircraft operation would be to fly the great-circle arc between departure and destination. The vertical profile of the aircraft would consist of a continuous climb out to the most efficient cruise altitude, then to cruise at constant speed, with the ability to cruise-climb continuously as the aircraft burns fuel and loses mass. In reality, the true optimal route can deviate considerably from the great-circle arc, instead taking the path which minimises the risk of bad weather encounters and collisions, abides by the fixed-route airspace structure, whilst also flying a route which is optimised with respect to wind and temperature. The magnitude and direction of wind and the localised variation in temperature experienced by the aircraft throughout flight, can have a drastic impact on route optimality, with tailwinds and colder temperature regions being favourable [139]. Ng et al. [144] found for a wide range of wind-optimal flight scenarios that domestic flights saved up to 3%, and international flights saving up to 10% on both fuel burn and travel time, despite flying longer routes. Furthermore, the vertical flight profile of the aircraft must adhere to flight level allocations, meaning that step climbs must be performed as fuel is burnt, further condensing air traffic and its corresponding emissions into narrow bands of altitude.

Airspace Capacity

The effective management of air traffic relies on the human cognition of air traffic controllers to make difficult decisions and carry out complex tasks in a time-critical dynamic environment. This includes ensuring safety through avoidance of poor atmospheric conditions and conflicts with other aircraft, maintaining order by flying along the fixed-route airspace network, and optimising air traffic flows with respect to wind and weather. As density and complexity levels of air traffic increase, so does the mental workload of the air traffic controller, up until a threshold level is reached where the controller can no longer safely handle the situation. The maximum number of aircraft permitted by the ATC team in charge of a particular airspace volume is known as the airspace capacity, and is driven by the airspace situation, state of equipment being used, and the controller's own mental state [122].

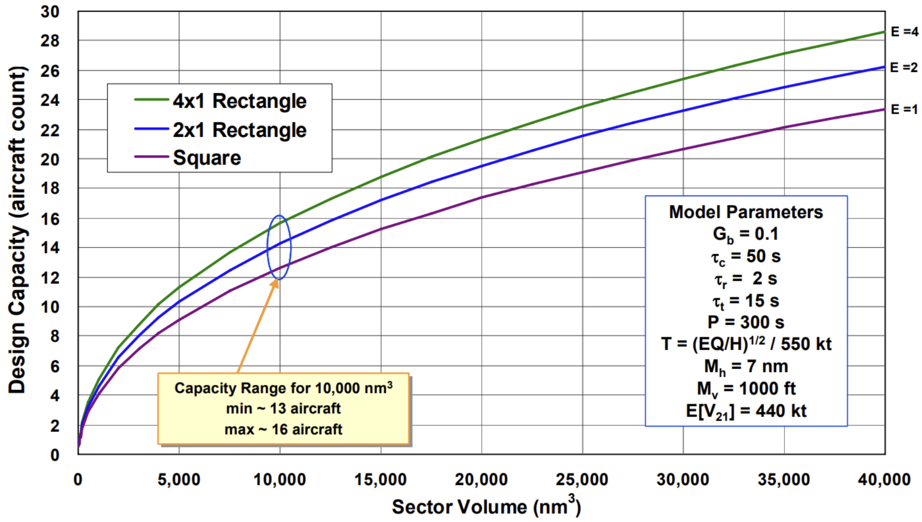


Figure 2.6.: Aircraft capacity estimation against sector volume for a range of airspace scenarios [211]. E refers to the length-to-width ratio of the sector, G_b, τ_C, τ_r and τ_r are all empirical parameters related to controller workload, P is the mean task recurrence period per aircraft, T is the sector transit time, M_h and M_v are the designated horizontal and vertical separation minima between aircraft within the sector and $E[V_{21}]$ is the typical aircraft closing speed [5].

Airspace capacity is limited more by controller workload than is it separation laws, meaning human cognition is the true limiting factor on the number of aircraft that can occupy a particular airspace volume at a particular time [13], [211]. Therefore, models of controller workload are often used to estimate airspace capacity, in which ATC tasks are modelled to determine a safe upper limit on workload. In Welch et al. [211], a macroscopic workload model is proposed which generalises ATC tasks into four distinct categories: background, transition, recurring and conflict tasks. This provides an objective basis for estimating capacity and enables the formulation of an analytical relationship between airspace capacity and sector volume, as seen in figure 2.6.

The capacity estimation model from figure 2.6 predicts that, for a 10,000 NM^3 rectangular sector of dimensions 156 NM (length, ratio 4:1) $\times$ 39 NM (width) $\times$ 10,000 ft (height), a maximum of 16 aircraft may be present at any one time. In a purely hypothetical situation where separation laws dictate capacity and all aircraft are travelling in one direction lengthways, the sector could support a maximum of 490 aircraft, assuming a separation of 5 NM laterally, 20 NM longitudinally and 1,000 ft vertically. This emphasizes the sheer extent to which human factors limit the ability to maximise capacity, and highlights the need for airspace modernisation to increase automation, integration and collaboration in the ATM system, enabling the further increase in capacity levels towards minimum separation capacity [63].

2.4.2. Global Air Traffic and Emissions Distribution

The nature of air traffic and emissions distribution was investigated in Olsen et al. [150] where a range of global aircraft emissions datasets are compared (NASA-Boeing 1992, NASA-Boeing 1999, QUANTIFY 2000, Aero2k 2002, AEDT 2006 and aviation fuel usage estimates from the International Energy Agency) to show distribution patterns in the latitudinal, longitudinal and vertical sense. Further to this, temporal variations with respect to both the diurnal (time of day) and seasonal (time of year) cycles are explored.

The climate sensitivity of the atmosphere is highly variable depending on the exact latitude, longitude and altitude combination, because of the spatially varying chemical and meteorological state of the atmosphere (e.g. [50], [84], [89]). Accurate accounting of spatial distribution patterns of air traffic is therefore very important in the estimation of aviation climate impact. Figure 2.7 shows the spatial distribution of fuel burn across the range of datasets in the longitudinal, latitudinal and vertical directions. The longitudinal distribution shows three emissions peaks around the densely-populated regions of the US, Europe and East Asia, with the largest situated above the North American land mass. It is evident in the latitudinal distribution, that the Northern Hemisphere dominates, with a strong peak in the northern mid-latitudes that appears due to high volumes of air traffic above the US and Europe, as well as along the connecting region of airspace, the North Atlantic flight corridor (NAFC). Contrarily, there are almost no emissions present in southern latitudes below 40° S, with the region between 40° S and the equator constituting only a small percentage. The altitudinal distribution on the other hand, experiences emissions peaks around both the low altitude LTO area and the high-altitude cruising regions between 9 and 13 km, with relatively low emission intensities at mid-altitudes. Furthermore, the peak around cruising altitude is discretised into peaks every other flight level, due to the vertical separation constraints and the allocation of aircraft to specific flight levels, thus owing to further increases in emissions intensities at these altitudes.

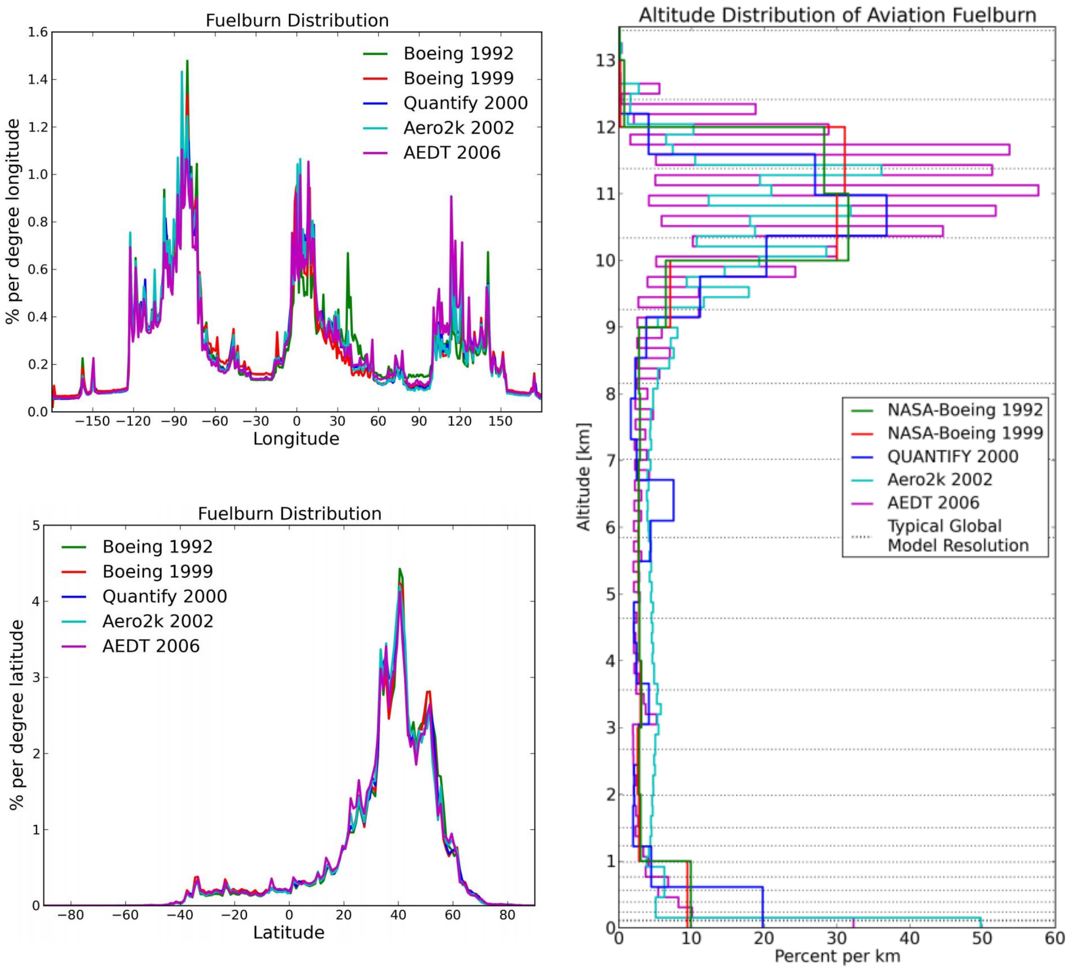


Figure 2.7.: Spatial (latitude, longitude and altitude) distribution of global aviation fuel burn from a range of aircraft emissions datasets [150].

The presence of diurnal and seasonal variations in key chemical and meteorological parameters throughout the atmosphere has been widely investigated in the literature (e.g. [167]). The diurnal and seasonal variation in aviation fuel burn from Olsen et al. [150] is displayed in figure 2.8. The temporal fluctuations in both the state of the atmosphere and the distribution of fuel burn and emissions allude to the fact that the climate sensitivity to aircraft emissions is always changing, and therefore these parameters must be under constant observation to ensure accurate determination of global climate effects from aviation. The diurnal cycle of global aviation fuel burn, as seen on the left-hand side of figure 2.8 displays a peak at around 15:00 UTC, which decreases through the night until around 09:00 UTC where total fuel burn begins to increase again. With regard to seasonal variation, there is significant variance between emissions datasets, however in general, all display a wintertime minimum between December and January, and a summer-time maximum between June and September.

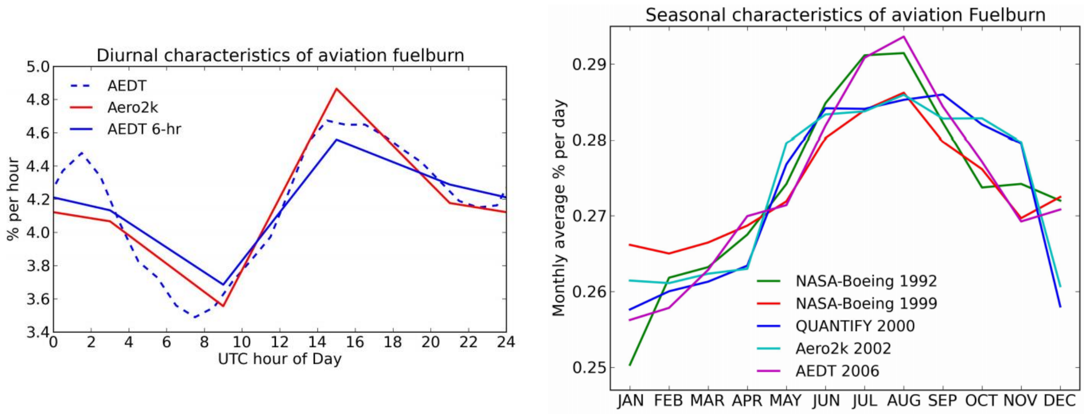


Figure 2.8.: Temporal (diurnal and seasonal) distribution of global aviation fuel burn expressed as percentage from a range of aircraft emissions datasets [150].

2.4.3. Local Air Traffic and Emissions Distribution

Due to the fixed-route nature of airspace, aircraft tend to fly along common airways or flight corridors, and pass common waypoints along their journey, leading to exceptionally high flux densities of aircraft through these regions at peak times. This has implications on the nonlinear chemical and physical effects occurring at the plume scale, due to the intersection of aircraft plumes and the elevated exhaust gas concentrations entrained within them. A prime example of a high-density airspace region is the NAFC, made up of a series of tracks that aircraft traversing the North Atlantic must follow, updated daily to allow for convective weather avoidance, tracking of the North Atlantic Jet Stream and favourable tailwinds to maximise efficiency [148]. The annually-averaged number of aircraft traversing the NAFC per day has increased from 800 in 1997 [102] to around 2,500 in recent years [23], owing to the tripling of passenger demand since then [2]. With North Atlantic air traffic being confined to a limited number of tracks (usually three or four), it can be assumed that aircraft separation distances and airspace capacities are pushed to their limit on a regular basis along this and other popular flight corridors around the world.

Previous experimental work on air traffic emissions in the NAFC was carried out in the late 1990's, through campaigns such as the Pollution from Aircraft Emissions in the North Atlantic Flight Corridor (POLINAT) and Subsonic Assessment Ozone and Nitrogen Oxide Experiment (SONEX) [176]. At least 20 follow up papers were published following these campaigns, in which POLINAT/SONEX data are utilised to provide insight on a number of major scientific issues [201]. A noteworthy publication regarding localised emissions impacts is Schlager et al. [168], which carries out an in-situ investigation of air traffic emissions signatures (nitrogen oxides (NO_x), sulphur dioxide (SO_2) and cloud condensation nuclei (CCN)) in the NAFC using experimental data from a POLINAT research flight. The research aircraft flew perpendicular to the major eastbound corridor tracks and took measurements of various chemical concentration fields and meteorological parameters throughout. The results show that the superposition of aircraft exhaust plumes led to peak concentrations of NO_x, SO_2 and CCN above background levels by factors of 30, 5 and 3, respectively. This is because plume dispersion timescales greatly exceed the daily frequency with which aircraft emissions are input into the flight corridor, resulting in an inhomogeneous concentration field with narrow and sharp peaks over a relatively low and smooth background level [176].

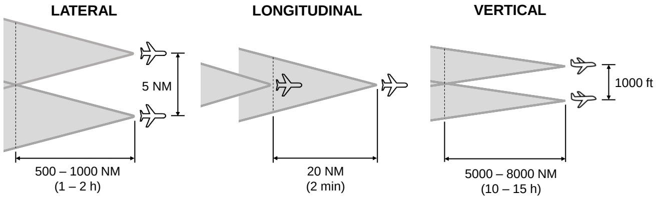


Figure 2.9.: Lateral, longitudinal and vertical maximum plume overlap scenarios, based on nominal separation minima [87]

In the observation of plume-scale climate effects in high-density airspace regions, aircraft separation minima determine the minimum possible distance between aircraft and hence the maximum possible overlap between aircraft exhaust plumes. The degree of plume overlap influences the magnitude of emissions saturation, further accentuating nonlinear plume-scale climate processes that occur, as elaborated on in section 2.6.4. Using plume dimension estimates from Kraabol et al. [112], maximum plume overlap scenarios can be inferred. Figure 2.9 displays the lateral, longitudinal and vertical maximum overlap scenarios for aircraft cruising at 550 kt. The longitudinal separation scenario proves to be the most effective formation for superimposing aircraft plumes, with the intersection time simply equalling the time taken to travel the separation distance between the two aircraft. For lateral and vertical superposition, the intersection time is much longer, as it is determined by the plume expansion rate in that particular direction, up until the midpoint between the two aircraft is reached. For lateral plume overlap to occur, the plumes must expand horizontally to a radius of 2.5 NM (15,190 ft), whereas for vertical, the distance is a mere 500 ft. Despite the drastic reduction in distance, the vertical plume overlap takes about an order of magnitude longer than lateral, because vertical plume expansion is substantially suppressed in comparison due to stable atmospheric stratification counteracting vertical motion [172]. In reality, air traffic flows are much more complex and the controlled and orderly formations shown in figure 2.9 are unlikely to occur naturally. However, the premise still holds that plume overlap occurs most frequently in congested flight corridors such as the NAFC, when aircraft travel along similar tracks and the vertical displacement between aircraft is minimal.

The importance of safety, order and efficiency in the air traffic management process and the characteristics of global and local air traffic flows have been discussed. As the following section will explore, the atmospheric response to aviation emissions is highly sensitive to time and location, because the instantaneous state of the atmosphere (i.e. the chemical composition and meteorological situation at that particular time and position) determines the production, loss and radiative response of key chemical species that induce climatic effects [123]. With insight into how air traffic and emissions are dispersed locally and distributed globally, the contribution of aircraft emissions to climate change can be more rigorously evaluated.

2.5. The Global Climate Impact of Aviation

Aircraft emissions induce climate effects by perturbing the flux of inbound shortwave (SW) solar radiation and outbound longwave (LW) terrestrial radiation emitted from the Earth's surface through absorption and scattering processes that give rise to warming or cooling of the atmosphere [31]. Climate metrics such as radiative forcing can be used to measure the climate contribution of each individual emission species, enabling the determination of the net global warming effect from aviation. This section explores the potential global impact of aviation deduced from measurement and modelling of atmospheric processes.

2.5.1. Aircraft Emissions in the Upper Troposphere and Lower Stratosphere

As figure 2.7 suggests, the vast majority (~60%) of aviation fuel burn and hence aviation emissions, occur at cruise altitudes, between 9 and 13 km vertically. The region of the atmosphere encompassing this volume of airspace is known as the Upper Troposphere and Lower Stratosphere (UTLS), with bounds of ±5 km above and below the conventional tropopause [67]. Around 20–40% of total aircraft emissions are released in the LS [81], [156] and the rest are released in the troposphere, extending from the surface at take-off to the UT at aircraft cruise altitudes. The greenhouse effect due to the release of chemically-active substances in the UTLS is considerably greater than that of emissions at the surface. This is because the climate in the UTLS is more sensitive due to increased residence times of pollutants, lower background concentrations (meaning emissions have a greater influence on the atmospheric chemistry), lower temperatures which drive reversible reactions in an unfavourable direction, and finally, a higher radiative efficiency which gives rise to more efficient photochemistry [178]. Johnson et al. [98] further validates this claim, with model results concluding that NO_x emissions constitute a 30 times greater climate impact in the UT compared to equivalent surface emissions, due to the absence of direct deposition and slower conversion to stable reservoir species at aircraft cruising altitudes.

Despite such a large proportion of air traffic emissions being released into the LS region, it is thought that the perturbation to the chemical and radiative state of the stratosphere is negligible, as the vast majority of species emitted into the stratosphere are transported downwards into tropospheric regions, where they interact with the atmosphere there [56],

[71], [117]. Henceforth, this review will primarily focus on the tropospheric response to aircraft emissions, except for water vapour, where the stratospheric climate response becomes particularly noteworthy.

2.5.2. Radiative Forcing of Aircraft Emissions

High altitude emissions from aviation impact the climate through a variety of climate forcing pathways. Some greenhouse gases such as CO_2 and H_2O are emitted directly, others are produced indirectly through chemical processing of aircraft emissions, such as the reaction of NO_x with atmospheric trace species to catalyse ozone production and methane destruction. Water vapour and PM emissions are responsible for the formation of high ice clouds known as condensation trails (contrails), which often trap outbound LW radiation within the atmosphere more efficiently than they reflect inbound SW radiation. PM emissions also have the potential to induce a climate perturbation through direct radiative processes due to aerosols, which may warm or cool the climate depending on the particle's optical and microphysical properties [117]. Diversity in the climate forcing pathways for each emission species means that the only reliable method of determining the severity of each climate contribution is to model the atmospheric response and determine the resulting impact on the radiative flux of the atmosphere [20].

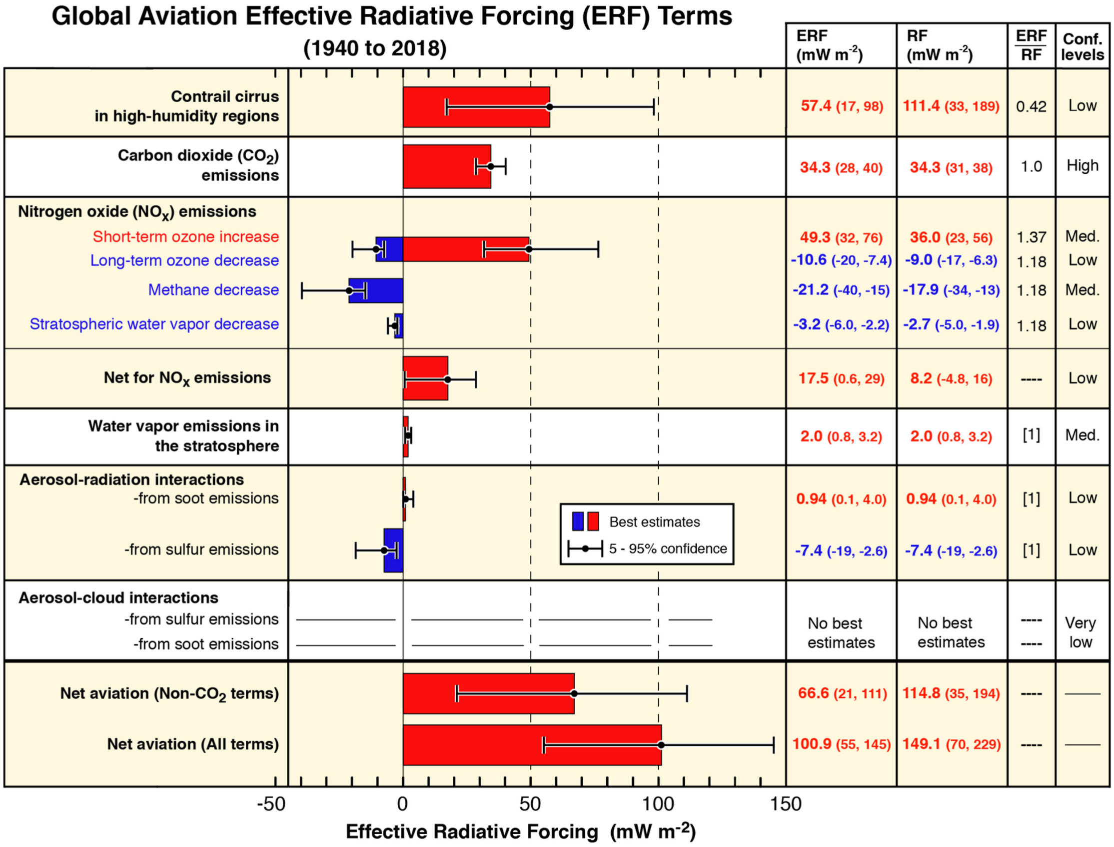


Figure 2.10.: Radiative forcing contributions from global aviation between 2000 and 2018 [116]. Error bars represent 5–95% confidence interval, with corresponding RF and ERF values shown in parentheses.

The most common climate metric used to compare the magnitudes of climate impact from a range of emission species is radiative forcing (RF). RF is defined as the perturbation to the net energy balance of the Earth-atmosphere system due to natural or anthropogenic factors of climate change, measured in watts per square metre [Wm^{-2}] [62]. A positive RF means that the climate forcing mechanism is inducing a warming effect and vice versa. Lee et al. [116] presents an updated analysis of the global effective radiative forcing (ERF) contributions for aviation-induced climate change. ERF is a newly proposed climate modelling framework that builds upon the RF concept by removing rapid atmospheric adjustments that bear no relation to the long-term surface temperature response that occurs over decadal timescales [31]. ERF serves as a more suitable equivalency metric to compare the global warming response of heterogeneously distributed short-lived climate forcers and uniformly distributed long-lived climate forcers. Figure 2.10 displays the ERF and RF contributions for each of the key aviation-induced climate forcers.

CO_2

The climatic effects of aviation CO_2 emissions are direct and well understood, with the thermal absorption of outbound LW radiation leading to warming through the planetary greenhouse effect [170]. Carbon dioxide from aircraft constitutes the second-largest ERF term in figure 2.10 at 34.3 mWm^{-2}, and the thermodynamic and photochemical stability of CO_2 means it has a relatively long atmospheric lifetime, on the order of 100 to 1,000 years [7]. This means that carbon emissions from aircraft simply serve to increase its atmospheric concentration, leading to the eventual distribution over global spatial scales. The ubiquitous and intuitive nature of CO_2-related warming deems it a suitable benchmark to compare warming from non-CO_2 climate forcers against. The assumption of instantaneous dilution in global models is sufficient for modelling the climate impact due to carbon dioxide, as it exists over vast spatial and temporal scales, meaning the climatic effects occurring on plume time scales are negligible compared with the impact induced over its entire lifetime [151].

Contrail Cirrus

When emissions of water vapour and PM are released into the aircraft exhaust plume in the cold and moist condition of the UTLS, the water vapour condenses around the particulates due to the high relative humidity (RH), then freezes due to the low external temperatures, permitting the formation of ice crystals. The accumulation of ice crystals in the aircraft wake gives rise to the generation of a condensation trail, also known as a contrail. The evolution of contrails, i.e. how they grow, disperse and persist with time, is largely determined by the ambient conditions of the surrounding atmosphere, with lower temperatures and higher humidities generally leading to more persistent and damaging contrails [179]. The formation and persistence of contrails can be predicted purely from the thermodynamic assumption that, as the hot and moist exhaust mixes with the colder and drier ambient air, a contrail will form if the plume exceeds water-saturation at any point and the temperature is low enough for ice nucleation to occur. A contrail will persist in the atmosphere if it mixes with air that is supersaturated with respect to ice, i.e. an ice-supersaturated region (ISSR), growing with time due to deposition of surrounding water vapour onto existing ice crystals [104], [179]. Persistent contrails can sometimes transition to contrail cirrus, either building onto existing cirrus clouds or forming new ones, spreading over vast swathes of the atmosphere [136] and inducing significant radiative effects. ISSRs occur frequently at aircraft cruising altitudes and often span hundreds of kilometres horizontally, however only reach depths of 100 to 1,000 m [189], [206]. Their shallow nature means that aircraft can avoid them by changing flight level by $\pm$2000 ft to minimise persistent contrail generation for a minor additional fuel penalty [180].

The optical characteristics of contrail ice crystals often exhibit significant levels of opacity, tending to absorb outbound terrestrial radiation more efficiently than they reflect inbound solar radiation. This induces a globally averaged ERF of 57.4 mWm^{-2}, meaning, despite their relatively short lifetime in comparison, the current contrail effect contributes more to climate change than the accumulation of aviation carbon, since the dawn of aviation [25]. Contrail evolution, i.e. how they grow, disperse and persist with time, is largely determined by the ambient conditions of the surrounding atmosphere, with lower temperatures and higher humidities generally leading to a more persistent and damaging contrails [179]. The formation and persistence of contrails can be predicted purely from the thermodynamic assumption that, as the hot and moist exhaust mixes with the colder and drier ambient air, a contrail will form if the plume exceeds water-saturation at any point and the temperature is low enough for ice nucleation to occur.

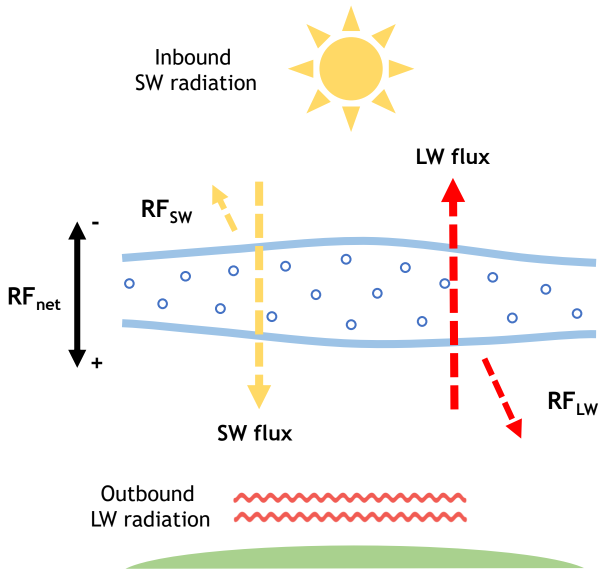


Figure 2.11.: Visual representation of contrail radiative forcing, adapted from Karcher et al. [104].

Contrails exhibit radiative forcing through the obstruction of both SW and LW radiative fluxes, with areal coverage and optical depth (opacity) being the key drivers of contrail climate impact [181]. A contrail's emissivity (ability to absorb and re-emit infrared LW radiation back towards Earth) and reflectance (ability to reflect inbound SW radiation back out to space due to scattering at visible wavelengths) is a function of contrail optical depth and ice particle microphysics. See figure 2.11 for a depiction of contrail RF. Contrails and other high ice clouds often warm the climate, as their thin optical depth means partial transparency to solar radiation, whilst their high ice density traps infrared radiation within the atmosphere effectively [104]. Contrail RF also displays a distinct diurnal trend; at night, contrails always induce a warming effect, as there is no SW scattering to counteract the LW absorption. During the day, contrails display a reduced net warming effect and perhaps even a net cooling effect, depending on the amount of solar radiation that is redirected back out to space [143]. The so-called Schmidt-Appleman criterion can be used to predict formation and persistence of contrails [6], [169], [177], however accurate quantification of contrail RF requires sufficiently detailed microphysical modelling, to deduce key information on the underlying formation mechanisms and physical and optical properties the contrail exhibits throughout its evolution [100]. See section 2.6.3 for elaboration on the microphysical processes that determine a contrail's radiative characteristics, and how these processes can be parametrised for modelling purposes.

Net-NO_x

Nitrogen oxides (NO and NO_2) released from aircraft are not radiatively active and therefore do not induce an immediate climate impact at the point of emission. Their chemical instability does however mean that they exhibit a number of indirect RF effects, due to the chemical reactions that occur following dilution into the ambient atmosphere. The chemical interactions between NO_x and trace species in the background atmosphere at aircraft cruising altitudes are highly non-linear and thus, the net-NO_x RF contribution is dependent on the instantaneous atmospheric state (time of day and year, latitude, background chemical composition, meteorological situation) [60], [113].

Emissions of NO_x in the troposphere initially leads to the short term local increase in ozone production efficiency (O_3) on the timescale of weeks to months. In addition, elevated NO_x and O_3 levels lead to increased hydroxyl radical (OH) production, which in turn, leads to the long term global destruction of ambient methane (CH_4) over the timescale of decades [140], [192], [213]. The short term increase in O_3 generates a strong positive ERF of 49.3 mWm^{-2}, whereas the long term CH_4 depletion causes a lesser negative ERF of -21.2 mWm^{-2} in comparison, however there are a number of secondary negative radiative effects arising from methane depletion that must also be accounted for. This is the reduction in stratospheric water vapour (15% CH_4 RF magnitude) and a decrease in long-term background ozone in the troposphere (45% CH_4 RF magnitude), resulting from reduced background CH_4 concentrations [83], [141]. Despite the long-term negative cooling effects, the short term warming from O_3 dominates, leading to a largely positive net ERF of 17.5 mWm^{-2} overall, as seen from figure 2.10. Section 2.6.1 explores the nonlinear NO_x-O_3 relationship further through explanation of the gas-phase photochemical processes that begin in the aircraft plume and are eventually distributed to regional and global scales.

Water Vapour

The direct radiative effects induced by aviation water vapour emissions in the troposphere are insignificant, because the influence on background concentrations is negligible when compared with the natural fluxes of the Earth's hydrological cycle [156]. Any tropospheric water vapour emissions tend to get lost through deposition, due to high humidity and precipitation in this region, leading to a lifetime of around 9 days. On the contrary, water vapour that is emitted into the stratosphere (without getting transported downwards into the UT) can induce a considerable effect on the surrounding environment, due to extreme dryness at these altitudes [97]. This is because increases in stratospheric water vapour (SWV) concentrations impact the climate directly through the greenhouse effect, as well as through influences on the gas-phase and aerosol chemical composition, leading to depletion of ozone and altering the formation and growth of polar stratospheric clouds [190]. The combined impact of the direct greenhouse effect and indirect impacts on ozone and polar stratospheric clouds due to increased SWV leads to an overall ERF of 2.0 mWm^{-2}.

As proposed in Lee et al. (2010) [117], the relatively small climate perturbation due to aviation-induced SWV has the potential to increase drastically as future flight concepts begin to take shape. This includes the environmental implications associated with the potential replacement of the current subsonic aircraft fleet with a supersonic high-speed civil transport fleet, that primarily operates in the stratosphere (e.g. [37], [73], [105], [134]). Furthermore, the prospect of transitioning the entire subsonic kerosene-based commercial fleet to a hypothetical fleet of Cryoplanes (hydrogen-powered aircraft with zero carbon emissions) would increase aviation H_2O emissions by a factor of ~2.5 [64]. In the high-humidity conditions of the UT, this brings about the potential for large increases in contrail production, whereas in the LS, this would induce significant perturbations to SWV concentrations.

Aerosol Effects

Aviation aerosol particles, either emitted directly post combustion, or formed in the wake of the aircraft, can perturb the energy balance of the atmosphere directly, as well as indirectly through the formation of contrail ice particles and heterogeneous chemical processing. The direct aerosol effect is primarily produced by the key non-volatile and volatile PM emissions: soot and sulphate aerosols.

Soot exhibits a direct radiative forcing as it has the strongest absorption of light at visible wavelengths per unit mass, more than any other abundant substance in the atmosphere, thus contributing to global heating through absorption of inbound solar radiation and light rebounded off reflective surfaces such as snow and ice [18]. The resultant heating of the atmosphere and reduction of sunlight can affect the hydrological cycle and large-scale circulation patterns, having potentially larger implications on the climate than previously thought. Despite its ability to strongly absorb sunlight, aircraft soot is responsible for only a few percent of total atmospheric black carbon, meaning the aerosol-radiation interaction brought about by aviation soot only constitutes a minor ERF of 0.94 mWm^{-2}.

Sulphur dioxide (SO_2) is formed when sulphur, which is present in hydrocarbon jet fuels, oxidizes during the combustion process [22]. SO_2 that is emitted from the aircraft exhaust can be oxidised to form gaseous sulphuric acid (H_2SO_4), which may condense on existing particles or contribute to new particle formation in low-condensation environments, resulting in sulphate aerosol formation. Sulphate aerosol is mainly composed of sulphuric acid and corresponding salts such as ammonium sulphate. The optical properties of sulphate mean that it tends to scatter inbound sunlight, thus leading to a net negative (cooling) of 7.4 mWm^{-2}, that sways the net direct climate impact of aviation aerosols towards cooling.

It is relatively well established that aviation-induced aerosols have a direct impact through radiative interactions, as discussed in this section, as well as indirectly through activation of water vapour on PM emissions through contrail formation. There are however, potentially large indirect consequences of aerosol particles interacting with cloud droplets and ambient ice particles nucleation on the aerosol surface. These effects are left without ERF estimates in figure 2.10, as there is great uncertainty around the accuracy of cloud process modelling and the ability to distinguish aircraft-induced clouds and natural clouds [157]. Figure 2.12 is a visual depiction of the typical climate forcing contributions due to aircraft emissions, as described in Lee et al. [116].

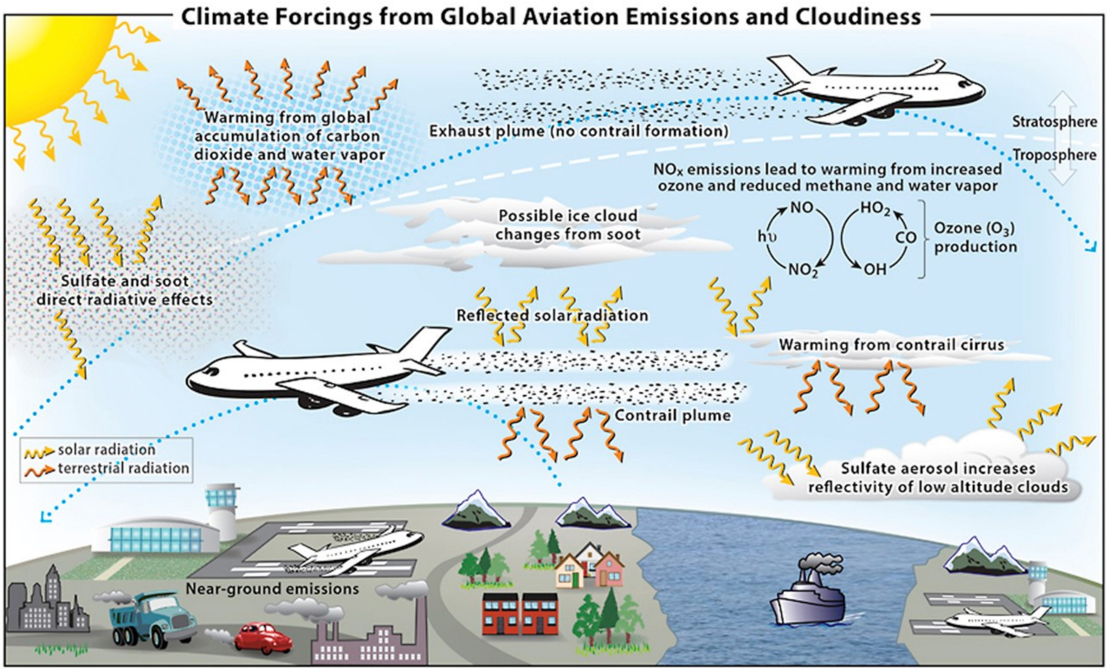


Figure 2.12.: Schematic overview of climate forcing mechanisms induced by aircraft emissions [116].

2.5.3. Global Climate Modelling

Quantifying aviation's global climate impact requires the use of computational climate models to predict the atmospheric response to the emission species released by aircraft. Climate models simulate the climate system through mathematical representation of established physical laws (e.g. conservation of mass, energy and momentum) and a plentiful supply of empirical data obtained through real world observation and measurement of physical and chemical quantities (e.g. chemical concentrations, meteorological parameters) [165]. Gridded aircraft emissions inventories are used as input to climate models [150], [214], in which emissions are homogeneously distributed throughout the entire grid cell that they are released into (i.e. the ID approach). The chemistry, physics and dynamics of the atmosphere are captured in the model, along with the perturbation to the atmospheric state due to emissions, providing an output of the spatial and temporal distributions of chemical species. Radiative transfer schemes can then be used to quantify the climate response to the perturbed chemical composition through instantaneous metrics such as RF and ERF. The variability in chemical and radiative properties of climate forcing species does however vary considerably with time. For example, CO_2 is distributed globally and affects the climate gradually over centuries, whereas contrails induce a severe yet short-lived, localised radiative impact [116]. Therefore, metrics such as Global Warming Potential (GWP, GWP*) and Global Temperature Potential (GTP) have been developed to better account for the temporal variation of certain properties, providing a more well-rounded analysis of the climate contributions of aircraft emission species [27], [61], [62].

There are two main climate model distinctions for use in aviation climate modelling: general circulation models (GCMs) and chemistry-transport models (CTMs). GCMs are highly sophisticated, yet very complex, as they are what are known as "online" models, calculating chemical composition, temperature and transport circulation simultaneously and in real time. This is often computationally expensive and may require very long processing times to run, depending on the simulation. CTMs on the other hand are reduced order "offline" models, calculating chemical composition based on pre-determined GCM results or empirically observed temperature and transport circulation data [156]. Example usage of GCMs in aviation climate modelling include the European Centre HAmburg Model (ECHAM) [191] for analysis of future contrail cirrus radiative forcing [16] and the Community Atmosphere Model version 5 (CAM5) [142] to comprehensively represent aviation aerosol climate impact [68]. CTMs appear much more frequently in the literature due to their versatility and computational efficiency. For example the UK Meteorological Office CTM STOCHEM [33] coupled with the Common Representative Intermediates (CRI) chemical mechanism [94] is used for modelling the global impact of aviation NO_x emissions on ozone production [209], MOZART CTM [49] for use in investigating the trade-off between CO_2 and NO_x emissions [57], and another six CTMs are presented in [156] for aviation climate impact modelling purposes. Various model intercomparison studies provide an up-to-date, elaborate review of the models available for UT and LS (and thus aviation) climate modelling [166].

2.6. Nonlinear Plume-Scale Climate Effects

As detailed in the previous section, the global contributions to aviation radiative forcing have been relatively well quantified. However, much of the modelling processes used to estimate these contributions are derived from global models assuming instant dispersion, thus excluding subgrid scale chemical and physical effects that occur in the first 2–12 h upon release into the atmosphere. This section reviews current scientific understanding of these plume-scale processes, and highlights their emerging relevance in increasingly high-density airspace regions, as the accumulation of chemically active emission species further propagates the nonlinear atmospheric response.

2.6.1. Gas-Phase Photochemistry

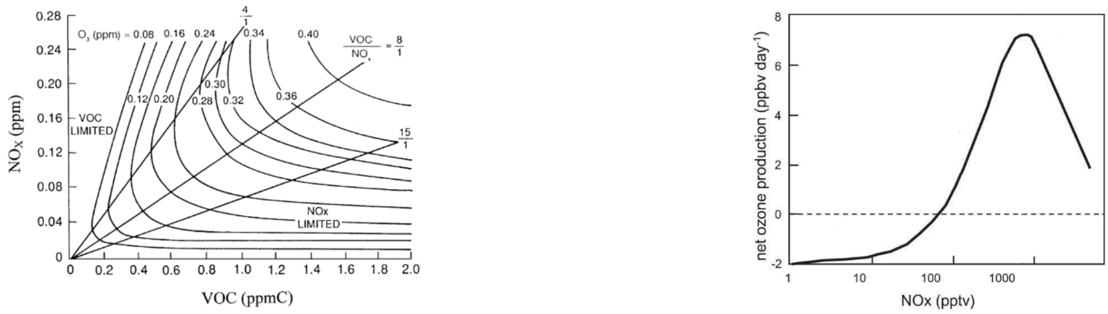
The ultimate chemical composition of the troposphere is largely influenced by the removal of natural and anthropogenic trace species (e.g. CH_4, CO, O_3) through oxidation reactions with atmospheric free radicals, primarily the hydroxyl radical (OH), which are relatively short-lived and in low concentrations owing to their fast reactivity [138], [193]. The relative abundance of OH in the troposphere controls the degree of removal of ambient trace species, otherwise known as the atmosphere's oxidative capacity. In the purely hypothetical situation where gas-phase radical chemistry was absent and hence the oxidative capacity of the atmosphere was zero, the levels of many harmful pollutants would continue to rise unabated, resulting in a drastic change in the chemical, biological and radiative state of the Earth-atmosphere system [163]. OH is produced via atmospheric photochemistry and photolysis; sunlight-initiated reaction of photolabile molecules to produce highly reactive species and/or radical species. Ozone is photolysed by ultraviolet light ($\lambda < 320$ nm) to produce excited singlet oxygen $O(^1D)$ (R 2.1), which then produces two hydroxyl radicals in the presence of sufficient levels of water vapour (R 2.2). The symbol M represents a non-reacting molecule in the equation, that has the role of absorbing reaction energy to create stable products.

$$O_3 + h\nu(\lambda < 320\,nm) \longrightarrow O(^1D) + O_2 \tag{R 2.1}$$

$$O(^1D) + H_2O \longrightarrow OH + OH \tag{R 2.2}$$

Understanding the influence of OH and the oxidative capacity of the upper troposphere is critical for the environmental analysis of aircraft emissions, as the closely-coupled chemical scheme involving hydroxyl and hydroperoxy radicals ($HO_x = OH + HO_2$) and nitrogen oxides ($NO_x = NO + NO_2$) determines the production and loss of key climate forcing species such as ozone and methane.

Most oxidation processes that occur in the troposphere are photochemical reaction schemes involving HO_x, and are therefore only applicable in daylight conditions due to the reliance on solar actinic flux [123]. There is however, a range of chemical processes mainly involving nitrate radicals, that are potentially important for nighttime tropospheric oxidation processes involving key climate forcing species. See Jenkin et al. [95] for further elaboration on nighttime tropospheric chemistry.



(a)

(b)

Figure 2.13.: **(a)** Example isopleth diagram displaying peak ozone concentrations calculated from various initial concentrations of NO_x and a specified VOC mixture, using the US EPA empirical kinetic model [42], [95]. **(b)** Schematic representation of the variation of net ozone production efficiency with NO_x concentrations, with magnitudes reflecting clean free tropospheric conditions (i.e. low VOC/NO_x ratio) [138].

Figure 2.13a from Jenkin et al. [95] is an exemplary ozone isopleth diagram that illustrates the O_3-NO_x-VOC relationship. At typical aircraft cruising altitudes (i.e. UTLS), where the emissions of NO_x have a significant influence on atmospheric chemistry, it is likely that VOC content is very low relative to the background NO and NO_2 concentrations. Therefore, it is likely that typical VOC/NO_x ratios are low in the UTLS, meaning that under the assumption of constant VOC concentrations, the NO_x-O_3 relationship takes a form similar to figure 2.13b.

Low-NO_x Regime

In unpolluted environments characterised by low NO_x concentrations, such as in regions where the ambient air is unperturbed by aircraft emissions, the dominant reaction pathway of OH is to react with CO (~75%), with the remainder reacting with CH_4 [200].

The dominant reaction pathway therefore involves the oxidation of CO to form CO_2, as in reaction (R 4.1) (for a detailed description of both the CO and the CH_4 oxidation cycles in low NO_x conditions, see Wasiuk [209]). In this CO oxidation cycle, produced atomic hydrogen (H) then reacts with oxygen to form HO_2 (R 4.2), which subsequently reacts with O_3 to form OH in the background troposphere (R 4.8). This initiates a chain sequence in which OH and HO_2 interconvert through the termination of ozone (R 2.6). This clean condition scheme therefore limits pollutant build up in the unpolluted upper troposphere and keeps ozone levels under control [91].

$$\mathrm{OH + CO \longrightarrow H + CO_2} \tag{R 2.3}$$

$$\mathrm{H + O_2 + M \longrightarrow HO_2 + M} \tag{R 2.4}$$

$$\mathrm{HO_2 + O_3 \longrightarrow OH + 2\,O_2} \tag{R 2.5}$$

$$\mathrm{OH + O_3 \longrightarrow HO_2 + O_2} \tag{R 2.6}$$

Net reaction:

$$\mathrm{OH + CO + 2\,O_3 \longrightarrow CO_2 + HO_2 + 2\,O_2} \tag{R 2.7}$$

Alternatively, HO_2 can react with itself to form hydrogen peroxide (H_2O_2) (R 2.8), or with organic peroxy radicals such as the methyl peroxy radical (CH_3O_2) to form organic hydroperoxides (R 2.9). These reaction pathways can become an effective sink for HO_x under most conditions, because the formation of peroxides prevents further HO_x interconversion [75].

$$\mathrm{HO_2 + HO_2 + M \longrightarrow M + O_2 + H_2O_2} \tag{R 2.8}$$

$$CH_3O_2 + HO_2 \longrightarrow CH_3O_2H + O_2 \tag{R 2.9}$$

NO_x-Limited Regime

In polluted environments, where NO_x levels are raised considerably above ambient concentrations (e.g. typical aircraft cruising altitudes), CO oxidation takes place through reactions (R 4.1) (R 4.2), however in the presence of nitrogen oxides, ozone is produced rather than depleted. Peroxide formation reactions (R 2.8) and (R 2.9) compete with the oxidation of NO to NO_2 (R 6.3) for available HO_2 concentrations. When the latter reaction prevails, NO_2 photolysis takes place, converting back to NO with ground state oxygen $O(^3P)$ forming as a byproduct (R 6.5). Subsequently, $O(^3P)$ reacts with atmospheric oxygen to produce O_3 (R 6.6).

$$HO_2 + NO \longrightarrow OH + NO_2 \tag{R 2.10}$$

$$NO_2 + h\nu(\lambda < 420\,nm) \longrightarrow NO + O(^3P) \tag{R 2.11}$$

$$O(^3P) + O_2 + M \longrightarrow O_3 + M \tag{R 2.12}$$

Net reaction:

$$CO + 2\,O_2 \longrightarrow CO_2 + O_3 \tag{R 2.13}$$

Additionally, the methane oxidation cycle leads to net ozone production in the presence of NO_x. The oxidation of methane by OH produces water vapour and the methyl radical (CH_3) (R 4.7), which further reacts with oxygen to form CH_3O_2 (R 2.15). The produced methyl peroxy radical can then react with NO to form NO_2 through reaction (R 2.16).

$$OH + CH_4 \longrightarrow H_2O + CH_3 \tag{R 2.14}$$

$$CH_3 + O_2 + M \longrightarrow M + CH_3O_2 \tag{R 2.15}$$

$$CH_3O_2 + NO \longrightarrow CH_3O + NO_2 \tag{R 2.16}$$

$$CH_3O + O_2 \longrightarrow HO_2 + HCHO \tag{R 2.17}$$

Net reaction:

$$CH_4 + 2\,O_2 \longrightarrow HCHO + O_3 \tag{R 2.18}$$

The methoxy radical (CH_3O) produced can form additional HO_2 and formaldehyde (HCHO) through reaction (R 2.17), which is then capable of reacting to form further NO_2 through reaction (R 6.3). The resultant NO_2 produced through reactions (R 2.16) and (R 6.3) consequently produces ozone through the same pathway as CO oxidation (i.e. NO_2 photolysis (R 6.5) followed by reaction of the $O(^3P)$ photoproduct (R 6.6)). The net reaction of methane oxidation in polluted environments therefore results in positive ozone production. Figure 2.14 is a visual representation of the NO_x-O_3-CO-CH_4 oxidation reaction scheme described thus far.

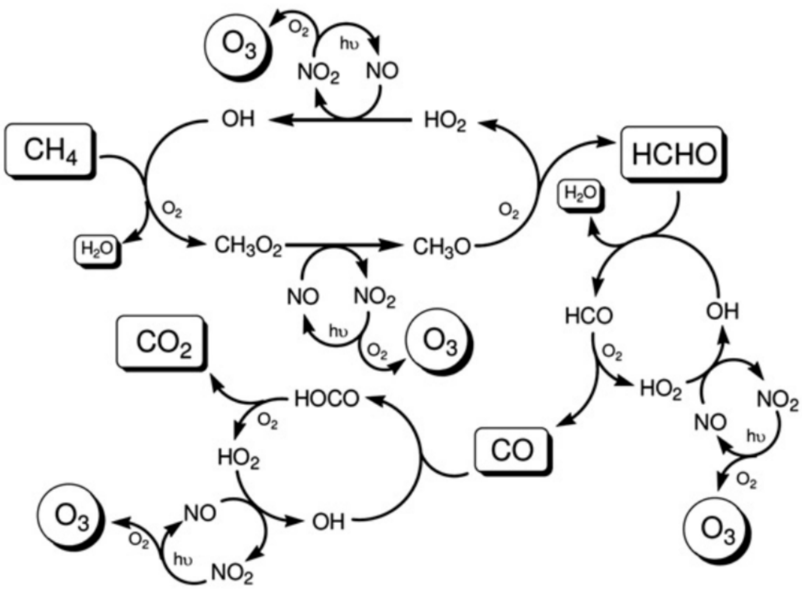


Figure 2.14.: Schematic representation of the OH initiated, NO_x-catalysed oxidation scheme for CO and CH_4 [94].

NO_x-Saturated Regime

Under very high NO_x conditions, such as inside aircraft exhaust plumes or in regions of the atmosphere where aircraft fly in close proximity and their corresponding emissions accumulate, the ozone production efficiency starts to drop off. This is because as NO and NO_2 surpass a threshold concentration, known as the compensation point (peak of the curve in figure 2.13b), they begin to compete with VOCs (e.g. methane) for reaction with hydroxyl (OH), organic peroxy (RO_2) and peroxide RCO_3 radicals [209]. The products of these reactions are examples of nitrogen reservoir species: nitrous acid (HONO), nitric acid (HNO_3), peroxynitric acid (HO_2NO_2), peroxy nitrates (RO_2NO_2) and peroxyacyl nitrates (RCO_3NO_2). Reservoir species are much less efficient at forming ozone as they are more stable and are also more likely to get washed out of the atmosphere through depositional processes [95]. They are also much more stable in the upper troposphere compared to the surface equivalent [107]. Thus, reactions (R 2.19) to (R 2.23) serve as termination reactions, removing radicals from the atmosphere and preventing additional ozone formation through conversion of NO_x to its more stable counterparts.

$$NO + OH + M \longrightarrow HONO + M \tag{R 2.19}$$

$$NO_2 + OH + M \longrightarrow HNO_3 + M \tag{R 2.20}$$

$$NO_2 + HO_2 + M \longrightarrow HO_2NO_2 + M \tag{R 2.21}$$

$$NO_2 + RO_2 + M \longrightarrow RO_2NO_2 + M \tag{R 2.22}$$

$$NO_2 + RCO_3 + M \longrightarrow RCO_3NO_2 + M \tag{R 2.23}$$

NO_x saturation effects become particularly important to observe in the case of diluting aircraft exhaust plumes, because the high NO_X-VOC ratio means that termination reactions are often favoured over catalytic ozone formation [188]. In the relatively fresh exhaust plume (first 10 minutes) where NO_x concentrations are significantly enhanced, ozone titration by NO results in large scale production of NO_2 (R 4.9), but decreases the formation of HO_x due to depleted ozone levels in the plume.

$$NO + O_3 \longrightarrow NO_2 + O_2 \tag{R 2.24}$$

The dilution of the plume results in reduced NO_x concentrations over time, and the in-plume chemistry transitions from NO_x-saturated to NO_x-limited. With ozone levels still depleted, the remaining HO_2 and RO_2 in the plume (formed from the oxidation of CO and VOCs by OH) react with the remaining NO to produce OH and NO_2 without further depleting ozone. This leads to increasing OH and NO_2 concentrations which give rise to a net ozone recovery due to NO_2 photolysis (reactions (R 6.5) and (R 6.6)), with full recovery to ambient concentrations within 1–2 h post emission. As ozone levels rise in the plume back towards ambient concentrations, the photochemical formation of OH becomes more common, through reactions (R 2.1) and (R 2.2). Newly formed OH can oxidise CO and VOCs to form peroxy radicals which catalyse ozone production, however it can also react with NO and NO_2 to form stable nitrogen reservoir species, through reactions (R 2.19) to (R 2.23) [60].

In the ID scenario inherent to large-scale climate models, ozone titration doesn't occur on the same scale due to lower mixing ratios of NO_x when instantly diluted. Therefore, initial ozone depletion and subsequent recovery in the exhaust plume is not properly captured, meaning that instead, ozone levels remain reasonably high throughout and hence HO_x production can remain stable. The stable HO_x levels mean NO_x to reservoir species conversion remains stable also, throughout the NO_x lifetime. On a global scale, it has been shown that inclusion of plume processes leads to a net reduction in ozone forming potential of aviation NO_x emissions. Vohralik et al. [207] summarises the estimates made for the degree of reduction in ozone forming potential when plume effects were included. Initial findings from Kraabol et al. [111] and Meijer et al. [129] estimated discrepancies in ozone production of 15–18%, however these studies only accounted for ozone depletion in the plume, and not the O_3 generated during plume expansion. Inclusion of both ozone depletion and production in the plume in Meijer [130], led to updated estimates in ozone formation changes 0% to -5% in January and +5% to -10% in July, indicating that plume processing can actually increase net ozone production when propagated to global scales.

2.6.2. Heterogeneous Chemistry

Reactions occurring in the atmosphere on either the gas-solid interface (e.g. aerosol particles) or the gas-liquid interface (e.g. cloud droplets) are referred to as heterogeneous reactions. The heterogeneous chemistry which can affect ozone concentrations through production and loss of HO_x and NO_x and the production of halogen radicals is extremely important in particle rich aircraft exhaust plumes and contrails [90]. The exhaust plume contains emitted soot particles and ultrafine aqueous aerosol particles which are either formed within the plume or entrained into the plume from ambient air, as elaborated on in the following subsection. In the case of contrail formation, heterogeneous chemistry becomes very efficient, due to the four-fold increase in particle surface area of contrail ice compared to typical exhaust and background aerosol surface area. Meilinger et al. [132] states that the heterogeneous reactions occurring on aerosol particles have a negligible effect on ozone, however contrail ice can influence the ozone response to aircraft emissions by $\pm 0.5\%$ on a macroscopic scale which varies depending on time of day and year. Due to the relatively minor impact heterogeneous chemical reactions have on aircraft-induced ozone perturbations, and hence on aviation climate impact, their effect will be acknowledged, but the chemical intricacies will not be discussed further. For more information, see the references contained within this paragraph.

2.6.3. Aerosol and Contrail Microphysics

Whilst the climate impact of aviation NO_x emissions is dependent on the photochemical processes catalysing ozone production and methane destruction, the evolution and radiative forcing of aerosols and contrails is predominantly controlled by microphysical processes that occur at the aircraft plume scale.

The Microphysical Formation of Aerosols and Contrails

Upon release into the atmosphere, aerosol particle formation occurs through one of two nucleation pathways:

1. The condensation of two distinct gas phase molecules to form a liquid phase droplet through what is known as binary homogeneous nucleation. This is the case for reactive sulphur emissions that get chemically oxidised into sulphuric acid (H_2SO_4),

which then condenses with water vapour to form H_2SO_4 / H_2O droplets [158].

2. The gas-to-particle conversion occurring on the surface of foreign particles is known as binary heterogeneous nucleation, often leading to a liquid coating that forms on the particle. For example, H_2SO_4 / H_2O droplets can form a partial liquid coating around chemically activated soot in the aircraft exhaust plume through heterogeneous nucleation, leading to soot aerosol formation; a process that plays an important role in the formation of contrails [101].

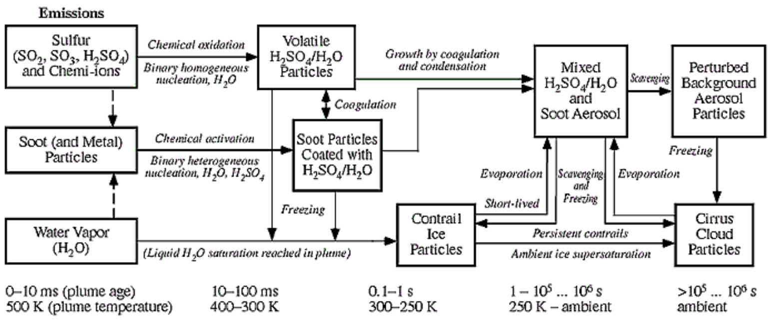


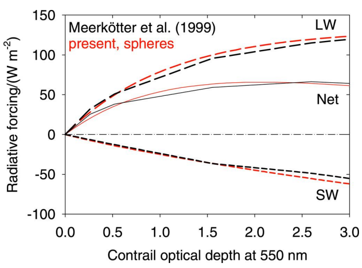
Figure 2.15.: The microphysical formation of aerosol and contrail particles in an aircraft plume. Displayed as a function of plume age and temperature [156].

Newly formed aerosol particles subsequently grow in the aircraft wake due to further condensation by uptake of surrounding water vapour, and a process known as coagulation. Coagulation refers to the collision between particles that results in the formation of larger particles, often initiated by graviational settling, turbulence or thermal motion. A process called scavenging occurs when two particles, one much larger than the other, coagulate and lead to the removal of the small particle from its particular size category, with its mass contributing slightly to the increase in mass of the larger particle. Coagulation commonly occurs in aircraft wakes between volatile particles such as H_2SO_4 / H_2O particles and soot aerosol, forming a mixed H_2SO_4 / H_2O-soot aerosol (see figure 2.15). This sulphate-soot aerosol may eventually become scavenged by background aerosol particles if it remains stable for up to a day [156]. Alternatively, if at any point during the mixing process of aircraft exhaust with air, water-supersaturation is exceeded, aircraft-induced particles and entrained background aerosol can be activated into water droplets through the uptake of surrounding water vapour [104]. If temperatures are below threshold according to the SA criterion, then these aerosol-activated water droplets will freeze to form contrail ice particles in the aircraft wake on the order of seconds post emission.

Contrail Microphysical Properties

The aforementioned microphysical processes (i.e. nucleation, condensation, coagulation, scavenging and freezing) determine the eventual composition and size distribution of particles in the aircraft wake, and if thermodynamic conditions permit, the formation and evolution of contrail ice particles [100], [164]. Hereinafter, this section will focus on the microphysics of contrail ice particles, as they induce the most significant radiative response out of all aviation climate forcers. Findings from Kärcher et al. [103] conclude that the predominant ice nucleation pathway for contrail formation is heterogeneous freezing of chemically activated soot aerosol particles, as these are the only remaining particle type in sufficient abundance at the time of freezing. However, simulation results from Kärcher et al. [102] strongly suggest that contrail formation is still likely in the absence of soot and sulphur emissions, through the activation and freezing of background aerosols.

The radiative forcing of contrails is thought to be determined by the product of optical depth and areal coverage [181]. In terms of areal coverage, contrail forcing is thus dominated by the presence of persistent contrails and contrail cirrus. However, the optical depth of a contrail is less dependent on its macroscopic properties and is instead determined by the optical and physical characteristics of the ice particles on the microscopic scale. The microphysical parameters deemed to be most responsible for inducing a radiative response are ice water content (amount of cloud ice per unit volume [104]), total ice particle number concentration, ice particle size distributions, effective radii, and ice particle shape [80]. Various studies have formulated methods to estimate RF and ERF from contrails, based on the parametrisation of these properties [12], [128], [173]. It is defined that the optical depth is proportional to the ice water content divided by the effective radius of entrained ice particles [173]. Contrails that tend to have more aspherical ice particle shapes are likely to have a stronger solar albedo, increasing the reflectance of the SW flux [128]. Ice number concentration is also thought to increase optical depth [99].



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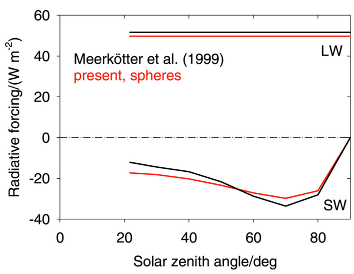


(b)

Figure 2.16.: **(a)** Daily mean instantaneous SW, LW and net components of RF assuming 100% contrail cover vs optical depth as calculated using the spherical ice particle model from Schumann et al. [182]. Results from Meerkötter et al. [128] shown for comparison. **(b)** RF vs SZA for spherical ice particles, with Meerkötter comparison shown. See [182] for more detail on assumed atmospheric conditions.

Optical depth of contrail cirrus is generally seen to increase radiative forcing, however the relationship is nonlinear and under assumed atmospheric conditions [128], [182], contrail RF increases with optical depth up to around 2.0, where it peaks, before decreasing at higher optical depths (see figure 2.16a). This is due to the lessening increase of positive LW forcing with increasing optical depth, whilst negative SW forcing continues to drop at a similar rate throughout. In addition to contrail optical properties, the magnitude and direction of contrail forcing is also directly determined by solar position, otherwise known as the solar zenith angle (SZA) (see figure 2.16b). Throughout the range of SZA values, LW forcing remains constant, because infrared emission from the Earth's surface is unaffected by solar flux. The SW flux on the other hand, goes further negative as SZA increases, up until a maximum at around 70°. Beyond this, the Sun begins to set and SW forcing returns to zero when the Sun is at 90° to the Earth's surface. This confirms the notion that contrails are solely warming at night, as the positive LW forcing always remains constant, whilst at night there is no chance of negative SW forcing.

2.6.4. The Saturation of Aircraft Emissions in High-Density Airspace Regions

In dense airspace regions, where the frequency of traversing aircraft is high, the resulting exhaust plumes may intersect and overlap, further affecting the nonlinear atmospheric response to chemical and microphysical processing occurring at the plume scale. Two outstanding saturation effects documented in the literature include the surpassing of NO_x-saturated conditions and the dehydration of surrounding water vapour due to contrail formation, leading to the mutual inhibition of ice particle growth in the aircraft wake.

NO_x-Saturated Conditions

The accumulation of nitrogen oxide emissions in the troposphere due to overlapping aircraft plumes has been observed empirically. For example, Schlager et al. [168] witnessed

NO_x concentrations of up to 30 times the average background concentrations in the NAFC, for an overlap of 2 to 5 aircraft plumes. As discussed in section 2.6.1, the net ozone production rate in the upper troposphere is a nonlinear function of the concentrations of NO and NO_2, with increasing NO_x leading to increasing O_3, up until a maximum, where any additional NO_x serves to reduce ozone production efficiency (e.g. figure 2.13b). The turnover point, sometimes called the "compensation point" is determined by competitive reactions involving nitrogen species and HO_x. In the NO_x-limited regime (left of the minimum ozone production efficiency $P(O_3)_{max}$), NO drives the production of ozone through reaction with the hydroperoxy radical (HO_2) [138]. However, it also drives the removal of HO_x through the reaction of OH with HO_2, HNO_4 and NO_2, thus limiting HO_x available for further O_3 production [212]. As NO_x levels increase up to compensation point, the increasing competition of HO_x removal processes begins to level off ozone production efficiency. Beyond this point, further increases in NO_x serve to decrease $P(O_3)$, thus signalling the start of the NO_x-saturated regime.

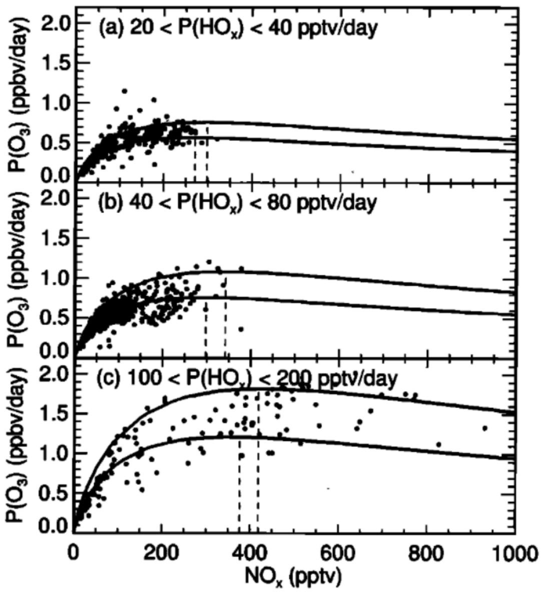


Figure 2.17.: Empirical observations of ozone production rate as a function of NO_x concentrations (parts per trillion volume, pptv), for three ranges of primary HO_x production: (a) 20–40 pptv/day, (b) 40–80 pptv/day, and (c) 100–200 pptv/day. Solid lines represent the range limits from model calculations. Dashed lines depict the NO_x level that induces peak ozone response [92].

During the POLINAT/SONEX campaign, Jaeglé et al. [92] presented the first empirical evidence of NO_x-saturated conditions in the NAFC. As seen in figure 2.17, $P(O_3)$ levels increased with increasing NO_x up until the saturation point, beyond which ozone production begins to drop off.

As seen in figure 2.17 at 100–200 pptv/day of HO_x production, $P(O_3)$ levels increased with increasing NO_x up until around 300 pptv, where it is clear that NO_x saturation has been reached. This means that any further emission of NO_x under those specific atmospheric conditions will potentially serve to decrease the net ozone production rate. Generally speaking, ozone production efficiency, and hence the threshold for

NO_x-saturated conditions, is largely dependent on just two variables: the ambient NO_x concentration (including the accumulation of NO_x from lingering aircraft plumes) and the HO_x production rate (which depends on the chemical composition of the atmosphere and the solar intensity [138]) [93].

Furthermore, model results from Kraabol et al. [113] in a study observing interactions between plumes revealed that, for aircraft flying along the same track, plumes of follower aircraft exhibit a considerably smaller ozone response than that of the leader aircraft. This observation on a local scale opens up the discussion for controlled saturation of emissions through formation flight, to take advantage of effects that are beneficial to the climate.

Dehydration Effects due to Contrail Formation

Contrails grow and persist in the atmosphere through deposition of water vapour from ice-supersaturated layers of the atmosphere. The conversion of atmospheric water vapour to solid phase ice particles has the potential to deplete local water vapour concentrations in regions conducive to persistent contrail generation [202]. It is thought that the depositional growth of contrail cirrus in regions of high airspace density can even lead to the dehydration of H_2O at typical flight levels, followed by the redistribution of humidity to lower levels, due to sedimentation and advection processes [174].

Contrail dehydration effects were first quantified by Burkhardt and Kärcher [25], in which a global atmospheric model was used to perform long-term integrations of the global impact of contrail cirrus on natural cirrus reduction due to ambient water vapour depletion. The paper found initial RF estimates to be -7 mWm^{-2}, lessening the contrail cirrus RF estimate by a factor of approximately one fifth, however the results were not conclusive. Schumann et al. [174] furthered this investigation through quantification of the impact of water exchange on contrail properties, large-scale humidity and the background climate. The results suggested that the drying at flight levels caused contrails to become thinner and last longer in the atmosphere, and the net reduction in contrail cirrus RF was deemed to be ~15% (not too dissimilar to the results of [25]). Furthermore, the results from a second run of the model, with aircraft emissions enhanced by 100 times, showing increasingly significant dehydration effects due to contrail cirrus formation and the redirection of humidity to lower flight levels. This implies that local dehydration effects will be far more significant in dense airspace regions where emissions accumulate and contrails can overlap, such as over mainland areas and in flight corridors. Work by Unterstrasser [202], [204] has elaborated on this concept of local dehydration, through numerical analysis of contrail growth in close proximity flight scenarios. The studies conclude that aircraft contrails compete for available water vapour, mutually inhibiting growth and leading to a saturation effect which diminishes the contrail of subsequent aircraft travelling through that region. Under proposed formation flight scenarios, the total extinction (optical depth) and total ice mass behind a two-aircraft formation are found to be reduced by 20–50% and 30–60% respectively.

The presence of plume-scale effects that are amplified in close-proximity flight scenarios begs the question; can climate beneficial saturation effects such as NO_x-saturated regimes and contrail-induced dehydration be exploited for mitigation purposes? Section 2.7 explores the concept of formation flight and how these additional climate benefits from emissions saturation may provide further impetus for real-world implementation of these concepts.

2.6.5. Parametrisation of Plume-Scale Effects into Global Models

Plume modelling methods serve as a useful tool for analysing the chemical and physical evolution of aircraft exhaust plumes throughout their lifetime. However, integrating high-resolution plume data into global atmospheric models is simply unfeasible when a large number of flights, on the order of 100,000–200,000 flights per day must be considered. To counteract this, simpler parametrisations of plume models are used, which capture so-called "effective emissions" that attempt to correct the ID response of the large-scale model, according to the predicted plume-scale effects and their influence on the eventual climate impact. Parametrising plume-scale climate effects into global atmospheric models that assume instantaneous dispersion is a topic covered extensively in the literature, for both ozone-NO_x chemistry [29], [129], [151], [153] and contrail and cloud processes [24].

Parametrisation of gas-phase chemical conversions

In Paoli et al. [151], three key concepts are reviewed which enable the parametrisation of nonlinear plume chemistry into ID global models; effective emission indices (EEIs), effective conversion factors (ECFs) and effective reaction rates (ERRs).

The EEI concept was first theorised in Petry et al. [159], in an attempt to account for the difference in concentration evolution of key chemical species between large scale models which assume ID and plume models which account for the entrainment of emissions throughout the plume lifetime. EEIs provide a suitable correction to the original EI of an emitted species, so that the concentration is the same in both models at the end of the plume "dispersion time" t_{ref}, when emissions are fully dispersed into the dimensions of the computational grid cell in which they were released.

Meijer [130] presents the concept of ECFs which are factors applied to the NO_x emission index to account for the increased chemical conversion rate of NO_x to nitrogen reservoir species in the plume. Since the total emitted reactive nitrogen (NO_y) is chemically conserved throughout the plume lifetime, the amount of NO_y throughout the plume lifetime is equal to the amount of NO_x emitted initially, and hence the sum of ECFs for NO_x and all reservoir species is unity [207]. Thus, the faster in-plume conversion rates lead to a decreased NO_x ECF, whilst the ECFs of reservoir species such as HNO_2

and HNO_3 increase considerably. As a result, the eventual net ozone production is affected as explained in section 2.6.1, however ECFs vary considerably depending on altitude, latitude and seasonal variation meaning the magnitude and direction of the ozone perturbation is also affected.

Despite the widespread use of EEIs and ECFs to parametrise plume-scale gas-phase chemical conversions in past literature, there are a number of known issues that affect accuracy such as mass conservation inconsistencies and poor accounting of local and regional variation in dispersion properties and turbulence. Cariolle et al. [29] attempts to overcome such issues with the ERR concept. ERRs reconstruct the concentrations of the emitted species in the plume by diluting the emission to the resolution of the computational grids cell, while modified reaction rates are used to model reactions with ambient chemical species that occur at the plume scale. Modified reaction rates are introduced by the determination of effective reaction rate constants, which are used to compute secondary species produced in the plume. As a result, ERRs build on the EEI and ECF concepts by ensuring full mass conservation by modulating pre-existing background chemical cycles instead of directly estimating mass change, and they account for chemical transport due to diffusion and turbulence processes that are entirely dependent on location.

Parametrisation of Heterogeneous Chemistry and Microphysics

Despite there being an extensive number of gas-phase chemistry parametrisations, the same cannot be said for parametrisations of heterogeneous chemistry and microphysics, for modelling aircraft plume-scale climate effects at a global scale. Kärcher et al. [102] was one of the first studies to explicitly attempt to parametrise heterogeneous reactions and aerosol microphysics in global models. The paper attributed the perturbations to the background aerosol layer (due to aerosol emissions from aircraft) to increasing surface areas and number concentrations of aerosol particles, which subsequently increase heterogeneous reaction rates. Findings from Meilinger et al. [131], [132] however, reveal that heterogeneous chemistry effects in dispersing aircraft plumes require special consideration of both the chemical and microphysical interactions, and the dynamical response of the plume itself. In [132], the Mainz Aircraft Plume Model is used to calculate the response of ozone and nitrogen reservoir species in the plume due to heterogeneous chemistry and microphysical effects. It is found from the modelling results that the impact of heterogeneous chemistry and microphysics on the ozone response at global scales is highly sensitive to local meteorology and the instantaneous state of the atmosphere. Therefore, parametrisation of these effects is much more convoluted than first thought, and due to the relatively minute impact on ozone response ($\pm 0.5\%$), it is not unreasonable to disregard these effects in global modelling efforts.

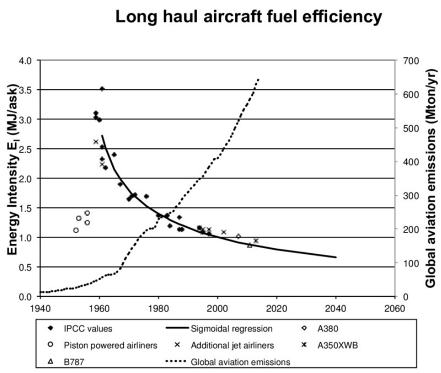
Contrarily, the parametrisation of contrail microphysics into global models is of crucial importance, as contrail radiative forcing is determined primarily by the optical properties of its constituent ice particles. Burkhardt and Kärcher [24] summarise the parametrisations necessary to include contrail radiative forcing in global models; this includes parametrising the factors affecting ice supersaturation, contrail formation and persistence, contrail spreading and ice water content. This led to the development of the process-based contrail cirrus module (CCMod), which was later implemented in the global climate model ECHAM4, for global contrail cirrus radiative forcing analysis [25], [26], [119]. More recently, a microphysical extension to CCMod was applied and implemented in ECHAM5 [12], [15]. Model outputs were used to determine the global effective radiative forcing of contrail cirrus in [116], finding that the ERF is more than 50% smaller than the RF equivalent. This is thought to be due to the reduction in natural cloudiness caused by contrail cirrus dehydration of the surrounding atmosphere, and due to the dependence of contrail climate impact on prevailing air traffic distribution patterns. See [116] for more info on the parametrisations of contrail and aerosol microphysics implemented to deduce global ERF estimates.

2.7. Aviation Climate Impact Mitigation

It is customary practice among aviation policymakers to solely focus on mitigating the greenhouse effect induced by aviation carbon emissions. This means that analysis of the climate impact of non-CO_2 emissions is largely neglected, despite being responsible for over two-thirds of aviation-induced climate change [116].

2.7.1. Conventional CO_2-Centric Mitigation Approach

The industry fixation on aviation CO_2 mitigation stems from the easily quantifiable nature of the substance and its impacts; the direct coupling to fuel consumption and its relative stability and long lifetime make it a prime target for mitigation. This is due to the mutually assured reduction in both fuel consumption and corresponding CO_2 emissions, providing an economic and environmental incentive. Conventional mitigation approaches have often taken the form of improvements to aircraft and engine design, aviation technology and infrastructure, and aircraft operations, all aimed at minimising aircraft fuel consumption. These changes have been largely incremental due to the prioritisation of safety and economic stability over rapid implementation. As a result, efficiency gains have stalled in recent years (1–2% per annum), whilst growth rates continue to rise unabated (4–5% per annum) [116], [155]. See figure 2.18a for a graphical representation of the trend in efficiency gains and aviation growth since 1940.



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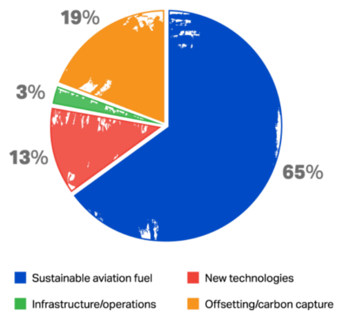


(b)

Figure 2.18.: **(a)** Efficiency gains versus absolute aviation emissions growth since 1940 and projected to 2040 [155]. **(b)** IATA Net Zero 2050 contributions [135].

At the International Air Transport Association (IATA) 77$^{\text{th}}$ Annual General Meeting in 2021, a resolution was approved for the global air transport industry to achieve net-zero carbon emissions by 2050, in accordance with the Paris climate agreement to limit global temperature rise to 1.5 ° C [135]. This plan relies on the following contributions, as displayed in figure 2.18b: sustainable aviation fuels (SAFs) are to provide 65% of the net carbon reduction, offsetting schemes and carbon capture are to deliver 19%, new technologies such as hydrogen and electric propulsion to provide 13%, and infrastructure/operational improvements to provide the final 3%. This means that 78% of the prospective net carbon emissions reduction comes from SAFs and alternative technology such as hydrogen and electric propulsion concepts.

The fundamental issue with reliance on SAFs and nascent technologies, is that they may take decades to bring about noteworthy reductions in aviation climate impact. This is due to the need for a fleet-wide overhaul of aircraft equipment and associated ground infrastructure, which would likely incur huge costs and require extensive certification and approval processes [14]. Not to mention, the environmental and ethical concerns have cast doubts over the prospect of a 4000-fold increase in global SAF production, to meet the IATA net-zero target of 449 billion litres per year by 2050 [8], [135].

Carbon offsetting schemes, such as ICAO's Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), have been proposed to alleviate aviation climate impact in the meantime. However, an EU study investigating the efficacy of carbon offsetting indicates that such schemes often fall far short on providing meaningful mitigation, because of questionable offset quality, perverse incentives taking away from the real need to decarbonise, a lack of participation from key markets, and a clear lack of overall transparency and enforceability [9], [38].

The remaining net-zero contribution is expected to come from improvements to operations and infrastructure, which will continually provide 1–2% per year. This will involve transitioning towards more efficient fuel management systems and improving overall technology and design of the aircraft, as well as advancing air traffic management systems and airspace modernisation [155]. However, as mentioned previously, improvements aimed at increasing aircraft fuel efficiency are largely outpaced by growth rates in passenger demand and hence emissions. Therefore, more must be done in the short term (i.e. the next 5 to 10 years), to reduce the dependency on nascent technologies which are still decades away from being impactful.

Whilst the proposed net-zero targets are solely focused on decarbonising the aviation sector, it is inherently assumed that the reduction in non-CO_2 emissions will follow a similar trend. This review paper has evidenced however that this is not necessarily the case, due to the sensitivity of non-CO_2 emissions to combustor conditions, as well as the dependency of the atmospheric response on the environmental conditions of the ambient air. For this reason, action must be taken simultaneously to tackle non-CO_2 climate impact, alongside the current decarbonisation plan, with the potential to provide substantial climate impact reductions on a much shorter timescale.

2.7.2. Alternative Non-CO_2-Focused Mitigation Approach

There are myriad solutions to mitigating aviation climate impact in the interim to low-carbon flight, however the focus here will remain on near-term operational mitigation measures that optimise aircraft routing by minimising non carbon-based radiative effects.

As alluded to in sections 2.2 and 2.5, non-CO_2 emissions are on the other hand, produced in varying quantities depending on combustor conditions. Also, when released into the atmosphere, they have a spatio-temporally sensitive climate response, which depends on the background atmospheric conditions into which they are emitted. As such, anthropogenic as well as natural perturbations to atmospheric chemistry need to be considered; the entrainment of emissions to the exhaust plume for several hours post emission, leads to locally elevated emissions concentrations. This causes reactive non-CO_2 species to experience nonlinear chemical and microphysical processing that occurs at the scale of the plume, which subsequently affects their net climatic response when propagated to global scales.

Measures such as climate-optimal aircraft routing and formation flight present the potential for a substantial climate impact reduction in a relatively short timeframe, whilst requiring minimal technology changes to the current fleet and ground infrastructure.

Climate-Optimal Aircraft Routing

The first measure proposed to tackle aviation's non carbon-based environmental impact is climate-optimal aircraft routing. This involves re-routing aircraft in flight to avoid regions of the atmosphere that are particularly sensitive to non-CO_2 climatic effects, such as where NO_x gives rise to excessive ozone production or where persistent contrails are formed. Simulation efforts have suggested that this method has the potential to reduce aviation climate impact by 10–20%, at the cost of only a few percent of additional fuel consumption [70], [121], [145]. Moreover, the non-CO_2 climate impact from aviation is not evenly distributed across all flight distances, meaning that operational mitigation can be targeted towards flights which induce the most significant impact. For example, in a case study on contrail avoidance strategy [198], it was found that diverting 1.7% of the aircraft fleet could reduce the contrail climate impact by 59.3%, with only a 0.014% increase in fuel burn and accompanying additional CO_2 emissions. This sheds light on the sheer potential for significant reduction in aviation climate impact, through a minimally invasive climate-focused route optimisation strategy.

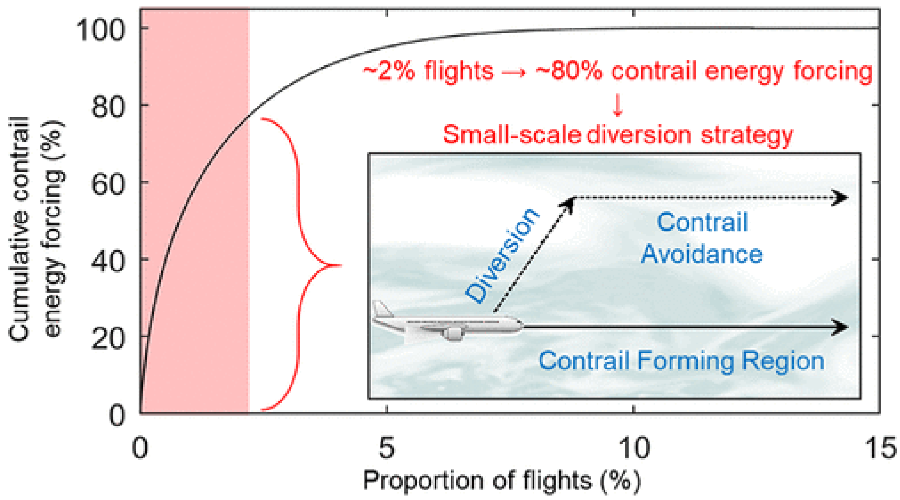


Figure 2.19.: Schematic representation of contrail avoidance strategy. Plot of cumulative contrail energy forcing against proportion of flights [198].

Implementation of this method in the real world does however require a number of issues to be addressed; Grewe et al. [72] lay out four key hurdles that must be overcome:

1. The accuracy and robustness in determining eco-efficient flight trajectories must be improved, and this must be possible near real time.

2. Consensus must be achieved on determining the extent to which cooling effects should be exploited (e.g. intentionally flying a route that generates a cooling contrail

could be seen as unnecessary intervention in nature).

3. The implications of fleet-wide climate-optimal routing on the air traffic management system must be assessed rigorously, to ensure safety, order and efficiency is maintained.

4. A non-CO_2 market-based measure or policy pathway must be adopted to incentivise this transition towards a climate-optimised air traffic network.

In the time since, a dedicated network of researchers from various institutions have been working towards bringing this concept to fruition on a large scale. The FlyATM4E research group has been developing methodologies to increase robustness in the determination of eco-efficient flight trajectories using so-called algorithmic climate change functions (aCCFs) [127]. Live trials are currently ongoing in the Maastricht Upper Area Control (MUAC) to investigate the operational feasibility of contrail prevention from an air traffic control perspective [48]. The inclusion of non-CO_2 effects of aviation in the European Union emissions trading scheme (EU ETS) and under CORSIA has also been discussed in great detail in a comprehensive report by Niklaß et al. [146]. Most recently, a comprehensive survey paper was published by Simorgh et al. [185], detailing the current operational strategies that are proposed in the state-of-the-art literature published in this field over the last few decades. The aim of this review was to collate methodologies used for aircraft performance modelling, climate modelling and optimisation, and to identify gaps to guide future research.

Formation Flight

Formation flight for wake energy retrieval involves the flight of two or more aircraft, with the follower aircraft positioned in the smooth updraft of the leader aircraft's wake. This has been demonstrated in simulation [10], [205] and flight testing [76] to reduce required lift and thrust of the trailing aircraft and offer a 5–10% reduction in fuel burn in paired formation. Most work to date focuses on two-aircraft configurations, however three [19] and more have also been considered. In figure 2.20, it is evident how this concept is used in practice to obtain aerodynamic benefits. The counter-rotation of the wake vortices of the leader aircraft leads to a stream of upwash on the outside of either rolled up vortex. Flying a follower aircraft in this region of upwash induces a vertical velocity, with aerodynamic benefits achieved at distances as much as 30 wingspans (i.e. ~1 km for Airbus A320).

Safety is a critical hurdle, particularly in close formation cases, along with regulatory and air traffic systems and processes. Airbus' fello'fly wake energy retrieval project has undertaken initial work to identify and address operational and safety challenges, proposing regulatory standard adaptations to facilitate adoption, and conducting a trial transatlantic flight in 2021 [53]. Claims in this document state that formation flight strategies could be deployed around the middle of this decade. Proposed longitudinal separation distances for formation flight range from 56 ft [76] to 1.5 NM [53], with lateral distances ranging from 10%-span overlap to 30 m separation showing benefit. Routing aircraft into formation has been considered in simulation [215], with net fuel (and hence CO_2 saving benefits of 5.8%) being identified for a single operator, rising to 7.7% for a transatlantic alliance. Significant benefits have been projected for long-haul and transatlantic routes, with more modest reductions for low-cost airline cases [106].

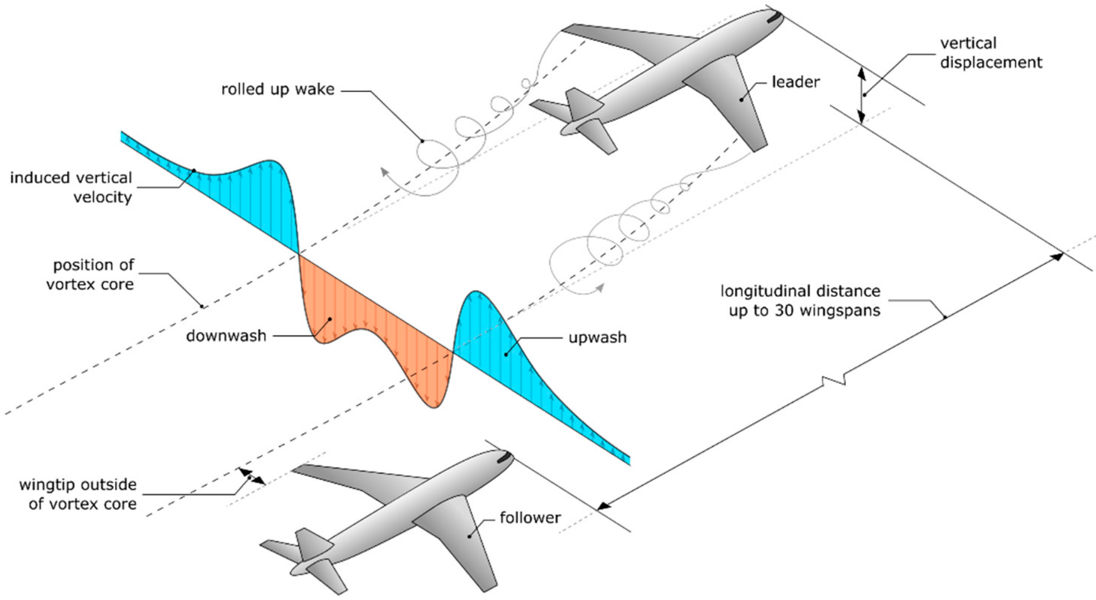


Figure 2.20.: Visual representation of the formation flight concept, depicting the fundamental principles of wake energy retrieval [124].

Non-CO_2 impacts have been neglected until recently, but a synergistic outcome of formation flight is realised via the saturation of emissions in the trailing exhaust plumes, leading to the aforementioned nonlinear chemical and microphysical response that reduces ozone and contrail formation, as detailed in section 2.6.4. This has been projected to offer net climate impact reduction for a two-aircraft formation of 13–33% [36], which is significantly higher than the CO_2-focused formation discount factors employed in the routing simulations mentioned above. Given the near-term feasibility demonstrated in industry-led trials, and the combined CO_2 and non-CO_2 benefits that produce an outsized impact when overall ERF is considered, formation flight offers a promising interim solution that can continue to be employed for drag reduction/wake energy retrieval once SAFs and hydrogen/electric propulsion are more widespread.

In this section, the prospects of non carbon-based operational mitigation measures have been discussed, and their potential to reduce aviation's net climate impact has been investigated. Gradual efficiency gains are unable to outpace the rapid growth in demand year upon year, meaning that more radical approaches must be implemented in order to achieve net-zero emissions targets. For this reason, two concepts have been proposed and discussed: climate-optimal aircraft routing and formation flight. The former aims to re-route the aircraft to minimise time spent in particularly climate sensitive regions of the atmosphere, whereas the latter aims to optimise atmospheric conditions artificially by overlapping aircraft plumes and saturating local concentrations of emissions to lead to climate-beneficial effects. For the industry to truly reap the benefits and maximise potential non-CO_2 reductions from implementation of these measures, there is the potential to perform them simultaneously, in a controlled and safe manner. This would involve flying aircraft in formation, along routes which are optimised with respect to minimum climate impact.

2.8. Conclusions

This review paper delivers a holistic overview of literature pertaining to the spatio-temporal climate sensitivity of the atmosphere to reactive non carbon-based species. This includes literature from a range of research areas: aircraft emissions, plume dispersion and dynamics, the distribution of air traffic and corresponding emissions, the climate impact of aviation on both a global and a local scale, and lastly the mitigation potential for both CO_2 and non-CO_2 climate impact.

A strong finding from this literature review, giving direction to fruitful future work on this research topic, is the effect of plume-scale nonlinear chemistry and microphysics on the eventual climate impact of reactive non-CO_2 aircraft emissions. The chemical and physical state of the atmosphere is affected by both natural variability in climate parameters, as well as by perturbations to atmospheric chemistry due to anthropogenic emissions. In the case of aircraft operating in the UTLS, the atmospheric state (and hence the chemical fate of reactive aircraft emissions) is influenced by both the background chemical composition of the surrounding air and the elevated concentrations of emissions within the aircraft exhaust plume. The sensitivity of the atmospheric response to emissions is heightened at these altitudes compared to at surface level, due to reasons elaborated on in section 2.5.1. Climate-optimal aircraft routing and formation flight can both be utilised to minimise time spent flying through atmospheric regions that are unfavourable in terms of climate impact.

Real world implementation of non-carbon based operational mitigation concepts does however require: knowledge of aviation fuel consumption, plume dispersion characteristics, and monitoring and forecasting of chemical and meteorological parameters along potential routes. Moreover, this information must be obtained to a sufficiently high-resolution and within a relatively short timespan, so as to capture adequate variability in climate parameters to make an informed and timely decision on routing. As a result, modelling assumptions will inevitably need to be made, to balance computational run time with accuracy. In particular, the incorporation of plume-scale effects into large-scale climate models is likely to bring about issues with resolving data between the two resolutions. Another key issue limiting feasibility at the current time is that of air traffic management. Further work should focus on these key challenges and work towards improving scientific understanding of the non-CO_2 radiative forcing multiplier. Achieving a general scientific consensus on this non-CO_2 multiplier would increase confidence amongst policymakers to include such effects in aviation climate policy, and hence accelerate the widespread adoption of interim mitigation approaches such as climate-optimal aircraft routing and formation flight.

The extensive background reading done to write this review chapter is used throughout the thesis, to inform the later chapters and to ensure work done is truly novel and relevant. The rest of this thesis focuses on a number of different studies conducted by my team and

I at the University of Bristol, across both Aerospace Engineering and the Atmospheric Chemistry Research Group (ACRG).

3. Investigating the Real-World Feasibility of Aircraft Exhaust Plume Intersections using Open Source Air Traffic Data

The content of this chapter is based on the following published conference article:

Tait, K. N.; Roome, S.; Hoole, J., "Investigating the Real-World Feasibility of Aircraft Exhaust Plume Intersections using Open Source Air Traffic Data", *Engineering Proceedings*, vol. 28, no. 1, pp. 16, 2022. DOI: `https://doi.org/10.3390/proceedings1010000`.

This chapter incorporates material from that work with permission from the publisher and co-authors, where necessary. I was lead author of this publication; I wrote the software code and conducted the simulation study with Stephen Roome, and wrote the paper in cooperation with Dr Josh Hoole (University of Bristol, Dept. of Aerospace Engineering). I also travelled to the Delft University of Technology, Netherlands, to present this research at the OpenSky Symposium 2022 **Opensky2022**.

This paper and associated research naturally followed on from the literature review publication that formed chapter 2. Throughout the process of reviewing the literature, it became apparent that only a handful of papers have investigated the prospect of aircraft plume overlap in real-world airspace **x**.

3.1. Introduction

Aircraft spend the majority of their flight at cruising altitudes (32,000–42,000 ft), where the air is thin enough to significantly reduce drag, whilst still dense enough to maintain sufficient oxygen flow to the engines. In atmospheric science terminology, this region is known as the Upper Troposphere Lower Stratosphere (UTLS), characterised by its low temperatures, low concentrations of atmospheric trace species and higher radiative efficiencies compared to ground level. Aviation is unique in that it is the only significant source of anthropogenic emissions in the UTLS.

3.1.1. Aviation Non-CO_2 Climate Impact

The two most significant contributions to aviation's high-altitude non-CO_2 climate impact are NO_x-based ozone production and condensation trails (contrails). Aircraft NO_x emissions do not contribute to the greenhouse effect directly, however they act as a precursor to chemical reactions which produce ozone and deplete methane, both of which are potent greenhouse gases. The warming due to tropospheric ozone production tends to outweigh the cooling due to methane reduction, thus resulting in a net warming effect, that constitutes 16% of aviation-related global heating [116].

In the exceptionally cold and humid conditions experienced in the UTLS region, water vapour can condense and freeze around particulate emissions in the aircraft exhaust to form ice crystals. The trail of visible ice crystals that forms behind the aircraft is known as a contrail. Contrails can be short-lived, dissipating over time with a lifetime of seconds to minutes, or they can persist for many hours, growing with time. The criteria for persistent contrail formation is ambient ice supersaturation. When aircraft generate contrails in ice-supersaturated regions (ISSRs), the high ambient humidity facilitates ice crystal growth, due to uptake of surrounding atmospheric water vapour. Persistent contrails can spread over hundreds of kilometres and transition to cirrus clouds throughout their evolution, resulting in a substantial man-made increase in high ice cloud coverage globally [104], [181].

The radiative properties of contrail ice crystals are such that they are typically more efficient at absorbing and reflecting infrared radiation emitted from the Earth's surface, than they are at scattering inbound solar radiation, resulting in a global heating effect. The warming induced by aviation contrails and contrail cirrus is on the order of 51% of total aviation climate impact [116]. This means that the transient, localised heating effect due to global contrail coverage on a day-to-day basis, is significantly greater than that of all the aircraft CO_2 emissions that have accumulated since the beginning of civil aviation [25].

3.1.2. Mitigating Aviation-Induced Climate Change by Modifying Flight Routes

At cruise altitudes, the degree to which aircraft non-CO_2 emissions cause warming or cooling is largely determined by the instantaneous atmospheric conditions (i.e. the local chemical and meteorological situation). The atmospheric state is itself, highly variable with respect to time and location, meaning that the same mass of emissions can have very different climate effects, depending on when and where they are released [127]. Climate-optimal routing is an operational mitigation concept, proposed to exploit this natural variability in environmental conditions throughout flight. It involves re-routing aircraft to avoid regions of the atmosphere that are particularly sensitive to aircraft emissions. For example, flying around an ISSR region to avoid persistent contrail generation, or flying through airspace in which the chemical composition is less conducive to tropospheric ozone formation, as illustrated in figure 3.1. Modelling efforts have demonstrated the ability to reduce the climate impact of flying by 10–20%, at the expense of only a few per cent additional fuel burn [126], [145].

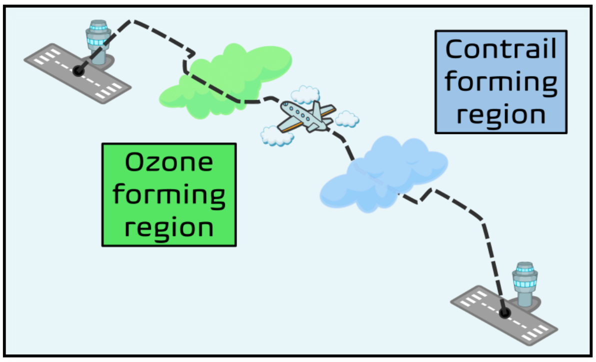


Figure 3.1.: Conceptual representation of an aircraft trajectory optimised for minimum climate impact.

Further to the notion that natural atmospheric fluxes influence aviation's non-CO_2 climate impact, there is evidence to suggest that artificial perturbations, due to emissions from lingering aircraft exhaust plumes, may also play a significant role [92], [168]. The overlapping of aircraft plumes can locally elevate chemical mixing ratios of key emission species, resulting in chemical saturation effects which are potentially beneficial from a climate change perspective. Firstly, the ability for NO_x emissions to produce ozone become hindered once concentrations surpass a threshold concentration. This means that additional NO_x input locally will only serve to reduce the NO_x-to-ozone conversion efficiency [96]. Persistent contrail formation is also subject to saturation effects. As a persistent contrail grows, it depletes water vapour from the surrounding atmosphere, thus impeding growth of any subsequent contrails formed in the same region [202]. Both these saturation effects lead to a reduction in the resultant non-CO_2 climate impact of associated flights, compared to the single aircraft scenario. In spite of this, climate-optimal routing studies typically do not consider more than one aircraft, thus neglecting saturation effects caused by residual aircraft emissions. Instead, a range of studies have explored the idea of flying commercial aircraft in formation, as a means of reducing both CO_2 and non-CO_2-related warming.

Formation flight involves flying one or more aircraft, one behind the other at a nominal separation distance of around 1–2 km. The follower aircraft flies in the upwash of the leader aircraft's wake vortices, which in turn reduces required lift (and hence thrust) for trailing aircraft. As a result, a fuel saving benefit of 5–8% can be achieved [10], [215]. The possibility of exploiting chemical saturation effects through controlled plume overlap has also been explored [36], [124]. However, no study to date has truly encapsulated the concepts of both climate-optimal routing and formation flight, as two parts of the same optimisation problem. Whilst air traffic data has been used in prior studies of aviation emissions [54], [194], including modelling of contrail formation [74], this is the first paper to the authors' knowledge, that will utilise such datasets to analyse the likelihood of aircraft plume overlap in high-density flight scenarios.

3.2. Methodology

The spatial and temporal distribution of global air traffic is highly inhomogeneous, due to variations in passenger demand, safety regulations, airspace organisation and flight route optimisation [69]. Due to the relative unpredictability and non-uniformity in air traffic flows and patterns, modelling and investigation should ideally be based around real-world data. In this paper, historical commercial aircraft trajectory data obtained from the open-source air traffic dataset OpenSky **Schafer2014**, is leveraged to explore how aircraft plumes typically overlap in airspace regions where air traffic densities are particularly high. The OpenSky Network sources data from Automatic Dependent Surveillance-Broadcast (ADS-B) ground-based receivers, which determine aircraft latitude, longitude and altitude, along with additional measurements such as flight track and ground speed, at a temporal resolution of one second. This dataset is used here to deduce information regarding air traffic flows and airspace density, as well as modelling of aircraft plume intersections, throughout the month of January 2022. All necessary computations were carried out using the Python programming language, coupled with the traffic API **Olive2019** to handle OpenSky data requests and all plots were generated using Python packages Matplotlib and Cartopy.

In flight air traffic sectors where demand is considerably high, or where two or more flight corridors intersect, air traffic can become significantly congested, with airspace capacities often approaching their upper limit [199]. Although undesirable from an air traffic safety and efficiency perspective, it so happens that these regions could provide excellent insight into the real-world practicality of formation flight and controlled plume overlap. Therefore, the first aim of study was to use OpenSky ADS-B data to locate the highest density flight regions during a typical busy time period. January 2022 was selected, as this represented a point in time when the aviation sector had recovered significantly from the COVID downturn, and was prior to the Russian invasion of Ukraine, which further disrupted the air traffic situation considerably. Trajectory data over this month were discretised into $1°$ longitude $\times$ $1°$ latitude bins, at a typical cruise altitude of 38,000 ft $\pm$500 ft. The categorised data were then used to construct a global heatmap of mean airspace density for each grid cell. To reduce computational intensity, snapshots of the air traffic situation were taken at the start of every hour, each day of the month.

Following the outcome of the global analysis, two key high-density hotspots were selected for further analysis. The flux densities through the box were calculated for each day and averaged over the whole month, to single out the time steps with the highest average aircraft count. The case study of these two hotspots continued at the subgrid scale, through higher resolution observation of trajectories and modelling of their associated exhaust plumes. Dispersion modelling of aircraft plumes has been covered extensively in the literature over the past few decades, with models of varying fidelities becoming widely accepted amongst the community [151]. For this paper, the aim was to select a relatively low fidelity model to track the plume dispersion from OpenSky aircraft trajectories, whilst also being computationally efficient, so that many trajectories could be easily tested. The model used here was that of the Multi-Shell Plume model from Kraabøl et al. [112]. Here, the plume's horizontal, vertical and skewed standard deviations (σ_h, σ_v and σ_s) and radii (r_h, r_v) of the outermost shell were modelled through equations (1) to (5).

$$\sigma_\mathrm{h}{}^2(t) = \frac{2}{3}s^2 D_\mathrm{v} t^3 + (2D_\mathrm{s} + s\sigma_{0\mathrm{v}}{}^2)st^2 + 2D_\mathrm{h}t + \sigma_{0\mathrm{h}}{}^2, \tag{3.1}$$

$$\sigma_\mathrm{v}{}^2(t) = 2D_\mathrm{v}t + \sigma_{0\mathrm{v}}{}^2, \tag{3.2}$$

$$\sigma_\mathrm{s}{}^2(t) = sD_\mathrm{v}t^2 + (2D_\mathrm{s} + s\sigma_{0\mathrm{v}}{}^2)t. \tag{3.3}$$

$$r_h = 3\sigma_\mathrm{h} \tag{3.4}$$

$$r_v = 3\sigma_\mathrm{v} \tag{3.5}$$

These equations derive from the two-dimensional diffusion equation, and hence contain initialised atmospheric parameters, according to the assumed turbulent conditions of Durbeck and Gerz [47]: wind shear (s $= 0.005$ s^{-1}), horizontal ($\mathrm{D}_h = 16.7$ $\mathrm{m}^2\,\mathrm{s}^{-1}$), vertical ($\mathrm{D}_v = 1.1$ $\mathrm{m}^2\,\mathrm{s}^{-1}$) and shear ($\mathrm{D}_s = 0.5(\mathrm{D}_v\ \mathrm{D}_h)^2$) diffusion coefficients, at an arbitrary time step t. The time step used in the plume simulations present throughout this paper is 10 s. This means that aircraft positional data were gathered every 10 s, and instances of plume segments were initialised, for active aircraft in the grid cell at that particular time step. For plume instances that already exist, they are expanded each time step, according to the above equations.

In addition to plume modelling, data were collected on aircraft-to-aircraft information across all flights that traverse through the observed grid cells over the specified timeframe. This includes relative distance, relative track and bearing between aircraft pairs. From this, criteria for scenarios where it was likely that two plume instances would overlap was determined, thus promoting climate-beneficial saturation effects, as elaborated on in section 3.1. The criteria to be met to achieve successful plume overlap were termed *significant plume encounters* (SPEs). Based on the E-folding time of NO_x and typical contrail growth rates at aircraft cruising altitudes, it was decided that a threshold overlap time for an SPE was around 2 hours **Shallcross2022**, [113]. Thus, the final objective of the study was to determine how many instances of aircraft pairs, at each time step, achieved a successful SPE throughout the observed duration.

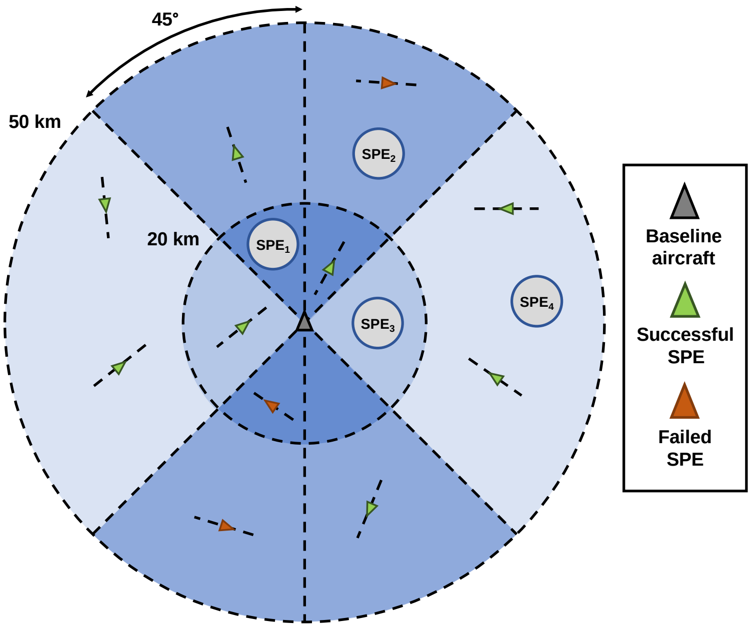


Figure 3.2.: Schematic drawing of SPE criteria used to deduce plume overlap cases and compare the relative importance of plume intersection events.

SPEs were ranked by significance from 1 to 4, as is visually represented in figure 3.2. Under assumed plume conditions, horizontal plume radius r_h is predicted to reach between 6 and 7 km at the 2 h threshold, whilst r_v approaches a mere 75–100 m. SPE ranks were designed in such a way as to capture all instances where plumes from aircraft pairs are likely to overlap, prior to the proposed 2 h threshold. SPE_1 is classed as any aircraft that is within a 20 km proximity of the baseline aircraft, has a bearing within $\pm$ 45° of the baseline track, and also, has a track that is within $\pm$ 45° of the baseline track. For SPE_2, the same rules on relative tracks apply, however relative distance is between 20 and 50 km. SPE_3 counts as any aircraft that is outside the $\pm$ 45° boundary, however is within 20 km. Relative headings were neglected here, as any aircraft crossing at such obtuse angles will likely experience plume overlap to some degree. Finally, SPE_4 represents any aircraft that does not satisfy any of the other conditions, however is still within a 50 km radius of the baseline aircraft. The ranks were deduced based on relative distance and tracks between all aircraft pairs at every time step, prioritising tracks over distance. This is because two aircraft flying a similar heading are extremely likely to experience significant and persistent plume overlap, even if they are a large distance apart longitudinally. This is because aircraft travel at high speeds and trajectories tend not to deviate too far from their heading, during the cruise segment of flight [196].

3.3. Results

This study brought to light some interesting findings regarding plume overlap occurrences, in real-world air traffic scenarios. Firstly, the results of the global analysis were plotted on a heatmap of mean airspace densities. This was subsequently overlaid onto a world map, to allow for visual identification of geographical biases in aviation demand. Figure 3.3 shows the heatmap, with magnification of the two overarching regions with the highest air traffic densities, the East Coast of the United States and Central Europe.

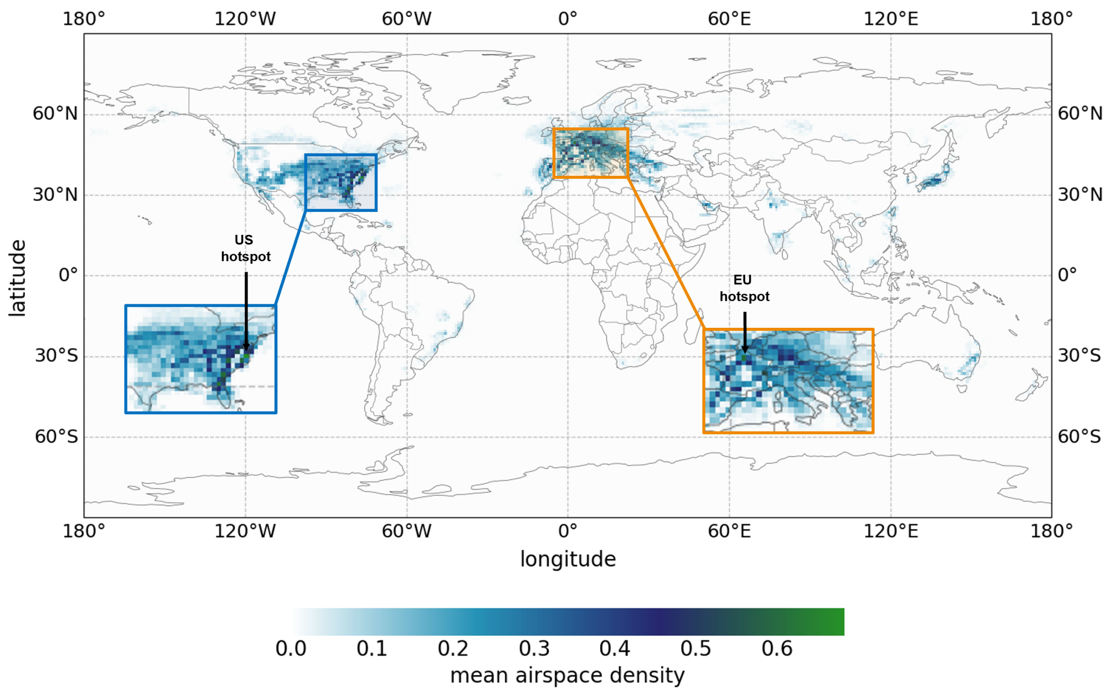


Figure 3.3.: Global heatmap of mean air traffic density. Snapshots of air traffic data were taken at the first second of every hour throughout 01/2022. Mean values are significantly lower than peak values due to large variations in aviation demand throughout the course of each day.

From these magnified regions, the two highest density grid cells were selected for a further subgrid scale analysis. These were the United States (US) hotspot, with a midpoint longitude of -76.5° and latitude of 36.5°, and the European (EU) hotspot at longitude 2.5° and latitude 49.5°. Again, both grid cells are at an altitude of 38,000 ft ±500 ft. Before conducting the high resolution subgrid scale study, it was first necessary to cherry-pick a suitable time interval to investigate, as running plume simulations over the entire month period was too computationally expensive. Instead, using trajectory data resampled at 1 minute intervals, the average aircraft count for each hour of the day was calculated over the whole month, for each hotspot grid cell. The results can be seen in figure 3.4.

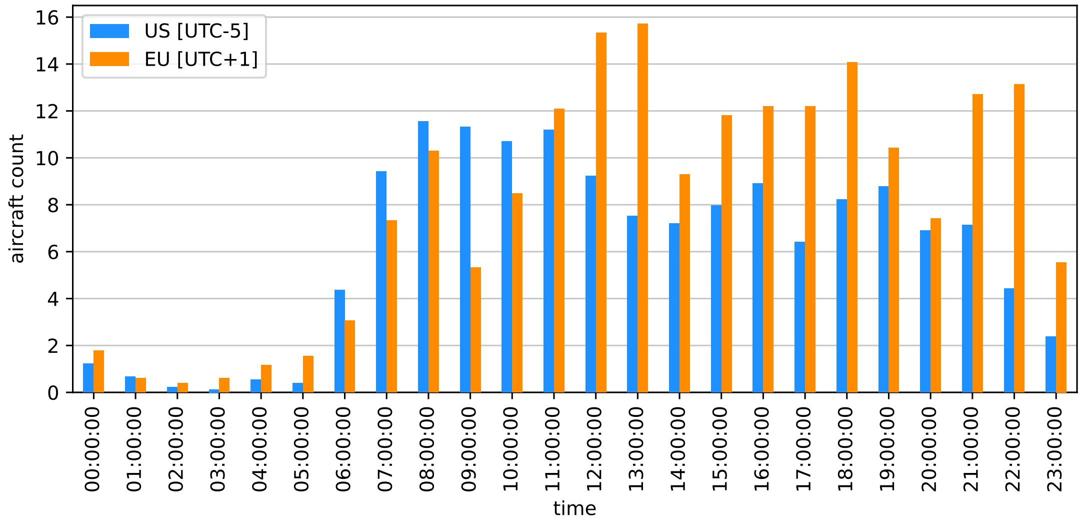
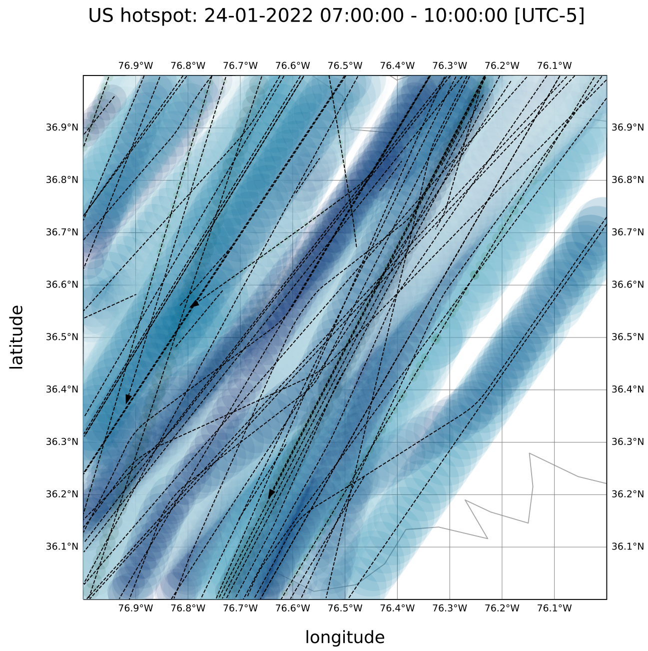
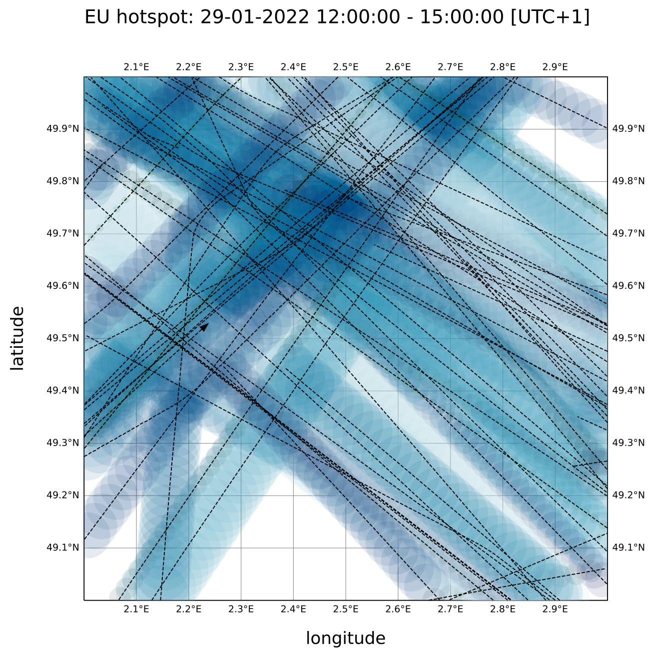


Figure 3.4.: Temporal variation in average aircraft count for the two hotspots, for each hour of the day, over all days in January 2022.

The US hotspot has notably less peaky data than the EU alternative, however it peaks earlier, at 08:00:00 EST (UTC-5), whereas EU peaks at 13:00:00 CEST (UTC+1). The absolute highest count for US was 26 aircraft, on 24-01-2022 at 08:00:00 EST. For EU, it was 31 aircraft, on 29–01–2022 at 13:00:00 CEST. Using these peak datetimes, a detailed plume analysis was then performed, using the plume modelling approach outlined in section 3.2. For each hotspot, the air traffic situation over three hours (peak time $\pm$ 1 h) was observed.



(a)



(b)

Figure 3.5.: Plume modelling simulation for high density hotspots. **(a)** US hotspot. **(b)** EU hotspot.

For both hotspots, over the specified duration, data were collected on distances, headings and bearings between aircraft pairs at each time step (every 10 s). Plume instances are modelled as circles on a 2-dimensional map, meaning it is not possible to visually distinguish altitude differences and vertical plume extent. Altitude was left out of the plots, because aircraft in cruise rarely tend to deviate far from their designated flight level.

Table 3.1 presents numerical results from the subgrid scale plume study, which collected data from 2114 aircraft pair instances for the US hotspot, and 2443 for the EU hotspot. For each, some key aircraft pair information was computed, and the SPE tests were conducted for each data point, including a test for no SPE occurrence at all (i.e. the data points which don't meet the SPE criteria).

Table 3.1.

	US hotspot	**EU hotspot**
Min. distance [km]	15.1	11.1
Min. relative track [°]	0.0258	0.0153
Avg. distance [km]	48.7	48.0
Avg. relative track [°]	15.2	148
SPE_1 [%]	2.05	3.74
SPE_2 [%]	30.3	11.1
SPE_3 [%]	0.420	0.223
SPE_4 [%]	9.18	3.48
No SPE [%]	57.6	81.4

3.4. Discussion

From the results presented, it can be deduced that there may well be huge potential to exploit chemical saturation effects to reduce the non-CO_2 warming from aviation. Intersecting plumes in high-density airspace regions can lead to a diminishing influence of both NO_x-based effects and persistent contrail growth. In this study, high-density hotspots were first investigated through a global heatmap analysis (figure 3.3). This plot of "mean airspace density", combined with the peak aircraft counts per hour, derived in figure 3.4, exemplify the temporal inhomogeneity of air traffic distribution, typically seen at aircraft cruising altitudes. Even for the highest density grid cells, the maximum average aircraft density is just above 0.6 aircraft per grid volume. Comparing this to a maximum flux of 31 aircraft traversing through the EU hotspot, at its peak hour shows that traffic patterns can be highly nonlinear and unpredictable.

In the plume plots of figure 3.5a and 3.5b, it is evident that, despite the EU hotspot having a higher number of aircraft pairings over the 3-hour period, the traffic in the US hotspot seems to be following a much more organised track structure. This has a large effect on the ability to achieve plume overlap, as can be seen in the results in table 3.1. Even though EU has 329 more instances of aircraft pairings recorded, the vast majority of its data points did not record an SPE occurrence (81.4%). US on the other hand, achieved a 42% SPE success rate, despite having a smaller aircraft count. As a result, this suggests that the US hotspot could have serve as a better candidate for controlled plume overlap, and that there are many other factors in play that may determine the effectiveness of this method (e.g. airspace complexity, temporal distribution, airspace organisation etc.).

3.5. Limitations

Whilst ADS-B trajectories enable aircraft exhaust plume intersections to be identified, there are limitations regarding usage for exhaust plume modelling. Firstly, identification of exhaust plume interactions is limited to global regions with ADS-B OpenSky Network ground receiver coverage, and therefore global estimates of aircraft exhaust plume intersections are not currently feasible. This limitation includes areas where sustained exhaust plume interactions are likely, such as oceanic crossings, particularly along the North Atlantic Flight Corridor, where densities are typically high [168].

Another significant limitation of relying on solely ADS-B derived-trajectories is the lack of meteorological information. The accurate modelling of exhaust plumes and contrails will of course be reliant on the pre-existing atmospheric conditions, along with an estimate plume drift due to wind conditions [74]. Consequently, the study presented in this paper has assumed that the plume follows the aircraft ground track. Future work could employ the meteorological information supplied my Mode-S data, or alternative approaches for estimating wind conditions via Mode-S data **Sun2018**, for the aircraft identified with plume intersections within the ADS-B dataset to generated improved plume modelling. When considering 3D plume modelling, further investigations into variations of reported GPS and barometric altitudes of operational aircraft is required, especially for the discrete 'steps' in aircraft altitude observed within ADS-B trajectories.

A final limitation with respect to considering solely ADS-B trajectories for identifying aircraft exhaust plume intersections is a result of the plume size and dispersion being related to aircraft operational parameters, including aircraft size (i.e. mass) and flying speeds. Both of these parameters will impact the engine exhaust flow and the wake generated by the aircraft, which are expected to impact the size and dispersion of the aircraft exhaust plume. Future work would aim to address this by coupling ADS-B trajectories with aircraft characteristics, prior to conducting plume modelling runs.

4. Potential to Offset Aviation NOx-induced Warming Through Formation Flight and Chemical Saturation

The content of this chapter is based on the following published journal article:

Khan, M. A. H.; Tait, K. N.; Derwent, R. G.; Roome, S.; Bacak, A.; Bullock, S.; Lowenberg, M. H.; Shallcross, D. E., "Off-setting climate change through formation flying of aircraft, a feasibility study reliant on high fidelity gas-phase chemical kinetic data", *Internation Journal of Chemical Kinetics*, vol. 55, no. 7, pp. 402–412, 2023. DOI: https://doi.org/10.1002/kin.21644.

This chapter incorporates material from that work with permission from the publisher and co-authors, where necessary. Dr Anwar Khan was lead author, and together with Prof. Dudley Shallcross, provided the theory and methodology for the study, and conducted the global chemistry analysis using STOCHEM-CRI. I co-wrote the paper with Dr Khan and Prof. Shallcross, carrying out further analysis and validation using the In-service Aircraft for a Global Observing System (IAGOS) dataset.

4.1. Introduction

Gas-phase kinetic studies underpin our understanding of key atmospheric processes such as stratospheric ozone formation, regulation and depletion **Rowland2006**, **Dameris2010**, and air pollution (e.g., NO_x, tropospheric ozone, peroxy acetyl nitrate (PAN), and aerosol) generation and reduction **VonSchneidemesser2015**, **Stockwell2019**. Research into this field is thus crucial for air quality and climate change assessment and mitigation. The aircraft sector is an important component in the assessment of air pollution and climate change, as it is one of only two non-surface sources of NO_x emissions [178], [196], [208], with the other being lightning **Tie2002**, **Schumann2007**, **Stockwell1999**. In addition to the direct emission of CO_2 through jet fuel combustion, aircraft emit NO_x which contributes to tropospheric ozone formation, as well as water vapour and particulates that give rise to contrail formation (Figure 1) [116].

Traditionally, mitigation efforts have gone toward incremental increases in efficiency, due to technological advancements and innovative design. However, due to the increasing urgency to rapidly reduce aviation climate impact, the focus has shifted toward more radical measures. This includes the development of new propulsion systems (e.g., liquid hydrogen) **Yusaf2022**, **Khan2022**, sustainable aviation fuels (SAFs) **Doliente2020**, and changes to aircraft flight routes and operations. Unlike the former two methods, modifications to flight routing pose the most viable solution for implementation this decade **Niklass2019**. This paper explores the potential of formation flying to reduce the ozone production efficiency of aircraft NO_x emissions.

Reducing NO_x emissions requires a lower temperature burn and this conflicts with trends toward higher temperature, more energy efficient jet fuel combustion. Formation flight may hold the key to maximizing the reduction in ozone production, due to any additional NO_x that is output from aircraft. The concept involves two or more aircraft flying no more than 1–2 km apart longitudinally, akin to energy-saving migratory flocks of geese. This enables follower aircraft to "surf" the wake of their leader, thus reducing required lift and hence thrust and fuel burn, on the order of 5% per aircraft [19], [215].

Further to the fuel saving achieved through this concept, there are also two notable chemical saturation effects that occur, due to the local accumulation of aircraft emissions. First, in airspace conditions conducive to persistent contrail formation, formation flight can help reduce the net contrail effect. This is because the generation of two or more persistent contrails in the same region will compete for available water vapor. Thus, they mutually inhibit growth and the combined warming effect is lessened compared with if they were generated individually. Unterstrasser et al. [204] explores this effect in great detail.

The second saturation effect to be observed is the accumulation of NO_x emissions, leading to NO_x-saturated conditions and a negative correlation with ozone production efficiency. It is well known in the urban environment, where there can be very high NO_x levels, that ozone formation is virtually zero as OH collapse occurs and NO_2 is converted to nitric acid (HNO_3) **Jenkin2020**, **Sillman2020**. HNO_3 eventually is incorporated into aerosol and removed from the atmosphere, representing a permanent sink for NO_x and preventing ozone formation. Can formation flying also deliver a reduction in ozone formation following NO_x emission? If NO_x levels are maintained at high enough levels, OH collapse could occur; this sounds bad, but along flight corridors, a significant fraction of NO2 may be sequestered to form HNO_3 and be unable to contribute to ozone formation. The chemical kinetics is in the aircraft's favour. In essence, when emitted, NO_x can, under high enough VOC levels, catalyse the formation of ozone, for example, using CO as a representative VOC, even though it is not strictly speaking one.

$$\mathrm{OH} + \mathrm{CO} \longrightarrow \mathrm{H} + \mathrm{CO_2} \tag{R 4.1}$$

$$H + O_2 + M \longrightarrow HO_2 + M \tag{R 4.2}$$

$$HO_2 + NO \longrightarrow OH + NO_2 \tag{R 4.3}$$

$$NO_2 + hv \longrightarrow NO + O \tag{R 4.4}$$

$$O + O_2 + M \longrightarrow O_3 + M \tag{R 4.5}$$

Therefore, through reactions (R 4.1)–(R 4.5), ozone is produced through NO_2 photolysis, and the longer the chain length converting NO to NO_2 (R 6.3) and back to NO (R 6.5) the more ozone is formed. However, if NO and NO_2 levels are high enough, the reaction of NO_2 with OH (reaction (R 4.6)) can start to compete with the reaction of OH with CO and CH_4 in the mid to upper troposphere (reactions (R 4.1) and (R 4.7)).

$$OH + NO_2 + M \longrightarrow HNO_3 + M \tag{R 4.6}$$

$$OH + CH_4 \longrightarrow \text{products} \tag{R 4.7}$$

The reason why reaction (R 4.6) may start to compete in the upper troposphere, where temperatures are far lower than they are at the surface, is that as an association reaction, the rate coefficient of the reaction (R 4.6), k6, increases with decreasing temperature **Atkinson2004**, **Butkovskaya2005** whereas rate coefficient of the reaction (R 4.7), k7 decreases with decreasing temperature **Atkinson2003** in a typical Arrhenius type behavior. Rate constant of the reaction (R 4.1), k1 has both a direct and association channel but appears to vary very little with temperature.

Aircraft NO_x perturbs the fast photochemistry of the upper troposphere–lower stratosphere (UTLS) by interfering with the balance between photochemical ozone production and ozone destruction. This balance is controlled by the NO_x-compensation point, the point at which with decreasing NO_x levels, photochemical ozone production through reaction (R 6.3) exactly balances photochemical ozone destruction through:

$$HO_2 + O_3 \longrightarrow OH + O_2 + O_2 \tag{R 4.8}$$

In the UTLS region, the NO_x compensation point, which is set by the rate coefficients for the reactions (R 6.3) and (R 4.8), is about 10 ppt as NO (calculated assuming the rate coefficient of reaction (R 6.3), k3 of $1.19 \times 10^{-11} cm^3 molecule^{-1} s^{-1}$, the rate coefficient of reaction (R 4.8), k8 of $1.13 \times 10^{-15} cm^3 molecule^{-1} s^{-1}$ based on Atkinson et al. **Atkinson2004** and O_3 100 ppb). That is to say, if the NO mixing ratio is less than 10 ppt, the fast HO_x photochemistry leads to ozone destruction. This is the general state of the UTLS region under typical conditions of temperature and pressure. In the presence of aircraft at cruise altitude, through their NO_x emissions, the NO_x compensation point may be exceeded and photochemical ozone production sets in, leading ultimately to an increase in global warming since the cold conditions of the UTLS favor enhanced radiative forcing from the increased ozone mixing ratios. The elevated OH concentrations driven by reaction (R 6.3) lead to elevated CH_4 destruction and hence decreased build-up, with its attendant negative radiative forcing and global cooling.

Most of the studies of the impacts of aircraft cruise emissions have been completed using sophisticated three-dimensional (3-D) Eulerian grid-based chemistry-transport models (CTMs). Despite a high degree of sophistication, typical spatial resolutions are no better than 1 °× 1 °(say, 100 km × 100 km in the North Atlantic air traffic corridor). A review of these studies is available elsewhere **Fuglestvedt1999**. Current CTM studies of these greenhouse gas impacts may well overstate the impacts of aircraft cruise emissions. This arises because of the current mis-match between the coarse spatial resolution of the CTMs and the fine-scale dimensions of the major air traffic corridors. A similar conclusion may be reached by consideration of the temporal resolution of these studies. The CTM studies may necessarily average aircraft cruise emissions over months or even years whereas on average flights last hours [208]. Together, the lack of spatial and temporal CTM resolution may ensure that aviation NO_x is present in excess of the NO_x compensation point at all times and in all grid squares through which a significant number of aircraft fly.

In this paper, first a kinetic model feasibility study was caried out to determine what NO_x threshold level is required to start reducing ozone formation. Second, an established 3D global CTM, STOCHEM-CRI, was used to identify regions of the atmosphere where the lowest NO_x thresholds exist. Thirdly, validation of the results from the global analysis of NO_x threshold concentrations was conducted by running the simulations with a much higher resolution dataset, and comparing these results with that derived from coarse STOCHEM-CRI model. The In-service Aircraft for Global Observing System (IAGOS) data (https://www.iagos.org/iagos-data/) is used in place of the 5 °× 5 °monthly global model fields from STOCHEM-CRI [108]. Finally, we develop a simple aircraft plume dispersion model which addresses the issues of inadequate spatial and temporal CTM resolution and understands how different the chemistry is likely to look under conditions of higher spatial and temporal resolution.

4.2. Methodology

4.2.1. Determination of Threshold NO_x from Kinetic Data

An estimate of the threshold level of NO_2 defining $[NO2]_{\mathrm{equiv}}$ as the concentration of $[NO_2]$ required to start OH collapse can be made by comparing the loss rates for OH with CH_4 (reaction (R 4.6)) and CO (reaction (R 4.1)) with that for OH with NO_2 (reaction

(R 4.6)).

$$k_6[\mathrm{NO_2}]_{\mathrm{equiv}}[\mathrm{OH}][\mathrm{M}] = k_1[\mathrm{CO}][\mathrm{OH}] + k_7[\mathrm{CH_4}][\mathrm{OH}] \tag{4.1}$$

A simple rearrangement of (4.1) provides an expression for $[\mathrm{NO_2}]_{\mathrm{equiv}}$:

$$\begin{aligned}[\mathrm{NO_2}]_{\mathrm{equiv}} &= \frac{k_1[\mathrm{CO}][\mathrm{OH}] + k_7[\mathrm{CH_4}][\mathrm{OH}]}{k_6[\mathrm{OH}][\mathrm{M}]} \\ &= \frac{k_1[\mathrm{CO}] + k_7[\mathrm{CH_4}]}{k_6[\mathrm{M}]}\end{aligned} \tag{4.2}$$

The temperature dependent rate coefficient of the reaction (R 4.1), $k_1 = 5.4 \times 10^{-14} \times ((T/298)^{1.5}) \times exp(250/T))$ and reaction (R 4.7), $k_7 = 2.45 \times 10^{-12} \times exp\left(\left(-1775 \pm 100\right)/T\right)$ have been obtained from Baulch et al. **Baulch1992** and DeMore et al. **DeMore1997**, respectively. The reaction (R 4.6) depends not only on temperature, but also on pressure. A decrease in pressure reduces the value of the rate coefficient and so there is competition between these two components with altitude: increasing altitude leads to a decrease in temperature and an increase in k_6, whereas pressure decreases, and this will decrease k_6. The term, $k_6[\mathrm{M}]$ was calculated as:

$$k_6[\mathrm{M}] = \frac{k_0[\mathrm{M}]}{1 + \frac{k_0[\mathrm{M}]}{k_\alpha}} \times 0.6^{\left(\frac{1}{1+\left(\log\frac{k_0[\mathrm{M}]}{k_\alpha}\right)^2}\right)} \tag{4.3}$$

Where $k_0 = (2.5 \pm 0.1) \times 10^{-30}(T/300)^{-4.4 \pm 0.3}$ and $k_\alpha = (1.6 \pm 0.2) \times 10^{-11}(T/300)^{-1.7 \pm 0.2}$, taken from DeMore et al. **DeMore1997**.

Temperature and pressure decrease with increasing altitude in the troposphere for a given latitude and longitude, but for a given altitude, temperature decreases from the equator to the poles. Although $\mathrm{CH_4}$ has a hemispheric gradient, higher in the Northern Hemisphere (~1800 ppbv) than the Southern Hemisphere (~1700 ppbv), and a slow inter-hemisphere transport with a temporal scale of approximately 1 year, mixing ratios of CH_4 are pretty constant throughout the troposphere as a result of its 10-year lifetime **Shallcross2020**. Whereas, CO, whose lifetime is around a month, has a strong gradient between Northern Hemisphere (120–180 ppbv) and Southern Hemisphere (60–70 ppbv), and displays mixing ratios that drop off considerably with altitude **Shallcross2020**. If we imagine, for simplicity, that CO is 1100 ppb at the surface and 100 ppb at 10 km and from surface to 10 km drops off by 100 ppb km^{-1}, we can start to make some preliminary calculations to scope out where in the atmosphere recreating the polluted urban atmosphere may be most likely.

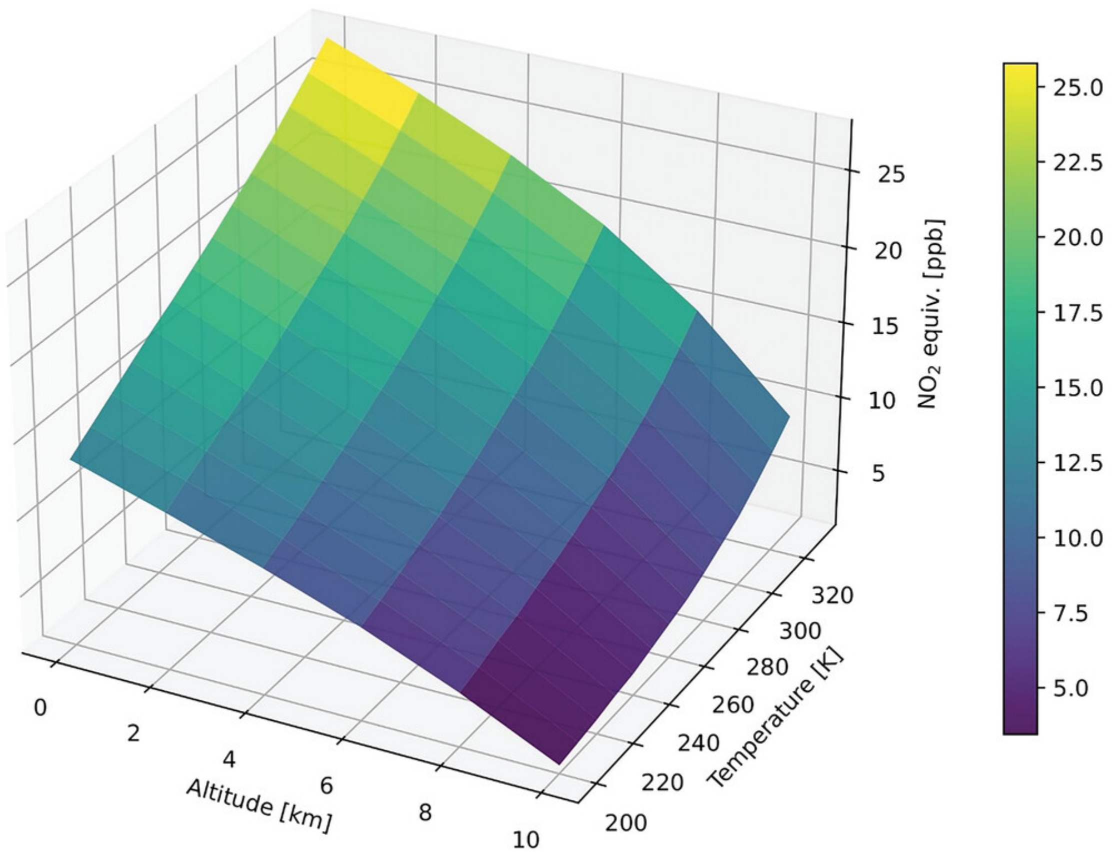


Figure 4.1.: Calculated $[NO_2]_{equiv}$ using simple kinetic equation (Equation (4.2)) at different temperatures and altitudes.

Therefore, the $[NO_2]$equiv was calculated as a function of temperature typical of the troposphere (200–330 K) and at pressures from the surface up to 10 km using the simple CO mixing ratio altitude profile noted earlier, with a fixed mixing ratio of CH_4, 1900 ppb.

4.2.2. Determination of Threshold NO_x from STOCHEM-CRI CTM Modelling Data and IAGOS Measurement Data

The same calculation as shown in section **??** was repeated using global model concentrations of CH_4, CO, and the rate coefficients as a function of temperature and pressure. These model fields were computed at a resolution of 5 ° longitude × 5 ° latitude with altitude binning corresponding to model output levels from Collins et al. [32]. The monthly global model fields of temperature, OH, CO, and CH_4 were obtained from the output of a 3-D chemistry and transport model, STOCHEM-CRI **Fuglestvedt1999**. These global fields provided a basis for calculation of $[NO_2]$equiv on a global scale to examine the spatial and temporal variations of this parameter. To further confirm the NO_x threshold trends seen in the global model analysis, the calculations were performed again, this time on measurement data from the three IAGOS projects: MOZAIC, CORE, and CARIBIC **Blot2021**. In these datasets, datapoints have been generated from measurements of key climate variables, using scientific instruments onboard civil aircraft. The MOZAIC project ran from 1994 to 2010 and collected data from over 33,471 flights. CORE collected data from 2011 up until present day, providing data on a further 7790 flights. Finally, CARIBIC has been operating since 1997 to present day, however, has only collected data from 541 flights. These datasets were used to create temperature, M and CO concentration fields, which are comparable to the corresponding STOCHEM-CRI model fields. Such fields were constructed by collating data in latitude, longitude, altitude bins, which match the resolution of the STOCHEM-CRI simulated fields. These fields are used in the calculation of NO_x threshold which are compared with that found from STOCHEM-CRI.

Measurements are taken at a nominal temporal resolution of 4 s throughout the entire flight profile. Anomalous or missing data led to sparsity between measurements; however, data quality has generally improved in recent years, as newer and more accurate equipment was deployed. To perform these calculations, all data points were removed from flights with either no useful compositional chemical data or where vital physical measurements such as temperature, pressure, or altitude were missing. These steps reduced our sample size down to 16,335 individual flights with 83.8 million useful data points, collected over a 28-year period.

The necessary computations were carried out using the C API to netCDF and Slurm workload manager on the BlueCrystal Phase 4 supercomputing facility (https://www.acrc.bris.ac.uk/acrc/pha Outputs were stored in CSV format and queried using an SQL database, where we binned data into the grid cell resolution of STOCHEM-CRI. Then averages of our estimated $[NO_2]_{equiv}$ values were calculated for each box in our world grid. The global distribution plots were generated using Python packages: Matplotlib and Cartopy.

4.2.3. Development of 1-D Box Model of Flight Corridor

A simple one-dimensional (1-D) model of aircraft cruise conditions was developed which addresses the issues of inadequate spatial and temporal CTM resolution. A set of grid boxes has been built perpendicular to an air traffic corridor in the UTLS region. The grid contains 999 boxes, each of 100 m × 100 m × 100 m and the contents are assumed to be instantaneously mixed. Each grid box contains a number of trace gases, that is, representing the chemistry of the UTLS region including CH_4, CO, NO_x, O3, H_2O, organic compounds, and their degradation products. The time-dependent behaviour of each trace gas is represented by a differential equation of the form:

$$\frac{\partial c_i}{\partial t} = P_i - l_i.c_i - \frac{\partial}{\partial x} K \frac{\partial c_i}{\partial x} \tag{4.4}$$

where c_i is the mass of species, i, t the time, x the axis perpendicular to the air traffic corridor, P_i the production rate of the species within the box, and l_i is the rate coefficient for the loss of the species within the box and the final term on the right-hand side of the equation represents the gain and loss of the species by exchange with neighboring boxes with the grid. Equation (4.4) was expressed in finite-difference form and integrated numerically with a dispersion time-step of 300 s and a chemical time-step of 20 s using a backwards-iterative Euler numerical method. K is the horizontal diffusion coefficient and was set at 10 m^2s^{-1} based on Paoli et al. [151] and Schumann et al. [172] for the UTLS region. The chemical production and loss rates were taken from the Common Representative Intermediate (CRI) mechanism **Watson2008**, **Utembe2009**, [94] which describes the chemistry of 220 species taking part in 609 reactions. Background conditions for the UTLS were taken from a previous run of STOCHEM-CRI for 40 ° N, June 21 at 150 mb and used to provide initial conditions for each of the 999 grid boxes.

4.3. Results and Discussion

4.3.1. $[NO_2]_{equiv}$ in the Troposphere Determined from Kinetic Data

Figure 4.1 illustrates that the $[NO_2]_{equiv}$ determined from kinetic data decreases with temperature and altitude and that the most likely place where the equivalent point may be reached is in fact around the cruise altitude of long-haul aircraft. Short-haul aircraft may not be able to attain such altitudes if the flight is very short in duration. The simple model presented in figure 4.1 suggests that as the flight reaches cruise altitude at about 10 km, the amount of NO_2 required for parity in OH loss is around 2 ppb. How feasible is it to generate such levels of NO_2? In a narrow flight corridor with frequent flights through it, it may well be.

4.3.2. Global Mapping of $[NO_2]_{equiv}$ at the Cruise Level

Figure 4.2 shows global $[NO_2]_{equiv}$ at model output level 8 which is determined by using $[CH_4]$, [CO], [M], and temperature fields simulated by STOCHEM-CRI. In this study, data at model output level 8 (9.2–11.8 km) were used, to best replicate the conditions experienced by civil aircraft in cruise. It is known that CO mixing ratios are very variable in space and time, for example, in the biomass burning season in the tropics, CO in the upper troposphere can be elevated considerably, which could be responsible for the spatial and temporal variations of global $[NO_2]_{equiv}$ level.

Figure 4.2 suggests that sub-ppb levels of $[NO_2]_{equiv}$ may exist in mid- and high-latitudes, but that several ppb (up to 2 ppb) are required in the tropics, where temperature is higher, and CO levels elevated through deep convection of CO from the surface sources (e.g., vegetation, biomass burning, anthropogenic emissions, VOC oxidation). The biomass burning season for Northern Hemisphere Africa and South East Asia is from November to March with peak burning in December and January; however, the biomass burning seasons for South America and Southern Africa show a peak in September to October with little burning between November and February, which is responsible for the monthly variations of $[NO_2]_{equiv}$ peaks at these areas (see Figure 4.2). There is little biomass burning over Central Europe and North America, but the strong anthropogenic source along with VOC oxidation leads to a higher level of CO in these areas, resulting in ~1 ppb $[NO_2]_{equiv}$. The vegetation emission of CO can also increase $[NO_2]_{equiv}$ at Amazon and South Africa forest regions during Austral summer months (December–January–February).

To further confirm the NOx threshold, termed $[NO_2]_{equiv}$, trends seen in the global model analysis in Figure 4.2, the results using IAGOS data are plotted in Figure 4.3. The $[NO_2]_{equiv}$ values obtained from both STOCHEM and IAGOS show a good agreement (see Figures 4.2 and 4.3) with the differences between all months averaged $[NO_2]_{equiv}$ at level 8 referred to by mean bias (STOCHEM-IAGOS) and the percentage change ((STOCHEM-IAGOS)*100/IAGOS) are found to be 0.147 ppb and 0.27%, respectively. The calculated global $[NO_2]_{equiv}$ from both model and measurement data (Figures 4.1 and 4.2) suggested that the northern mid-latitudes of the atmosphere are the most favourable region existing with the smallest NOx thresholds (0.5 ppb) needed before reduction in ozone formation is likely to occur at cruise altitude of aircraft.

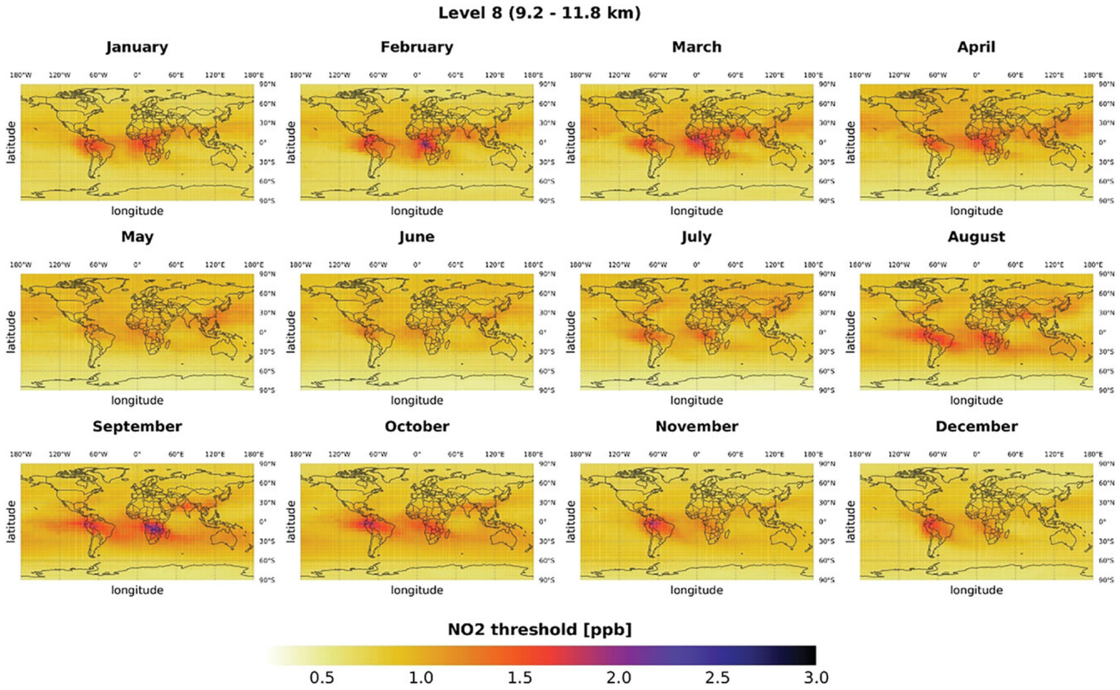


Figure 4.2.: Global $[NO_2]_{equiv}$ at model output level 8 (9.2–11.8 km) estimated using $[CH_4]$, [CO], [M], and temperature fields simulated by STOCHEM Common Representative Intermediate (CRI).

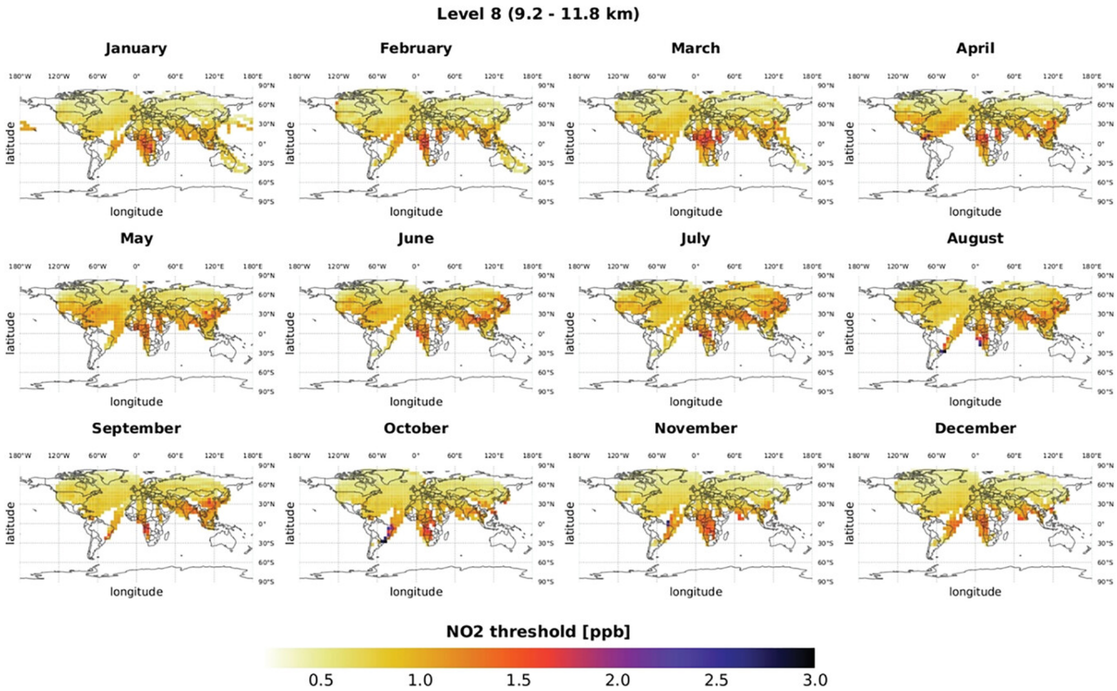


Figure 4.3.: Global $[NO_2]_{equiv}$ at model output level 8 (9.2–11.8 km) estimated using [CO], [M], and temperature fields from In-service Aircraft for Global Observing System (IAGOS) dataset and $[CH_4]$ from STOCHEM Common Representative Intermediate (CRI) model simulation.

Assessing how well NOx is sequestered by OH radicals to form HNO_3, terminating ozone production, will depend on $k6$ (function of pressure and temperature) and this itself will change with altitude, latitude, and time of year. As altitude increases for a given location, temperature decreases and so $k6$ will increase. However, at the same time pressure is decreasing and so $k6$ will decrease, determining which process wins and whether $k6$ ultimately decreases or increases with altitude requires high fidelity kinetic information. An increasing $k6$ with altitude may mean that this termination reaction is more effective at high altitudes for a given $[NO_2]$. These reactions are hard to study at low temperature and as a function of pressure and so uncertainties increase. The rate coefficients ($k1$, $k6$,and $k7$) were perturbed upwards and downwards by considering the uncertainties of the rate coefficients and used to calculate global annual average $[NO_2]_{equiv}$ (see Supplementary information Figure A). The uncertainty analysis shows that $[NO_2]_{equiv}$ can be decreased and increased by up to 2.5% and 5% in the northern mid-latitudes due to the using lower limit and higher limit rate coefficients, respectively (Figure A).

The calculation of $[NO_2]_{equiv}$ did not consider VOCs contributions in the study. The methane mixing ratios have been rising in the troposphere since ca. 2010 but most VOCs are decreasing. For ethane and propane **Pozzer2022**, two of the longest lived VOCs, and species that will have elevated concentrations in the upper troposphere, current measured concentrations are indeed decreasing **Pozzer2022**, and suggest that these VOCs would contribute less than 5% of the OH loss compared with CH_4. Given that loss of OH with CH_4 is typically 30% of that for CO, VOCs other than CO and CH_4 can be ignored in any initial analysis such as carried out here. Even for acetone, another long-lived species with elevated levels in the upper troposphere higher levels are seen in the summer months (800 ppt) but drop to around 200 ppt in winter**Dufour2016**. So, for short periods of time in a year, levels approach that required to be around 5% of that for CH_4 but for the Northern Hemisphere, where it is expected the formation flying to be most effective, levels are much lower.

4.3.3. Aircraft Plume and its Chemistry Study using a 1D Dispersion Model

In a first box model experiment, an aircraft emits a pulse of NOx into the centre box of the model grid at 12:00 z. The pulse size was set based on a global aircraft cruise NOx emission of 3.9 Tg NOx per year and a distance travelled **5** of 36.4 billion km that is

107 g NOx km^{-1}. This NOx was instantaneously mixed into the centre box volume and the time development of the model species was then followed. After several seconds, the NO-NO_2-O_3 chemistry was rapidly established, a strong depletion in ozone formed and then NOx began to disperse horizontally into the neighbouring boxes over the period of hours and days. After 1 h the horizontal distribution of NOx showed an aircraft plume with a width of 3 km, as shown in Figure 4.4, which expanded to 6.8 km after 6 h and 13.1 km after 24 h. Plume width expanded with the square root of lapsed time, see Figure 4.5, exactly as predicted by the Einstein–Smoluchowski equation.

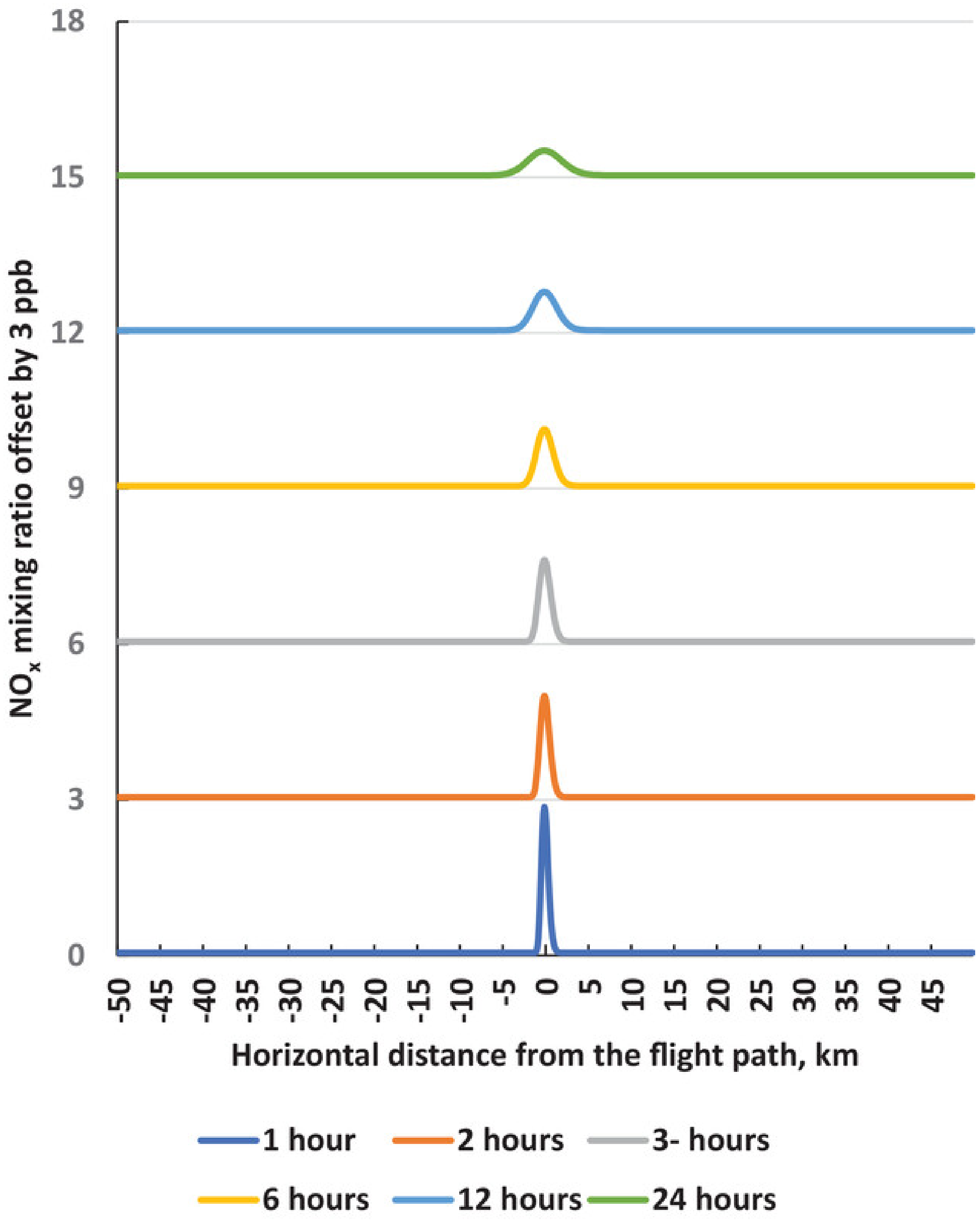


Figure 4.4.: NOx mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb.

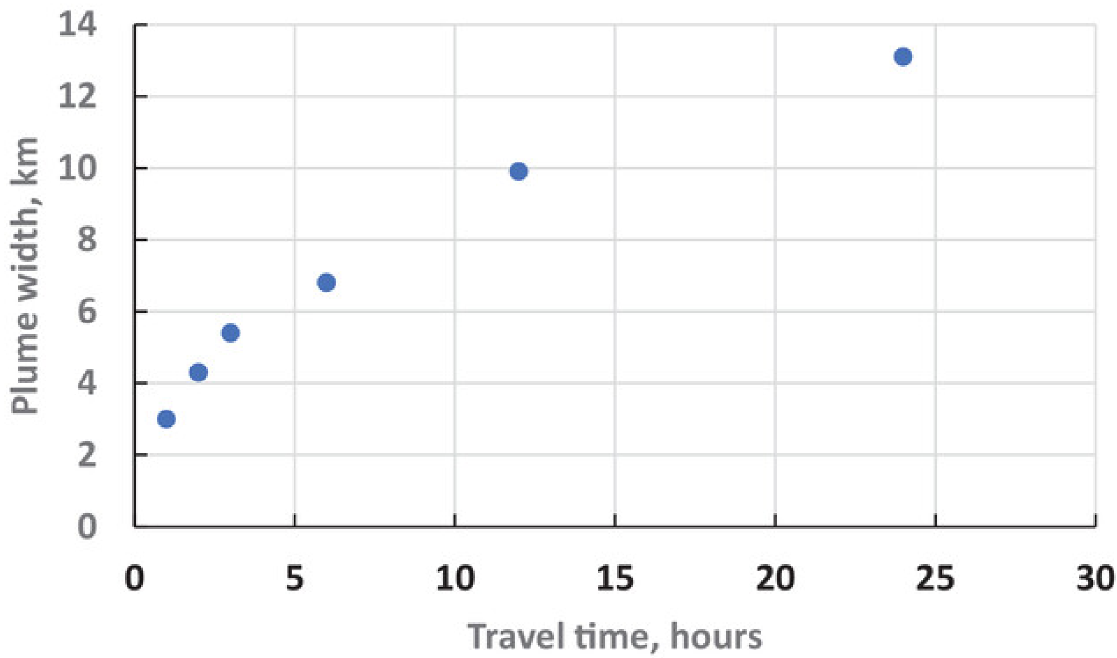


Figure 4.5.: Aircraft plume width as a function of travel time.

Initially all of the ozone would have been completely wiped out in the centre of the flight-path by its rapid reaction with NO:

$$NO + O_3 \longrightarrow NO_2 + O_2 \qquad \text{(R 4.9)}$$

However, during daytime, the product NO_2 is rapidly photolysed to form ozone. Overall, the reaction sequence (R 6.3)–(R 4.5) rapidly returns the mixture of NO–NO_2–O_3 to a steady state in a few minutes in which the ozone levels are largely restored to their values before the passage of the aircraft.

The distribution of ozone perpendicular to the flight path is shown in Figure 4.6 for various times after the passage of the aircraft. One hour after the aircraft passage, ozone levels show the presence of a slight dip close to the center of the aircraft plume demonstrating that the stationary state in the NO–NO_2–O_3 system has been re-established.

After 3 h, the central dip is still evident but photochemical ozone production is starting to appear in the wings of the aircraft plume. As time passes, the production in the wings becomes more evident and the central dip is filled in. After 24 h, photochemical ozone production is well on its way and an ozone plume becomes apparent with a width of about 16 km and a height of just over 0.4 ppb at its center. At this stage, the ozone plume is about 3 km wider than the NOx plume as horizontal dispersion moves the photochemically produced ozone into the background atmosphere. After 4 days, see Figure 4.6, the plume is about 30 km wide and has a height of about 2.7 ppb.

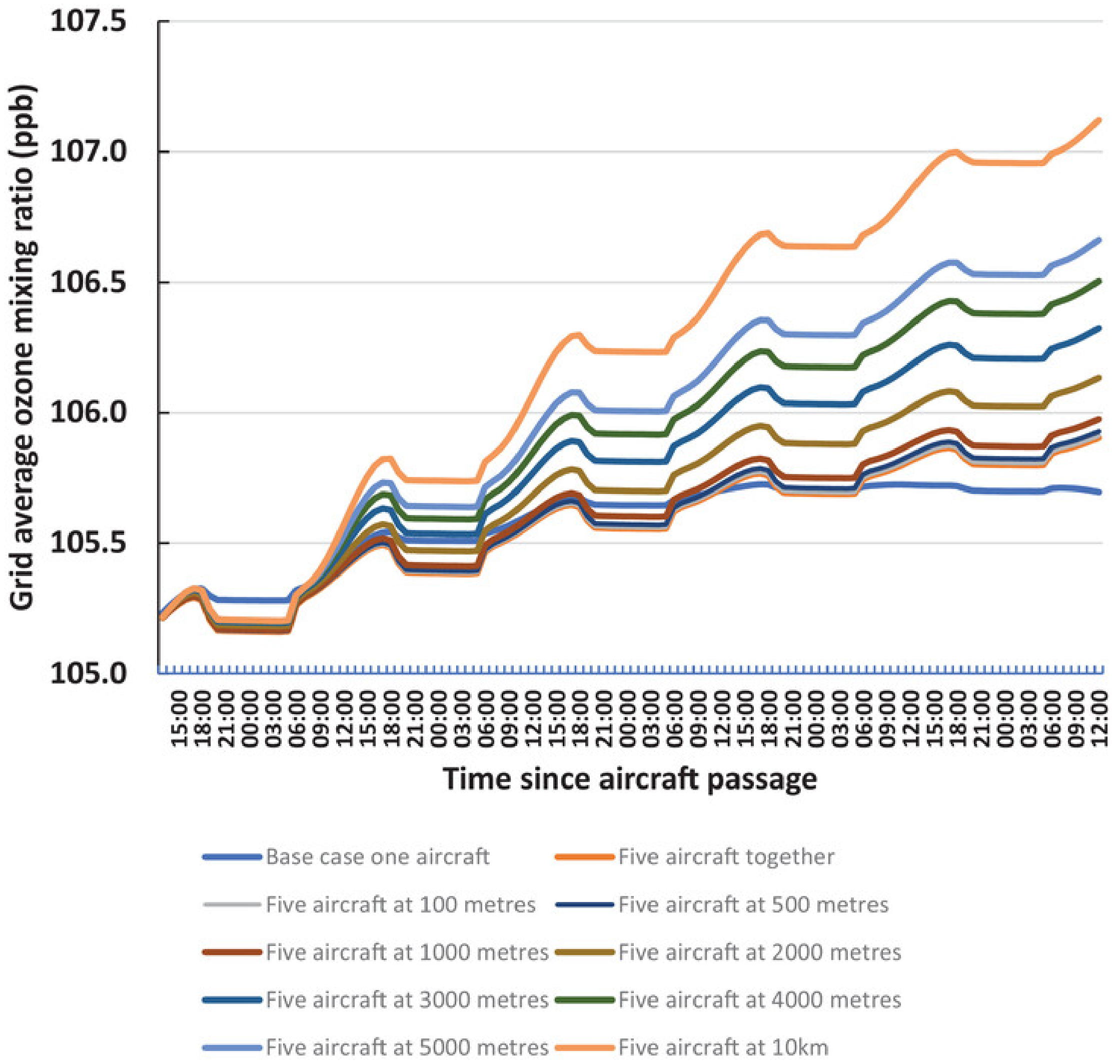


Figure 4.6.: O_3 mixing ratios as a function of distance from the flight path at increasing times after the passage of an aircraft, each offset by 3 ppb.

In the next application of the aircraft plume dispersion model, the case of five aircraft flying at the same time but separated in space are considered. To begin with the five aircraft flying together without any spatial separation are reported and then further cases where they are separated by 100 m, 500 m, and so on up to 10 km are investigated.

Figure 4.7 shows the time development over 5 days of the 99.9 km grid average O_3 for a base case with one aircraft flying, five aircraft flying together and five aircraft with spatial separations increasing from 100 m to 10 km.

In each plot, the grid average O_3 perpendicular to the flight path shows a steady increase over the 5 days. This increase has small depressions during night-time when the NO–NO_2–O_3 system adjusts to the lack of sunlight. The steady increase is driven by photochemical ozone production because the chosen initial conditions for these experiments have NO levels above the local NOx compensation point. Initially the plots for the five aircraft lie below the one aircraft case because of the more intense O_3 depletion found with more aircraft. However, the situation reverses after 1 day and O_3 levels in the five aircraft cases then lie above the one aircraft case. After further time has elapsed, the five aircraft cases begin to diverge with the cases with wider separations diverging the greatest.

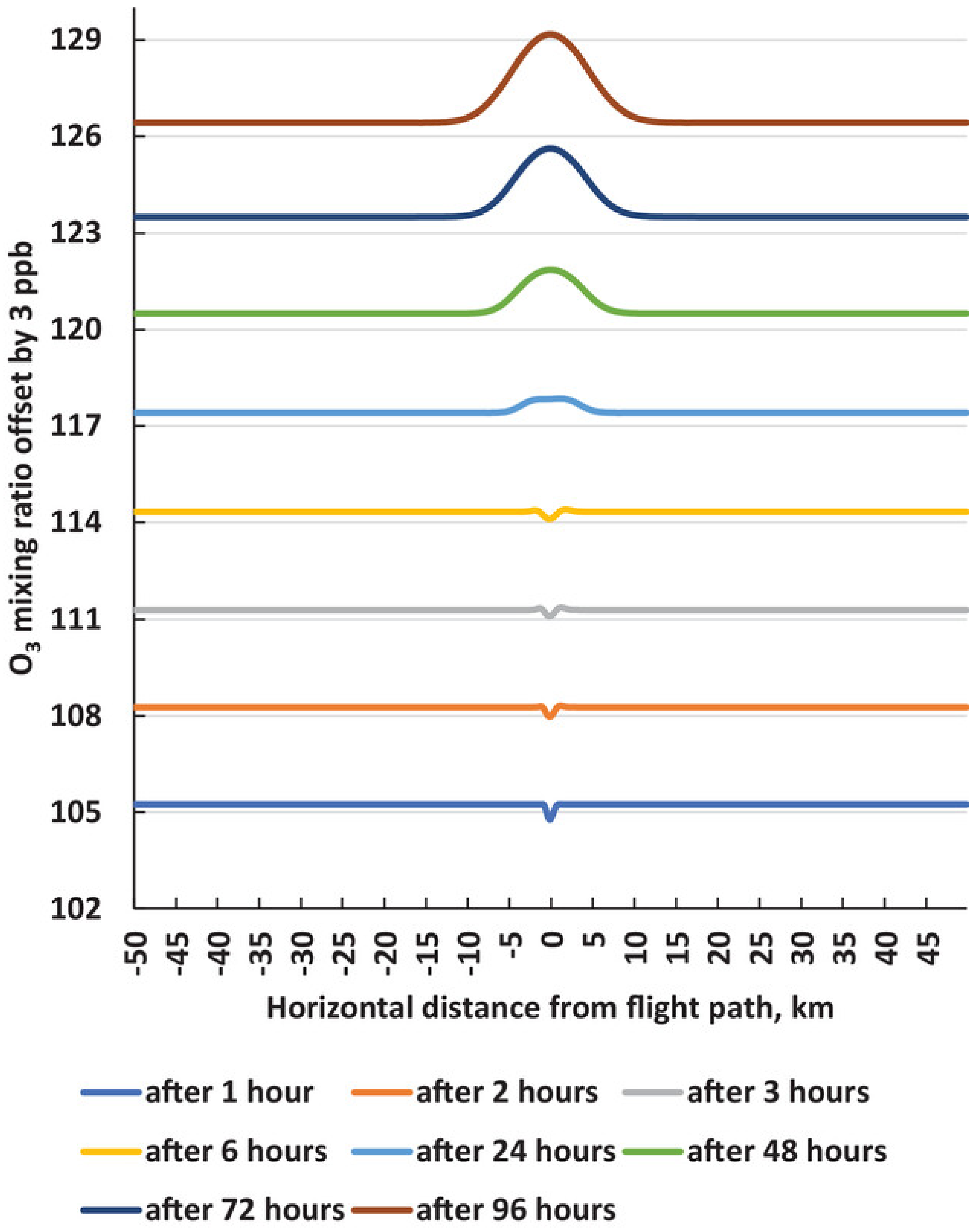


Figure 4.7.: Time development of the average O_3 mixing ratios across the 99.9 km grid for a base case with one aircraft, five aircraft flying together and flying with horizontal separations increasing from 100 m to 10 km.

After 5 days, the case with the 10 km separation has a grid averaged O_3 mixing ratio which is 1.2 ppb higher than the no-separation case, see Figure 4.7. This excess O_3 drops to just under 0.8 ppb with a horizontal separation of 5 km and to 0.2 ppb with a separation of 2 km. On this basis, the impact of aircraft emissions on spatially-averaged O_3 could be halved if the aircraft could maintain separations inside 4 km.

Clearly, this is a first illustrative example and our aircraft plume dispersion model is a highly simplified representation of the complex interplay between aircraft cruise and the local environment in an air traffic corridor. However, the results are encouraging enough for the continuation of model development, increasing spatial resolution, examining more complex scenarios, and assessing further proposals for the formation flying of aircraft.

4.4. Conclusion

When aircraft fly close to one another, their exhaust plumes can overlap, leading to the accumulation of emissions in local airspace. In the UTLS, where aircraft typically spend most of their flight, high emissions concentrations of NOx and water vapour can lead to reductions in both ozone production and persistent contrail generation.

This paper initiated an investigation into the chemical response of the atmosphere to aircraft emissions saturation, in high-density airspace. This included a simple kinetic comparison, a coarse resolution analysis on a global scale, and a higher resolution box model study. The calculation using simple kinetic model shows that at cruise altitude, the amount of NO_2 required for parity in OH loss with loss due to reaction with CO and CH_4 is around 2 ppb. Similar calculation using the STOCHEM-CRI global model, and IAGOS measurement data shows that 0.5 ppb of $[NO_2]_{equiv}$ may exist in northern mid-latitudes. Northern mid-latitudes would be most likely to benefit from formation flying to suppress ozone production. The box model results show that the impact of aircraft emissions on spatially averaged O_3 could be halved if the aircraft could maintain separations inside 4 km rather than 10 km.

The results generated from these simulations indicate a clear trend toward a climate beneficial outcome due to emissions saturation. In future papers, we aim to increase sophistication of the box model by integrating higher level plume modelling methods and a contrail model into the analysis. In addition to this, the aim is to integrate air traffic trajectories into the study, such that emissions from either real world or simulated flights can be modelled more accurately.

5. Development of a Novel Aircraft Plume Modelling Framework

In this chapter, a robust methodology is presented for estimating the chemical saturation effects that occur, as a result of overlapping aircraft exhaust plumes in busy airspace. This is done through explicitly modelling aircraft trajectories, simulating their flight performance to ascertain fuel burn and emissions data, then simulating plume dispersion to investigate how those emissions mix into the surrounding atmosphere. Then aggregating these data to a common Eulerian grid for multiple flights in the same region enables further chemical and microphysical modelling, to explore a range of responses to aircraft emissions in high-density airspace.

5.1. Introduction

A number of studies to date have conducted climate-chemistry simulations, to investigate the plume overlap effect [36], [113], [124]. Through modelling of local plume dispersion and the ensuing chemical and microphysical transformations, they have found a general trend towards reducing NOx- and contrail-related warming under increasing emissions concentrations. This is due to saturation effects which occur in high-emission airspace, as elaborated on in **??**. A case study in Dahlmann et al. [36] showed that the mitigation potential from formation flight could be around 22% on average per flight, due to fuel burn reduction and NOx saturation. Such studies do not however, provide far-reaching conclusions on the range of saturation effects that occur, under different formation flight configurations and atmospheric conditions. In chapter 3, the practicality of controlled plume intersection was investigated, through real-world air traffic data analysis. Under the "significant plume encounter (SPE)" criteria laid out, it was found that nearly 40% of all aircraft plumes in the high-density hotspots over mainland US and Europe, exhibited some degree of plume overlap. Chapter 4 approaches the problem from a different perspective, focusing primarily on the chemical saturation effects that would result from high NOx emissions. A metric called NO_2 threshold was developed, to represent the propensity of a region for NOx saturation in the upper troposphere. The chapter concluded with a first-order photochemical box model experiment, which went some way towards quantifying the chemical saturation effect in terms of formation flight and plume intersection. The motivation behind this research was to begin to identify suitable regions of the atmosphere to exploit chemical saturation effects resulting from formation flight and plume overlap.

Chapters 5 and 6 serve as a direct continuation of this body of research, aiming to provide further real world context through higher-fidelity aircraft plume modelling and local atmospheric chemistry analysis. The literature survey conducted throughout the duration of this PhD revealed a lack of pre-existing modelling architectures, suitable for simulating the emissions, dispersion and local chemical and physical evolution of overlapping aircraft exhaust plumes. One of the most influential papers on aviation climate impact in recent years, Lee et al. (2023) [118] cited the journal article version of chapter 2, Tait et al. [196]. This paper claimed that uncertainties remain in chemistry modelling of aircraft NOx due to a lack of sufficient subgrid scale modelling of aircraft plumes. It was, therefore, deemed imperative to construct a novel framework to fit these requirements and to address insufficiencies highlighted in recent literature. This led to the development of the Gridded Plume Analysis Tool (GPAT). The following sections will delineate the underlying models, input parameters and datasets contained within the GPAT architecture.

5.2. GPAT Overview

The Gridded Plume Analysis Tool (GPAT) derives from work done by the University of Bristol Atmospheric Chemistry Research Group (ACRG), Professor Ulrich Schumann and colleagues at Deutsches Zentrum für Luft- und Raumfahrt (DLR), as well as Breakthrough Energy and their recent development of the *pycontrails* open-source repository for aviation climate impact modelling [184]. Each group's respective work was integral to the development and application of GPAT, as will become apparent throughout this chapter.

This tool was built out of a development version of the pycontrails library (v0.53.2), with the majority of the models and interlinking software being written in Python. The only exception to this being the photochemical box model (BOXM v2.0) written in FORTRAN 90, which is an adaptation of a legacy model (BOXM v1.0) [40] built in FORTRAN 77. All libraries, packages and tools used throughout the software development phase will be outlined explicitly where relevant.

The primary purpose of GPAT is to simulate a spatial domain in which to plot aircraft trajectories, estimate their emissions, model the dispersion and advection of the exhaust plumes carrying the emissions, then to compute the evolution of atmospheric chemistry and physics across the domain. The software architecture of the modelling framework is presented in figure 5.1. This portrays the input parameters, constituent models and the input and output datasets related to each model.

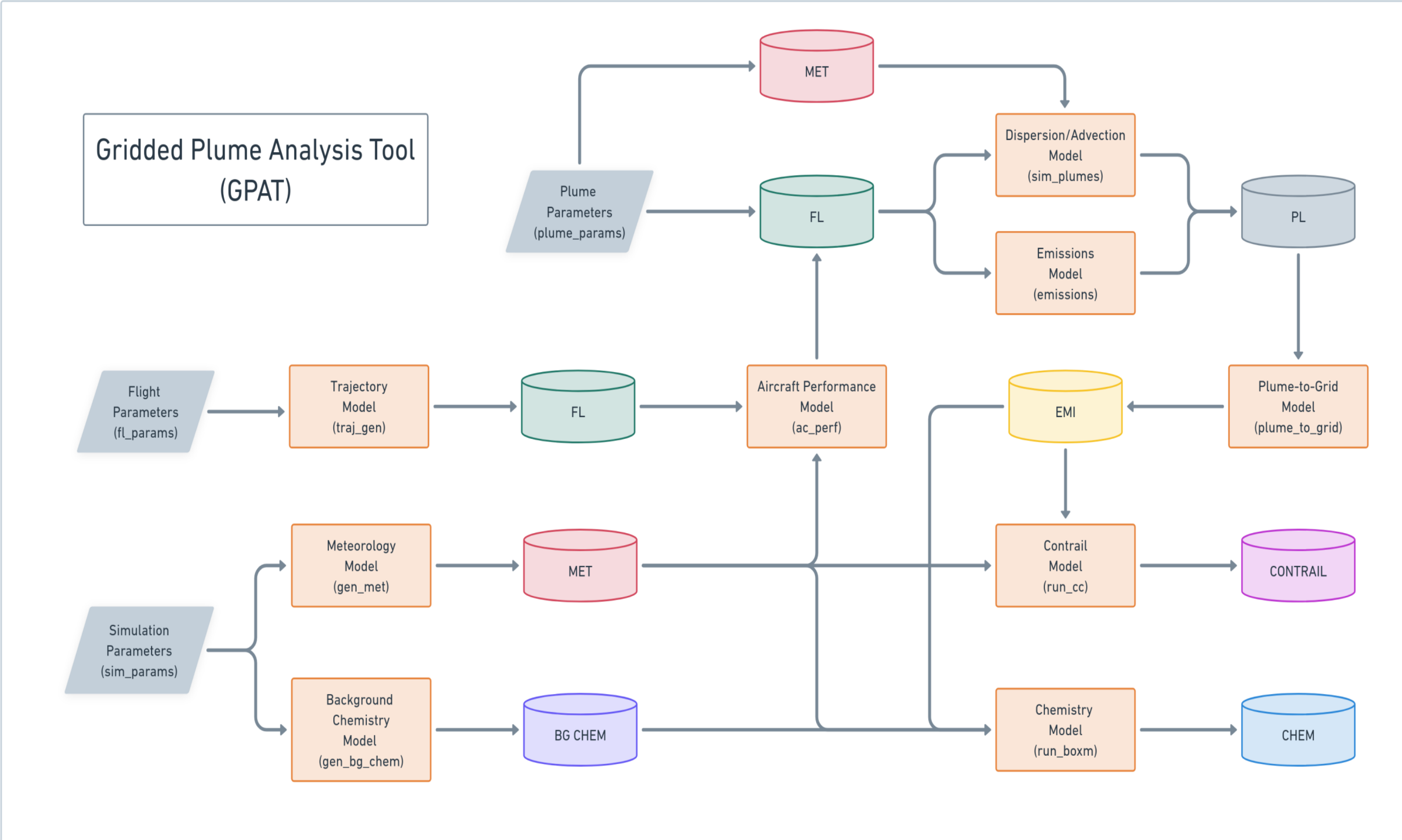


Figure 5.1.: GPAT process flow schematic portraying the input parameters (rhombuses), constituent models (boxes) and flow of data through a range of datasets (cylinders).

5.3. Input Parameters

The Gridded Plume Analysis Tool requires three sets of user-defined input parameters, to provide the necessary information to conduct a model run. This comprises flight parameters related to aircraft and flight properties, plume parameters pertaining to the characteristics of the associated aircraft exhaust plumes, and lastly the simulation parameters, required to instantiate the physical and chemical modelling environment.

5.3.1. Flight Parameters (fl_params)

The flight parameters give information on the flights taking place in the simulation. This includes the start time, duration and time step of the flight data to be generated, the aircraft types, speed, heading and initial coordinates of the first aircraft. For formation flight scenarios, it also contains the longitudinal (dx), lateral (dy) and vertical (dz) separation minima between the leader and follower aircraft, as well as the number of aircraft in the simulation. It is assumed that follower aircraft are travelling at the same speed and along the same heading. Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

Parameter	Description	Example value
t_0_fl	Flight start time	2022-01-01 12:00:00
rt_fl	Flight run time [min]	60
ts_fl	Flight time step [min]	2
ac_type	Aircraft type	A320
fl_0_speed	Aircraft speed [m/s]	100
fl_0_heading	Aircraft heading [deg]	45
fl_0_coords$_0$	Leader aircraft initial coordinates [lat. deg, lon. deg, alt. m]	(45.0, 45.0, 12,000)
sep_dist	Separation minima [dx m, dy m, dz m]	(1000, 100, 0)
n_ac	Number of aircraft	5

5.3.2. Plume Parameters (plume_params)

These are the parameters that determine how the exhaust plumes, and emissions contained within them, disperse and advect spatially over time. This includes the integration time step, max age of each plume, initial dimensions, wind shear, horizontal and vertical resolution of the plume simulation, as well as the number of Gaussian slices for the plume aggregation step (for more on this, see section 5.8). Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

Parameter	**Description**	**Example value**
dt_integration	Integration time step [min]	2
max_age	Maximum age of plume segments [hours]	2
depth	Initial plume depth [m]	50
width	Initial plume width [m]	50
shear	Wind shear [1/s]	0.01
hres_pl	Horizontal resolution of plume aggregation [deg]	0.05
vres_pl	Vertical resolution of plume aggregation [m]	500
n_slices	Number of Gaussian slices to aggregate to grid	10

5.3.3. Simulation Parameters (sim_params)

This set of parameters defines the spatial and temporal domain of the simulation run, specifying the geographical bounds of the region of interest, the time frame over which the atmospheric chemistry and physics is simulated, the spatial and temporal resolution of the domain, wind effects, and species in and out of the model. Table ?? shows these parameters as they are read by the software, along with a description of their use, units and an example value.

Parameter	**Description**	**Example value**
t_0_sim	Simulation start time	2022-01-01 12:00:00
rt_sim	Simulation run time [days]	5
ts_sim	Simulation time step [s]	20
lat_bounds	Latitude bounds [deg]	(40, 50)
lon_bounds	Longitude bounds [deg]	(40, 50)
alt_bounds	Altitude bounds [m]	(12000, 13000)
hres_sim	Horizontal resolution of simulation [deg]	500
vres_sim	Vertical resolution of simulation [deg]	10
eastward_wind	Eastward bound wind speed [m/s]	0
northward_wind	Northward bound wind speed [m/s]	0
vertical_velocity	Vertical wind velocity [m/s]	0
species_in	Emission species concentrations aggregated to sim. [ppbv]	("NO", "NO2", "CO")
species_out	Output species concentrations [ppbv]	("O3", "NO2", "NO")

5.4. Datasets

The datasets contained within the GPAT modelling framework help maintain a consistent flow of data between models, providing information about the specific flights in the run and the associated fuel burn and emissions masses throughout those flights (FL). The dispersion and advection of their corresponding plumes is captured in the Plume Dataset (PL). To investigate the ensuing climate effects of those emissions through gridded chemical and microphysical analysis, the Background Chemistry (BG_CHEM) and Meteorology (MET) Datasets is defined. The Emissions Dataset (EMI) is generated through aggregation of 1D emissions and plume geo-vector data to a 2D Eulerian grid.

The Contrail (CONTRAIL) and Chemistry (CHEM) Datasets are then generated by concatenating background chemistry, meteorology and emissions datasets, followed by chemical and microphysical processing through the run_cc and run_boxm models. This final step provides insight into how the emissions present in the model affect the composition of the surrounding atmosphere, over the specified runtime of the simulation. All data structures contained within the GPAT framework are created using the Numpy, Pandas, Xarray and pycontrails Python libraries [77], [85], [184], [197].

5.4.1. Flight Dataset (FL)

The Flight Dataset is first generated when the user-defined Flight Parameters (fl_params) are passed through the Trajectory Model. The latitude, longitude, altitude coordinates as well as the waypoint number are calculated for each time step throughout the flight, according to t_0_fl, ts_fl and rt_fl. Flight attributes are also assigned to this dataset as metadata; the flight's unique identifier (UID), aircraft type, number of engines, wingspan, maximum Mach number, maximum altitude and engine UID.

The Flight Dataset is passed to the Aircraft Performance Model, which estimates aircraft fuel flow using the aircraft positional data, aircraft-specific characteristics and the instantaneous meteorological situation. The down-selected meteorology data is assigned prior to the performance run. Aircraft performance is then simulated, and the dataset is updated with columns for true airspeed, thrust, thrust setting, rate of climb or descent, engine efficiency, fuel flow, fuel burn and aircraft mass at each flight waypoint. Total fuel burn is also added to the FL dataset's attributes. An exemplary Flight Dataset is provided in table A.1.1.

5.4.2. Meteorology Dataset (MET)

MET provides information on the meteorological situation over all time steps and grid cells of the simulation region, derived from the outputs of the GPAT Meteorological Model (gen_met). The air temperature, water vapour concentration and wind field data are taken from the UK Met Office Unified Model (UM) for the year 1998 at a resolution of 5° × 5° [35]. The pressure is estimated according to the pressure levels and associated altitude bands from the model output levels in Collins et al. [32], presented in table 5.1.

Table 5.1.: Model output levels from Collins et al. [32] used throughout the GPAT model architecture.

Model output level	**Mean pressure [hPa]**	**Approximate height [km]**
1	1013–912	0.0–1.0
2	912–810	1.0–2.0
3	810–709	2.0–3.0
4	709–607	3.0–4.2
5	607–505	4.2–5.6
6	505–404	5.6–7.2
7	404–302	7.2–9.2
8	302–201	9.2–11.8
9	201–100	11.8–16.2

The Meteorological Model then calculates additional parameters, including specific humidity, relative humidity, the number density of air (M) as well as atmospheric concentrations of oxygen (O_2) and nitrogen (N_2). The model also calculates the Solar Zenith Angle (SZA) across all grid cells and time steps in the simulation domain.

5.4.3. Background Chemistry Dataset (BG_CHEM)

This dataset contains the necessary chemical concentration data for all the background trace species initialised in the BOXM Photochemical Trajectory Model (see **??**). These data are obtained from the STOCHEM Common Representative Intermediates (CRI) v2.2 R5 model outputs presented in Khan et al. [108]. The Background Chemistry Model downloads the species data based on the simulation start time. The STOCHEM-CRI model outputs are at a $5° \times 5°$ resolution, and are monthly averaged, requiring further interpolation to fit the simulation domain. Table **??** shows all species and example values.

Table 5.2.: Example background chemistry concentrations generated from Background Chemistry Model, for North Atlantic Flight Corridor at 13 km altitude in January.

Background species	**Full chemical name**	**Concentration [ppbv]**
NO2	Nitrogen dioxide	4.605×10^{2}
O3	Ozone	6.763×10^{1}
NO	Nitric oxide	2.150×10^{-2}
CO	Carbon monoxide	8.912×10^{1}
H2O2	Hydrogen peroxide	6.962×10^{-2}
HNO3	Nitric Acid	2.107×10^{-2}
CH4	Methane	1.714×10^{3}
C2H6	Ethane	4.769×10^{-1}
C3H8	Propane	1.120×10^{-1}
NC4H10	n-Butane	2.811×10^{-1}
C2H4	Ethylene	2.447×10^{-3}
C3H6	Propylene	3.175×10^{-5}

TBUT2ENE	t-But-2-ene	7.439×10^{-9}
HCHO	Formaldehyde	3.314×10^{-2}
CH3CHO	Acetaldehyde	2.920×10^{-2}
C5H8	Isoprene	4.947×10^{-8}
C2H2	Acetylene	1.554×10^{-1}
BENZENE	Benzene	2.392×10^{-2}
TOLUENE	Toluene	2.014×10^{-3}
OXYL	o-Xylene	4.037×10^{-4}
C2H5CHO	Propionaldehyde	1.391×10^{-3}
CH3COCH3	Acetone	2.599×10^{-1}
CH3OH	Methanol	2.243×10^{-1}
MEK	Methyl ethyl ketone	3.481×10^{-3}
CH3OOH	Methyl hydroperoxide	6.970×10^{-3}
PAN	Peroxyacetyl nitrate	2.292×10^{-1}
MPAN	Methyl peroxyacetyl nitrate	2.225×10^{-6}

5.4.4. Plume Dataset (PL)

The Plume Dataset comprises all data associated with the plume waypoints from every flight in the simulation. These plume waypoints advect and disperse according to wind and wind shear present in the model, meaning their position, heading and radial dimensions evolve over time. Each row in the dataset contains a unique combination of flight UID, waypoint number and time. Data associated with each waypoint include position (latitude, longitude and altitude), waypoint heading angle, waypoint age, Gaussian covariances in the lateral (σ_{yy}) and vertical (σ_{zz}) direction and plume width. The Flight Dataset is passed through the emissions model, resulting in the estimation of emission masses for each plume waypoint. Corresponding data for fuel flow, fuel burn and the mass of speciated emissions are provided in the dataset. Section **??** gives a detailed explanation of the emissions estimation process, as well as the emissions included in the model. See table A.1.2 for an example Plume Dataset from the GPAT modelling framework.

5.4.5. Emissions Dataset (EMI)

EMI is an output dataset generated from the Plume-to-Grid Model. Geo-vector plume and emissions data across all flights, contained in the Plume Dataset, is passed into this model. The model aggregates the 1D Gaussian plume data to a 2D latitude-longitude grid for further processing. The Emissions Dataset includes all emissions data in concentration form (molecules/cm^3), enabling the summation of background chemistry and emissions concentrations on the same grid, required for the subsequent contrail and chemistry processing.

5.4.6. Contrail Dataset (CONTRAIL)

5.4.7. Chemistry Dataset (CHEM)

The CHEM dataset is an Xarray dataset which serves as a container for all the inputs fed into, and outputs generated from, the BOXM photochemical trajectory model. The inputs include air temperature, M, water vapour, O_2 and N_2 concentrations, SZA, emissions (EMI) and key background species concentrations (BG_CHEM). BOXM creates a number of new variables that are written to the Chemistry Dataset. These are photolysis parameters (J) and coefficients (DJ) for all 96 photochemical reactions in the model, and thermal rate coefficients (RC) for the 512 thermal reactions. The output concentrations (Y) of all 219 species are included, across all time steps and grid cells in the simulation environment. Table **??** shows the structure of the GPAT Chemistry Dataset

5.5. Models and Methods

The input parameters and input datasets to and output datasets from all models have been outlined above. This section delves into the underlying methods and practices used in the GPAT modelling framework, explaining what each model is responsible for, how it functions, as well as why it is necessary in the context of the thesis objectives. The novelty of this framework lies in the plume aggregation approach, and the application of a photochemical trajectory model to simulate aircraft plume evolution on the subgrid scale. This unlocks new possibilities to investigate the cumulative chemical and physical effects that occur locally in aircraft plumes, especially in high-density airspace where two or more plumes intersect and saturate the atmosphere with emissions.

5.5.1. Trajectory Model (traj_gen)

The Trajectory Model generates sampled flight trajectory data from the input Flight Parameters. Firstly, total flight distance is calculated from the product of aircraft speed (fl$_0$_speed) and flight runtime (rt_fl). The leader aircraft's initial coordinates (fl$_0$_coords$_0$), heading (fl$_0$_heading), speed and distance flown are then fed into an external function that carries out forward geodetic computations to estimate the final coordinates of the aircraft at the end of its runtime [186]. A pycontrails Flight object is then created, which resamples and fills the trajectory with intermediate waypoints which correspond with the flight timestep (ts_fl). These waypoints are an important component of the flight data, as they serve as the release points for the ensuing emission pulses created by the aircraft. Lastly, the trajectory data is filtered to only include trajectory points in the simulation domain.

If there is more than one flight present in the simulation, the same process is applied except the initial coordinates are shifted according to the user-defined separation minima (sep_dist). This informs the model of the latitude, longitude and altitude offset to start subsequent flights. Figure 5.2 portrays the data flow for the Trajectory Model.

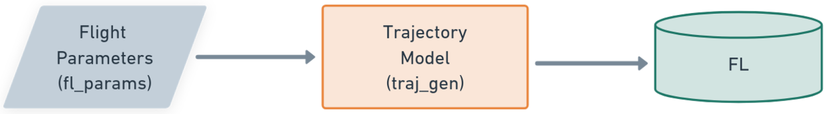


Figure 5.2.

5.5.2. Meteorology Model (gen_met)

This model generates meteorological data across all grid cells and time steps in the modelling environment. It takes in arrays of latitudes, longitudes, altitudes and time and downloads 1998 UK Met Office UM archive data for air temperature (T), water vapour concentrations $[H_2O]$ and wind fields. Pressure (p) is estimated from the model level outputs in table **??**. The density of dry air (ρ_d) is calculated first (5.1), where R_d is the gas constant of dry air. This is then used to quantify specific humidity (q) through (5.2), where M_d is the molar mass of air and N_a is Avogadro's constant ($N_A = 6.022 \times 10^{23}$).

$$\rho_d = \frac{p}{R_d \times T} \tag{5.1}$$

$$q = \frac{[\mathrm{H_2O}] \times [\mathrm{M}]}{N_A \times \rho_d \times 1 \times 10^6} \tag{5.2}$$

M_d and ρ_d are then used to calculate the number density of air [M], where $[O_2] = 0.208 \times [\mathrm{M}]$ and $[N_2] = 0.781 \times [\mathrm{M}]$:

$$[\mathrm{M}] = \frac{N_A \times 1 \times 10^6}{M_d \times \rho_d} \tag{5.3}$$

Solar zenith angle is then calculated across all grid cells and time steps using the pycontrails in-built SZA calculator function **pycontrails_sza**. Knowledge of these meteorological parameters is crucial for estimating the photochemical and microphysical evolution of an aircraft plume and its associated emissions masses.

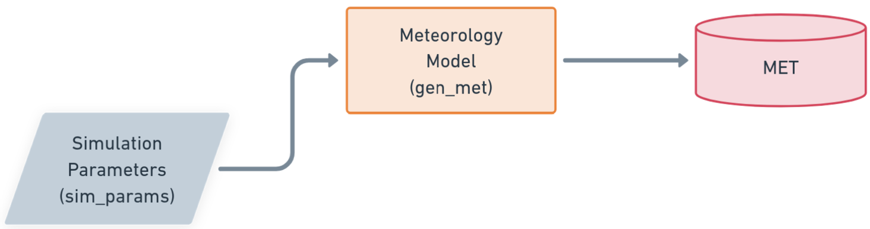


Figure 5.3.

5.5.3. Background Chemistry Model (gen_bg_chem)

The gen_bg_chem model sets an initial baseline for the atmospheric chemistry in the simulated environment, focusing on the dominant species of the upper troposphere where aircraft cruise. This includes nitrogen oxides (NOx), ozone (O_3) and volatile organic compounds (VOCs). The species concentrations are derived from monthly-averaged STOCHEM-CRI global model outputs [108]. These data are converted from molecules/cm^3 to parts per billion by volume (ppbv) and then interpolated to the grid resolution specified in the GPAT Simulation Parameters. The BG_CHEM dataset contains these gridded background concentrations, as shown in figure **??**.

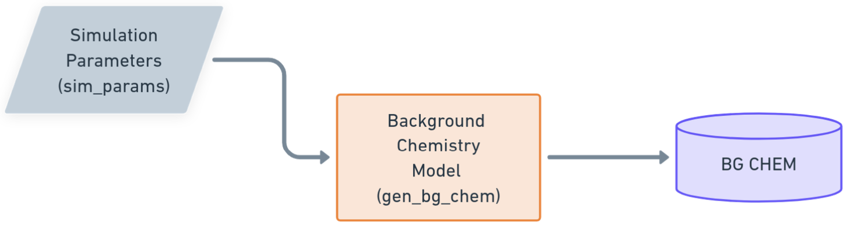


Figure 5.4.

5.5.4. Aircraft Performance Model (ac_perf)

The Aircraft Performance Model computes estimates of thrust, fuel flow, fuel burn and aircraft mass throughout flight, using the Poll-Schumann aircraft performance methodology [161], [162]. These performance calculations are based on aircraft-specific parameters and trajectory data from FL (figure 5.5), as well as temperature and pressure from MET, interpolated and down-selected to fit the flight waypoint positions and time steps.

Prior to passing FL data to the Poll-Schumann performance model, MET data is down-selected to the waypoints along the flight path and true airspeed (TAS) is calculated.

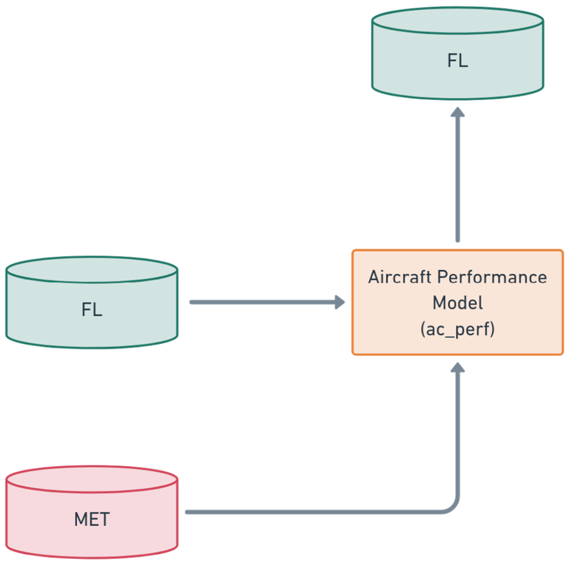


Figure 5.5.

5.5.5. Emissions Model (emissions)

The emissions model utilises aircraft performance outputs from FL to quantify the emissions speciation in the plume, in terms of the emission indices (EIs) and emissions totals for each waypoint. The mass fractions of unburnt hydrocarbons were estimated from empirical data, presented in figure 9 of Masiol et al. [125]. The GPAT Emissions Model is responsible for the emissions masses of carbon dioxide (CO_2), water vapour (H_2O), sulphur dioxide (SO_2), nitric oxide (NO), nitrogen dioxide (NO2), carbon monoxide (CO), formaldehyde (HCHO), acetaldehyde (CH3CHO), ethylene (C2H4), propene (C3H6), acetylene (C2H2), benzene and non-volatile particulate matter (nvPM). Emission species and their corresponding emission indices are provided in table **??**. NOx, CO and CO EIs are calculated from interpolation of the International Civil Aviation Organisation's Engine Emissions Databank [88].

Total emission masses for each plume waypoint in the model are calculated as follows, for primary, secondary and hydrocarbon combustion emissions. Emission masses are denoted by E [kg_{emi}], emission indices are EI [kg_{emi}/kg_{fuel}], fuel flow is FF [kg_{fuel}/s] and emission time step is TS [s].

$$E = EI \times FF \times TS \tag{5.4}$$

Accurate estimation of emission masses at each flight waypoint are a crucial step for subsequent local atmospheric modelling. An aspect as equally important for investigating local atmospheric effects from aviation, is the dispersion and advection of aircraft exhaust plumes and the emissions contained within them.

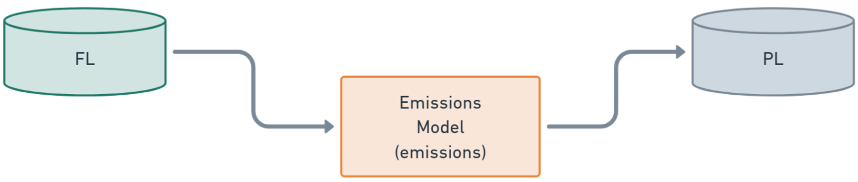


Figure 5.6.

5.5.6. Dispersion/Advection Model (sim_plumes)

The sim_plumes model captures the spread and movement of emissions by simulating the dispersion and advection of all aircraft exhaust plume segments present in the simulation environment. The pycontrails Dry Advection scheme is utilised to approximate the dynamical evolution of all aircraft plumes, which derives from the Contrail Cirrus Prediction Tool Gaussian dispersion methodology **pycontrails_DryAdvection2024**, **Schumann2012**. It takes input of the plume integration time step (dt_integration), plume max age (max_age), intial plume width and depth. Using knowledge of the wind and wind shear fields, the Dry Advection module estimates the change in plume cross-sectional dimensions and geographical position, at each time step.

The output parameters from this model include new latitude, longitude, altitude values, width, height and the covariance matrix (σ), which defines the Gaussian dispersion characteristics of the plume at each waypoint. These parameters make up the majority of the data contained in PL. The FL dataset is effectively merged into this new Plume Dataset, meaning primary flight and emissions data is carried over for further plume calculations.

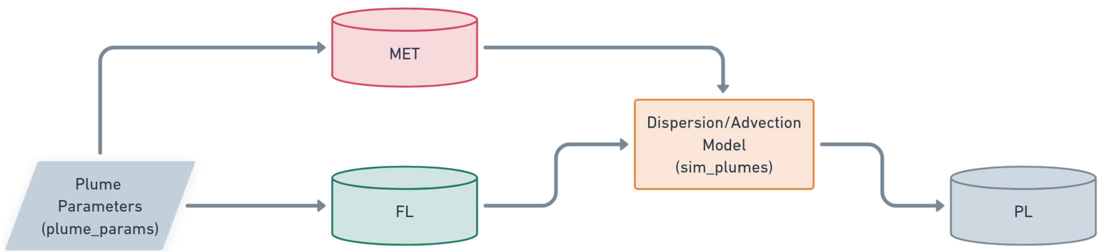


Figure 5.7.

5.5.7. Plume-to-Grid Model (plume_to_grid)

The PL dataset generated previously provides 1D vector data on the temporal evolution of plume waypoints, regarding position, heading, plume cross-sectional dimensions and emissions masses. To simulate the regional-scale chemical and physical processes which occur, and hence capture the cumulative effects due to plume intersections, it is necessary to aggregate the vector data to a common grid.

Gaussian Plume Aggregation

The Plume-to-Grid Model uses a novel conservative remapping method, which remaps Gaussian plumes onto a 2D Eulerian latitude-longitude grid. It does so by dividing the plume into cross-sectional segments which connect plume waypoints. Each segment is subsequently divided into slices which correspond to

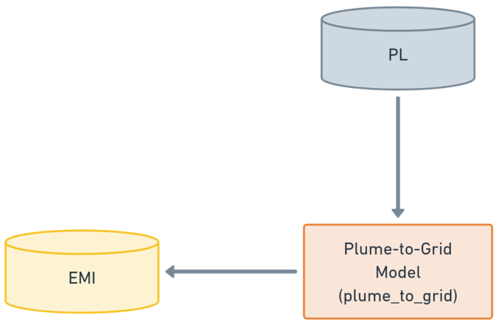


Figure 5.8.

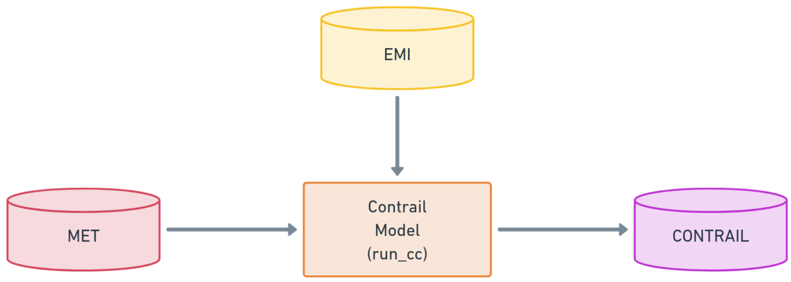


Figure 5.9.

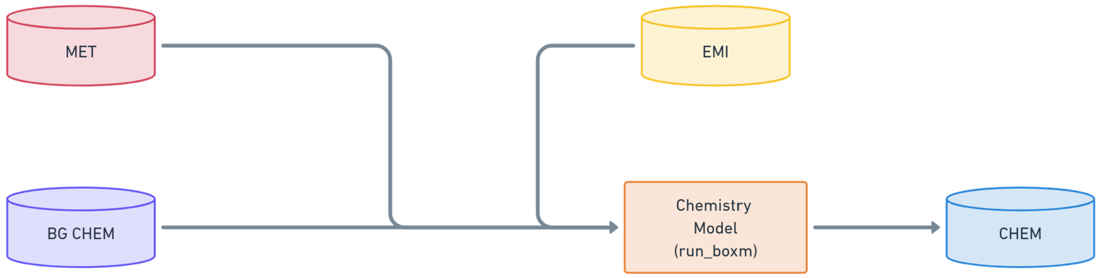


Figure 5.10.

5.5.8. Contrail Model (run_cc)

5.5.9. Chemistry Model (run_boxm)

5.5.10. Validation

Mass Conversation

Box Model Outputs

5.6. Assumptions and Limitations

6. Investigating Plume Overlap as A Climate Solution

6.1. Introduction

In this chapter, GPAT is used to conduct a series of studies to explore the potential for climate benefits through controlled formation flight of commercial aircraft, and the resulting overlap of their exhaust plumes. As explained in detail in section 2.6.4, microphysical and gas-phase photochemical saturation effects can occur under high emissions concentrations in the Upper Troposphere (UT), where aircraft typically cruise. This means that inside a fresh exhaust plume, or in a region where many aircraft plumes are clustered, nonlinear plume-scale saturation effects can occur, influencing the fate of key chemical species and aircraft contrails.

Aircraft nitrogen oxide ($NOx = NO + NO_2$) emissions are responsible for around 16 % of aviation climate impact, primarily due to the short-term production of ozone, a potent greenhouse gas. Emissions of water vapour and particulates from the aircraft exhaust can generate condensation trails (contrails), which are thought to have a net warming effect that is responsible for around 50 % of aviation-induced climate change. There are two primary saturation effects that occur in high-emissions regions at a particular flight level in the UT, which can reduce the climate impact of aviation NOx and contrails; when critical NOx threshold concentrations are surpassed, ozone production efficiency decreases with increasing NOx. It is thought this could reduce the net-NOx Furthermore, when persistent contrails are formed in a close proximity to one another, they experience mutually inhibited growth due to competing for finite ambient water vapour. In both cases, this means that flying aircraft close together and overlapping their exhaust plumes can provide climate benefits over the alternative scenario, where plumes evolve independently of each other.

The following series of studies presented in this chapter focus on NOx saturation, building on many years of experience and previous studies published by the University of Bristol ACRG (e.g. [41]). It is also a direct continuation of the research presented thus far in this thesis, and serves as a prime use case for the GPAT modelling framework.

6.1.1. NOx-Saturated Chemistry Primer

When aircraft plumes overlap, NOx emissions accumulate, and can tip the local domain from a NOx-limited to a NOx-saturated (or VOC-limited) regime. In the NOx-limited regime, hydroxyl radicals (OH) typically react with volatile organic compounds (VOCs), driving the daytime odd hydrogen cycle (HOx = OH + HO_2). This process generates organic peroxy radicals (RO_2), which contribute to ozone formation through the following mechanism:

$$\mathrm{RH + OH \longrightarrow R + H_2O} \tag{R 6.1}$$

$$\mathrm{R + O_2 \longrightarrow RO_2} \tag{R 6.2}$$

Aircraft NOx emissions are composed mostly of nitric oxide (NO $\approx$ 95 %), with the remainder being nitrogen dioxide ($NO_2 \approx$ 5 %) [109]. Aircraft NO speeds up the catalytic oxidation of VOCs and reduces ozone destruction by HOx through reaction with peroxy radicals to form NO_2:

$$HO_2 + NO \longrightarrow OH + NO_2 \tag{R 6.3}$$

$$RO_2 + NO \longrightarrow RO + NO_2 \tag{R 6.4}$$

NO to NO_2 conversion is followed by photolysis of NO_2, which results in the production of atomic oxygen, and subsequently ozone [55] through (R 6.5) and (R 6.6).

$$NO_2 + h\nu(\lambda < 420\,\mathrm{nm}) \longrightarrow NO + O(^3P) \tag{R 6.5}$$

$$O(^3P) + O_2 + M \longrightarrow O_3 + M \tag{R 6.6}$$

When NO_2 concentrations exceed a threshold level, the local system transitions from NOx-sensitive to NOx-saturated conditions due to a high NOx-VOC ratio meaning that termination reactions of NO and NO_2 to reservoir species (e.g. HONO, HNO_3, HO_2NO_2) are often favoured over catalytic ozone formation. These reactions act as a net sink for NOx due to low reactivity and eventual loss through deposition. NO titration then becomes a dominant pathway for ozone destruction, reducing the net O_3 production efficiency (see section 2.6.4: NOx-Saturated Regime for more on this).

6.1.2. Exploiting NOx Saturation through Plume Overlap

The dynamical evolution of aircraft exhaust plumes is a phenomenon often overlooked in global and regional studies of the atmospheric effects resulting from aviation emissions. The dispersion and advection of the exhaust mass happens over timescales of hours, meaning emissions concentrations can remain much higher than background inside the plume during this time. Global and regional models assume instantaneous dispersion (ID), in which dynamics are reduced to homogeneous distribution across the occupied grid cell at the time of emission [151]. This assumption precludes any attempt to model saturation processes which occur on the subgrid scale. Such effects are exacerbated when the aircraft transit frequency through a region is high. The following studies begin to address the nuance in the ozone mitigation potential of formation flight due to plume intersections. When many aircraft plumes overlap, the NOx emissions in a localised region of the atmosphere can reach saturated levels, but this highly depends on a number of factors such as time, location, number of aircraft, separation minima, and metrics used.

Firstly, five representative case scenarios are presented, which show NOx saturation potential for high traffic density hotspot locations. A Background Study is then carried out, which tests a range of metrics indicative of ozone sensitivity across all locations and months of the year. The ID Vs Plume Study addresses the difference in outputs between an instant dilution modelling approach, and the GPAT plume aggregation approach, to examine the differences in ozone output for a range of aircraft configurations. In the Sensitivity Study, five case scenarios are tested under a single aircraft Vs. five aircraft configuration to explore the region's response to NOx saturation. Finally, a Parametric Study is conducted to investigate the variation in saturation benefits under a range of unique aircraft configurations (no. of aircraft and separation distances).

6.2. Background Study

6.3. ID Vs Plume Study

6.4. Sensitivity Study

7. Conclusions and Further Work

7.1. Conclusions

7.2. Further Work

A. Appendix

In this section, any additional tables of data and figures are included, to aid the reader in understanding the models and methods used and results presented throughout this PhD thesis.

A.1. GPAT

A.1.1. Flight Dataset Example

A.1.2. Plume Dataset Example

Table A.1.: Flight Dataset Example (1/3)

Flight UID	**Time**	**Lat. [deg]**	**Lon. [deg]**	**Alt. [m]**	**Pressure level [hPa]**
0	13:00:00	47.500	-32.900	12,500.0	178.645
0	13:02:00	47.499	-32.741	12,500.0	178.645
0	13:04:00	47.498	-32.581	12,500.0	178.645
0	13:06:00	47.498	-32.422	12,500.0	178.645
0	13:08:00	47.497	-32.263	12,500.0	178.645
0	13:10:00	47.496	-32.104	12,500.0	178.645
1	13:00:00	47.500	-32.887	12,500.0	178.645
1	13:02:00	47.499	-32.727	12,500.0	178.645
1	13:04:00	47.498	-32.568	12,500.0	178.645
1	13:06:00	47.498	-32.409	12,500.0	178.645
1	13:08:00	47.497	-32.250	12,500.0	178.645
1	13:10:00	47.496	-32.090	12,500.0	178.645

Table A.2.: Flight Dataset Example (2/3)

Fuel burn [kg]	**Aircraft mass** [kg]	**ROCD**	**Air temp.** [K]	**Spec. hum.** q
125.326	6.098e+4	0.0	219.286	4.700e-5
125.220	6.086e+4	0.0	219.334	4.600e-5
125.115	6.073e+4	0.0	219.383	4.600e-5
125.046	6.060e+4	0.0	219.415	4.500e-5
125.016	6.048e+4	0.0	219.429	4.400e-5
125.016	6.035e+4	0.0	219.443	4.400e-5
125.317	6.098e+4	0.0	219.290	4.700e-5
125.211	6.085e+4	0.0	219.339	4.600e-5
125.106	6.073e+4	0.0	219.387	4.600e-5
125.044	6.060e+4	0.0	219.416	4.500e-5
125.014	6.048e+4	0.0	219.430	4.400e-5
125.014	6.035e+4	0.0	219.444	4.400e-5

Table A.3.: Flight Dataset Example (3/3)

True airspeed [m/s]	**Thrust** [N]	**Thrust setting**	**Engine eff.** η	**Fuel flow** [kg/s]
99.706	9.529e+4	0.940	0.099	1.044
99.707	9.493e+4	0.940	0.099	1.044
99.709	9.457e+4	0.940	0.099	1.043
99.710	9.420e+4	0.940	0.099	1.042
99.712	9.383e+4	0.940	0.099	1.042
99.712	9.383e+4	0.940	0.099	1.042
99.706	9.529e+4	0.940	0.099	1.044
99.707	9.493e+4	0.940	0.099	1.043
99.709	9.457e+4	0.940	0.099	1.043
99.710	9.420e+4	0.940	0.099	1.042
99.712	9.383e+4	0.940	0.099	1.042
99.712	9.383e+4	0.940	0.099	1.042

Table A.4.: Plume Dataset Example (1/4)

Flight UID	**Waypoint No.**	**Time**	**Lat. [deg]**	**Lon. [deg]**	**Alt. [m]**	**Pressure level [hPa]**
0	0	13:00:00	47.500	-32.900	12500.0	178.645
1	0	13:00:00	47.500	-32.887	12500.0	178.645
0	0	13:02:00	47.500	-32.900	12500.0	178.645
0	1	13:02:00	47.500	-32.741	12500.0	178.645
1	0	13:02:00	47.500	-32.887	12500.0	178.645
...	...	...	...	...	...	...
0	5	19:06:00	47.500	-32.104	12500.0	178.645
1	4	19:06:00	47.500	-32.250	12500.0	178.645
1	5	19:06:00	47.500	-32.090	12500.0	178.645
0	5	19:08:00	47.500	-32.104	12500.0	178.645
1	5	19:08:00	47.500	-32.090	12500.0	178.645

Table A.5.: Plume Dataset Example (2/4)

Plume age [hh:mm]	**Heading angle [deg]**	**Plume width [m]**	σ_{yy} [$\mathbf{m^2}$]	σ_{zz} [$\mathbf{m^2}$]	**True airspeed [m/s]**
00:02	359.618	8.107e+1	8.215e+2	4.355e+2	99.706
00:02	359.618	8.107e+1	8.215e+2	4.354e+2	99.706
00:04	359.618	1.438e+2	2.585e+3	5.584e+2	99.706
00:02	359.618	8.106e+1	8.214e+2	4.353e+2	99.707
00:04	359.618	1.438e+2	2.585e+3	5.584e+2	99.706
...	...	...	...	...	...
05:58	359.618	5.298e+4	3.508e+8	2.218e+4	99.712
06:00	359.618	5.344e+4	3.569e+8	2.233e+4	99.712
05:58	359.618	5.297e+4	3.508e+8	2.218e+4	99.712
06:00	359.618	5.342e+4	3.566e+8	2.231e+4	99.712
06:00	359.618	5.341e+4	3.566e+8	2.231e+4	99.712

Table A.6.: Plume Dataset Example (3/4)

Fuel flow [kg/s]	**Fuel burn** [kg]	**CO2** [kg]	**H2O** [kg]	**SO2** [kg]	**nvPM** [kg]	**NO** [kg]	**NO2** [kg]
1.044	125.326	396.029	154.150	0.105	0.009	1.723	0.091
1.044	125.317	396.001	154.140	0.105	0.009	1.723	0.091
1.044	125.326	396.029	154.150	0.105	0.009	1.723	0.091
1.044	125.220	395.696	154.021	0.105	0.009	1.721	0.091
1.044	125.317	396.001	154.140	0.105	0.009	1.723	0.091
...	...	...	...	...	...	...	...
1.042	125.016	395.051	153.770	0.105	0.009	1.717	0.090
1.042	125.014	395.043	153.767	0.105	0.009	1.717	0.090
1.042	125.014	395.043	153.767	0.105	0.009	1.717	0.090
1.042	125.016	395.051	153.770	0.105	0.009	1.717	0.090
1.042	125.014	395.043	153.767	0.105	0.009	1.717	0.090

Table A.7.: Plume Dataset Example (4/4)

CO [kg]	**HCHO** [kg]	**CH3CHO** [kg]	**C2H4** [kg]	**C3H6** [kg]	**C2H2** [kg]	**BENZENE** [kg]
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
...	...	...	...	...	...	...
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000
0.061	0.001	0.000	0.001	0.000	0.000	0.000